Riedel Router Control Software (RRCS)

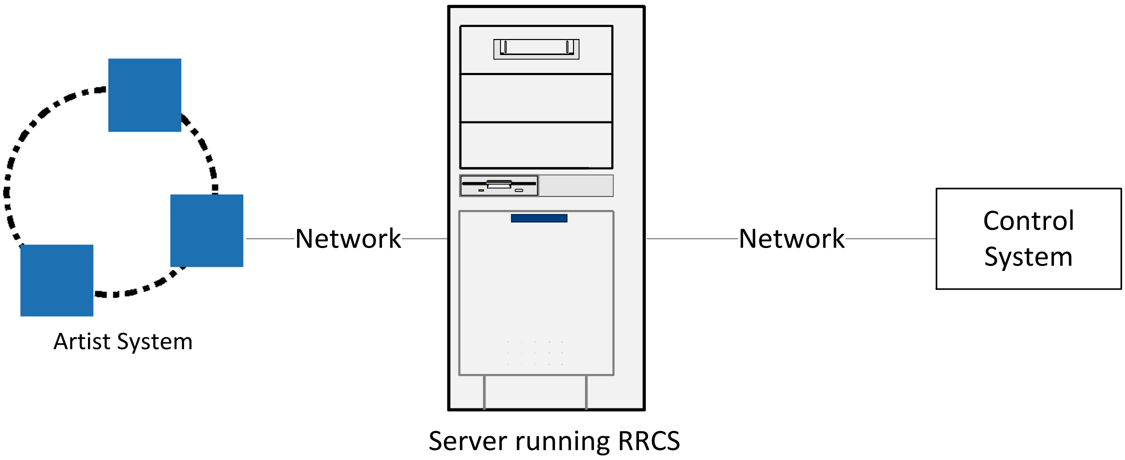
Interface protocol specification

RRCS 8.8.1.Rev1

1 CONTENT

2 Overview

This document describes RRCS (Riedel Router Control Software), the 3rd party gateway to control the Artist intercom system. RRCS is a software running on Windows XP, Vista, 7 or Win10. On one side it is connected to one node of an Artist ring, on the other side it provides a network socket with a disclosed protocol described in this document.



2.1 Feature list

As a rough overview, RRCS provides the following features to external control systems:

- Setting/Removing crosspoints
- Crosspoint volume control and notifications
- Input / Output gain control
- Set alias / label transfer
- GPI Input change notifications
- Switch GPI outputs
- Logic source control and notifications
- System Status / Alarms
- IFB table control
- Panel key configuration
- Send String notifications
- Using multiple Riedel-panels as external keyboards to control other systems (panel-spy-feature)
- Port Cloning

3 Document History

Version 8.8.1

- Introduced support for NSA Devices (NSA-003A, NSA-004A, NSA-005A, NSA-006A, NSA007A, NSA-010C), NSA connections (NSAConnection, NSAConnectionSplit) and NSA GPIOs
  - ConfigurationChangeEx / BufferConfigurationChangeEx→device: You can create, edit and delete NSA Devices. → See 8.10.4.36
  - ConfigurationChangeEx / BufferConfigurationChangeEx→portex: You can now create, edit and delete NSA connections and NSA split connections. See → 8.10.4.6
  - ConfigurationChangeEx / BufferConfigurationChangeEx→clientcard: You can now edit Allocated GPIOs on AES67 SICs. See → 8.10.4.34
  - ConfigurationChangeEx / BufferConfigurationChangeEx→gpio: You can now create, edit and delete GPIOs. See → 8.10.4.37
  - GetAllDevices : Now lists all configured NSA Devices including all properties. → See 8.8
  - GetAllPort, GetPort: Now lists all configured NSA connections and NSA split connections incl. all properties. See → 8.8
  - GetPortExTypeList: Now also lists the two new NSA connection types. See → 8.8
  - Introduced helper type TGPIOAddress. See → 6.7
  - Introduced APIs GetAllGpIns and GetAllGpOuts which returns lists of all configured GPInputs and GPOutputs. See → 8.6
  - SetGpOutput, GetGpInputState and GetGpOutputState support NSA GPIOs
- Introduce Feedback Suppression property for all 12xx Panels connected to Artist-1024 on ConfigurationChangeEx / BufferConfigurationChangeEx →PortEx API. See → 8.10.4.6
- Introduce Auto Dial properties for Codec Connections on ConfigurationChangeEx / BufferConfigurationChangeEx →PortEx API. See → 8.10.4.6
- Alarm types and messages for missing and non-redundant audio are unified.
- Added the ability to enable a keep-alive mechanism on notification channels. See → 9.8

Version 8.7.1

- Introduced support for ‘Stage’ Ports.
  - Introduced new PortType ‘Stage’ for the ObjectType ‘PortEx’ on the ConfigurationChange, ConfigurationChangeEx and BufferConfigurationChange APIs that supports creating, editing and deleting ports of type ‘Stage’. → See 8.10.4.6
  - GetAllPorts / GetPort now returns port of type ‘Stage’ including their specific properties. → See 8.8
  - GetPortExTypeList now returns ‘Stage’ port type. → See 8.8
  - Introduced API ‘GetStageNetAddr’ and ‘SetStageNetAddr’ to get and set Stage net address. → See 8.24
  - Introduced API ‘GetStageRegistryUrl’ and ‘SetStageRegistryUrl’ to get and set the Stage Registry URL. → See 8.24
- Introduced new Alarm “SICFailed” to be used as trigger to start N+1 Redundancy processes.
  - The new Alarm gets raised by RRCS within RegisterForEventsEx, type ‘General’, when a SIC drops out or both of its media interfaces are down. See 9.7
- Please Note: The KeyMode integer values between the APIs ConfigurationChange->Edit->PanelKey and GetKeyConfiguration behave differently. The integer values for ‘Auto’, ‘Momentary’ and ‘Latching’ are increased by 1 on the GetKeyConfiguration API compared to ConfigurationChange->Edit->PanelKey.

Version 8.6.1

- Introduced support for ConnectVoIPDevice
  - Introduced API GetAllDevices that returns all configured Devices of type CodecVoIPDevices including the VoIPDevice specific properties and their linked ports. → See 8.8
  - Introduced new ObjectType ‘device’ on ConfigurationChange, ConfigurationChangeEx and BufferConfigurationChange that supports creating, editing and deleting of devices. → See 8.10.4.36
  - GetAllPorts / GetPort now returns ‘CodecConnection’ port types including their specific codec properties. → See 8.8
  - GetPortExTypeList now returns ‘CodecConnection’ port type. → See 8.8
  - Introduced new PortType ‘CodecConnection’ for the ObjectType ‘PortEx’ on the ConfigurationChange, ConfigurationChangeEx and BufferConfigurationChange APIs that supports creating, editing and deleting ports of type ‘CodecConnection’ → See 8.10.4.6

- GetObjectList, GetObjectPropertyNames and GetObjectProperty now supports devices. → See 8.9
- ConfigurationChange, ConfigurationChangeEx and BufferConfigurationChange now support VoIP specific properties on the object type 'call-to-port-cmd'. → See 8.10.4.12
- GetObjectPropertyNames, GetObjectProperty, GetCommandList and GetPortsCommandList now return the VoIP specific property values for CallToPort commands. → See 8.9

- Introduced support for Trunkline settings.
    - GetAllPorts / GetPort now returns 'TrunkLine' port types including their specific trunkline specific properties. → See 8.8
    - ConfigurationChange, ConfigurationChangeEx and BufferConfigurationChange on PortEx ObjectType now support editing trunkline specific settings. → See 8.10.4.6

Version 8.5.1 05/06/2023

- Introduced API ResetAllNodes to reset all online nodes within the ring. → See 8.22
- Introduced API DeletePortCommands to delete either all commands of a port or all commands of a specific command container. → See 8.23
- Introduced APIs GetAllConferences and GetAllGroups to retrieve a list of all configured conferences / groups including all their properties. → See 8.8
- The properties 'DimLevel' and 'IsTrunkEnabled' are now supported on IFBs and can now be pulled with GetAllIFBs → See 8.8 or being set with ConfigurationChange. → See 8.10.4.4
- Introduced APIs GetKeyConfiguration and GetAllKeyConfiguration to retrieve the current configuration of a specific or all keys. → See 8.8
    - Also introduced TKeyPostion Helper Type. → See 6.5
- When creating commands, RRCS drivers can now control whether to fail or not to fail on duplicate commands with the new optional property 'ForceFail'. → See 8.10.4.8 to 8.10.4.32
- Introduced APIs ConfigurationChangeEx and ApplyConfigurationChangeEx. In contrast to the APIs ConfigurationChange and ApplyConfigurationChange where always the whole request is handled atomar, the new APIs handle individual instructions from a request as an atomar operation. Only when application errors occur within an individual instruction of a request, the configuration changes of the whole request won´t be applied. Please note that multiple instructions that build on one another may fail if one of the first instructions of this sequence fails. → See 8.10.1, 8.10.3, 7
- Introduced support for Dante UICs
    - ConfigurationChange client-card now supports Dante. DanteDeviceName can now be set Dante Client-Card and Dante UIC. → See 8.10.4.34
    - ConfigurationChange portex now supports creation, editing and deletion of 2-wire In/Out, 4-wire and 4-wire split on Dante UICs. → See 8.10.4.6
- Introduced support for Permanent Volume Bars on ConfigurationChange PortEx and GetPort/GetAllPorts APIs for 1200 series panels and 2300 series panels connected to Artist-1024. → See 8.10.4.6, 8.8
- Introduced support Lever-Up key presses on 1200 series Panels with PressKeyEx API (→ See 8.11), GetCommandList/GetPortsCommandLists (→ See 0, 8.9.3) and ConfigurationChange for Remote Key Command (→ See 8.10.4.18).
- Introduced support for Incoming Call Signaling on ConfigurationChange PortEx and GetPort/GetAllPorts API. → See 8.10.4.6, 8.8
- Introduced support for getting and setting the longname of a Net. → See 8.8

Version 8.4.1 01/09/2022

- GetLevelMeterValues API now supports addressing PoolPort inputs. → See 8.5
- PressKey and LockKey APIs now supports addressing PoolPorts → See 8.11
- Introduced APIs GetLTC and SetLTC to retrieve or set the current local trunk controller role on a Node/NIC. → See 8.16
- NetworkSpeed property can now be set on AES67 SICs. → See 8.10.4.34
- Introduced APIs ConnectToArtist and DisconnectFromArtist. → See 8.21

Version 8.3.1 08/04/2022

- Query Port properties: See → 8.8
    - GetAllPorts now returns all properties supported by the port type for each configured port
    - Introduced new API GetPort to query all properties of an individual port

    - Note: The new properties have been added to the well known xml response structure and thus is backward-compatible
- Query command properties: See → 0, 8.9.3
    - GetCommandList now returns all properties for each configured command on the specified command container
    - Introduced new API GetPortsCommandLists to query all configured commands including their properties of an individual port
    - Note: The new properties have been added to the well known xml response structure and thus is backward-compatible
- Introduced GetPortExTypeList to query all supported port types on the CfgChange->PortEx API. See → 8.8
- 1200 series Smartpanels can now be configuration on a AES67 2022-7 interface. See → 8.10.4.6
- S/MUX highspeed option for MADI SICs. See → 8.10.4.34
- Reply Command now supports to enable or disable the Scrolling through the reply history. See → 8.10.4.19.1

Version 8.2.1 12/08/2021

- Global ScrollList: Introduced the ability to configure KeypadShortcut properties on Ports and Groups. See → 8.10.4.6 and 8.10.4.3
- 2300 SmartPanel Support on Artist-1024:
    - The 2300 series SmartPanels now can be created on Artist-1024. See → 8.10.4.6
    - Properties 'UseGroupColorOnPanel' and , GroupColorIndication' can be applied now. See → 8.10.4.6
- Bolero Beltpack Support on Artist-1024
    - Bolero Beltpacks now can be created on Artist-1024 (Media 1 & 2) See → 8.10.4.6
    - Property 'MulticastPortToBolero' now can be set in order to configure the Bolero port Artist connects to.
- ConfigurationChangeRequest error desciption field now includes the index of the failing ConfigChange
- Please Note: The N + 1 Redundancy support is implemented by the API ConfigurationChange-> client-card -> swap. See 0

Version 8.1.1 23/03/2021

- Introduced new ConfigurationChange API ('portex') with enhanced User Interface in regards to 'port' to create, edit and delete ports. → See 8.10.4.6
    - This new API replaces the ConfigurationChange port (see 8.10.4.5). This means the old API is now marked as deprecated and Riedel reserves the right to discontinue (bugfix-)support from version 9.0 on.
    - The creation of ports is more explicit by introducing the parameter 'PortType' instead of the usage of specific sub-structs for certain types. This also reduces the amount of the various specific sub-structs to a few sub-structs that are now working independent from the specific port type to be configured.Furthermore all types of ports can now be created.
    - Please Note: The speration of the port type from the sub-structs implicitly made the whole interface more generic. For instance a user can now define a GroupColor values for every port type, whilst only specific port types really support such parameters to be set. This actually has not changed to behavior in ConfigurationChange port, but users might now get more often into such situations. In the example you will find comments that describe briefly the ruleset that RRCS applies.
    - Also the creation and editing of 2-Wire and 4-Wire types has been simplified. The specific Client-Card related split types has been merged to one generic split type. Users can now also define input and ouput specific sub-structs at once when creating split types or editing a 4-Wire instead of defining them in 2 separate requests. Only when editing a split type, users still have to define settings in 2 separate requests, since the addressing of ports (PortAddress) always points either to an input or to an output.
    - The new interface also supports all new features and enhancements that have been added to ConfigurationChange port interface in this version.
- Introduced Monitoring functionality on certain APIs
    - Added property 'Monitoring' to ConfigurationChange call-to-ifb that allows to configure the monitoring behavior. → See 8.10.4.10.1
    - Added property 'Monitoring' to ConfigurationChange call-to-port that allows to configure the monitoring behavior. → See 8.10.4.12.1

  - Added property 'MonitoringState' to ConfigurationChange panel-key that allows to configure the initial monitoring state of a Panel Key. → See 8.10.4.33.1
- Introduced new operation 'swap' to ConfigurationChange 'client-card'. Note this new operation can only be performed with Artist-1024 SIC configurations. → See 8.10.1, 8.10.4.34
- Introduced the ability to enable and disable ports, groups and IFBs for trunking
  - Introduced Get-/SetTrunkingNetAddr method → See 8.16
  - GetAllPorts now returns information of the current trunked status of each port → See 8.8
  - Added properties 'IsTrunkEnabled', 'IsTrunkLineEnabled' and 'TrunkingPortAddr' to ConfigurationChange port that allows setting up a port for trunking. → See 8.10.4.5.1
- Introduced the ability to configure up to 80 channels on Artist-1024 SICs → See 8.10.4.5; 8.10.4.6
- Added property 'DuplexEnabledForCall2Port' to ConfigurationChange reply-cmd that allows to enable / disable the Duplex function → See 8.10.4.19.1
- Added property 'UseScrollListEntryKeyMode' to ConfigurationChange panel-key that allows to enable / disable the usage of the keymode set up in a scoll list. → See 8.10.4.33.1
- Added property 'PlayMode' to ConfigurationChange port 'PortAES67' struct that allows to configure the play mode for AES67 Panels. → See 8.10.4.5.1
- Added property 'LongName' to ConfigurationChange client-card that allows to configure the long name of SICs. → See 8.10.4.34.1
- Introduced new function 'SetSystemTimeOnAllNodes' that synchronizes all nodes with the current system time, RRCS is running on. → See 8.20
- GetCommandlist now returns uniquely identifiable description for group and conference commands. See → 0
- Bugfix: Renamed parameter 'DestinationUsesSecondChannel' to 'LocalUsesSecondChannel' in ConfigurationChange listen-to-port-cmd. See →8.10.4.13.1

Version 8.00.21 07/09/2020

- RRCS now supports to create, edit and delete a RSP-1216HL Panel (Rsp1216HL). See 8.10.4.5
- Support of PoolPorts on certain APIs
  - Please Note: The PoolPort property on all APIs in RRCS are 0-based. The property is generally treated as an optional parameter that is initialized with '-1' (undefined) by default and has a value range from -1 to 32.
  - {TPortAddress}-, {TGroupPortAddress}-, {TConferencePortAddress}- and {TMemberChangeList}-Struct now has a further optional member 'PoolPort' which generally enables addressing of PoolPorts in both directions, requests and replies. → See 6.1, 6.2, 6.3, 6.4

    This is accompanied by the following APIs:

    - GetAllIfbs → See 8.8
    - GetCommandList → See 0
    - GetObjectList, GetObjectProperty → See 8.9.1
    - ConfigurationChange Group → See 8.10.4.3
    - ConfigurationChange Ifb → See 8.10.4.4
    - ConfigurationChange Port → See 8.10.4.5
    - ConfigurationChange virtual-key → See 8.10.4.35.1
    - ConfigurationChange dim-level-command → See 8.10.4.11
    - ConfigurationChange call-to-port-cmd → See 8.10.4.12
    - ConfigurationChange listen-to-port-cmd → See 8.10.4.13.1
    - ConfigurationChange route-cmd → See 8.10.4.14
    - ConfigurationChange switch-gpio-out-cmd → See 8.10.4.15
    - ConfigurationChange select-audiopatch-cmd → See 8.10.4.16
    - ConfigurationChange control-audiopatch-cmd → See 8.10.4.17
    - ConfigurationChange remote-key-cmd → See 8.10.4.18
    - ConfigurationChange dim-panel-speaker-cmd → See 8.10.4.22
    - ConfigurationChange beep-panel-cmd → See 8.10.4.23
    - ConfigurationChange set-input-output-gain-cmd → See 8.10.4.29

        - ConfigurationChange clone-output-port-cmd → See 8.10.4.32
    - {TCmdPosition}-Struct now has a further optional member 'which enables addressing addressing places of commands on a PoolPort. → See 6.6

      This is accompanied by the following APIs:

        - GetCommandList → See 0
        - ConfigurationChange command-container → See 8.10.4.7
        - ConfigurationChange for all types of commands → See 8.10.4.8 up to 8.10.4.32
    - GetAllPorts, GetAllCaps and GetAllIfbs now enumerates PoolPorts and returns them in the corresponding reply. See → 8.8
    - Introduced GetPoolPortInfo to collect PoolPort information of a specific port. → See 8.19
    - ConfigurationChange Port now allows to also to create and delete PoolPorts at Ports, generally supporting PoolPorts (SipPhonePort, TelephoneCodec) → See 8.10.4.5
- Added several missing consistency checks which which may have corrupted configuration, which may which led to error messages in various Director dialogues or configuration migration issues. Several creation, update and deletion rules for certain confuration items have been moved from Director´s UI layer into the business logic layer and are now shared between Director and RRCS. This essentially applies to the following areas:
    - Commands are not allowed on every Panel, every command container (key, virt. Key, virt. Function, etc)
    - All configurationproperties that can be found in the tabs Details1 and -2 in property dialog of a port are dependend on the port type
    - AES67 Input and Output configuration has several properties that are dependend among themselves
- ConfigurationChange Port allows to set InitialIfbVol → See 8.10.4.5
- Get/Set- PortAlias/PortLabel now support Split-4Wire by optional parameter IsInput. It´s default is 0/false. -> See 8.3 and 0

Version 8.00.12 14/05/2020

- Support of Artist1024.
- When addressing a configuration item like a Port on an Artist1024 node, you now need to use the SIC address instead of the Node address as value in the node property (of for instance in TPortAddress or TCmdPosition structs, ...). In Director you can set/view the SIC addresses in the property dialog of a Artist1024 node.
- With introducing the redundancy feature 2022-7 in Artist1024, we also introduced a special port addressing scheme for AES67 SICs. Generally ports on media 1 are to be addressed from 0 to 127, ports on media 2 from 512 to 639, ports on -7 from 1024 to 1151. Ports on Madi SICs are to be addressed from 0 to 127 (0..63 = Interface1, 64-127 = Interface2).
- Along with Artist1024 support, licensing features have been added → see 8.18
- ConfigurationChange for client-cards now allows to change the amout of allocated ports per media interface on Artist1024 SICs. Furthermore Madi & AES67 Sic has an additional optional property 'MediaInterface' (Default Media Media1) to assign specific other properties to a media interface. → see 8.10.4.34
- On Ports configured on Madi SIC you can now set the Madi channel individually. Director sets a 1:1 mapping when migrating configuration from an older confoiguration scheme/version to the v8.0.11 configuration scheme, but for newly created ports, Director users now have to set the channels individually. In RRCS per default a 1:1 mapping is still set up if not defined by RRCS users individually. → see 8.10.4.5 InputChannel & OutputChannel on the Wire family structs
- ConfigurationChange for AES67 client-card now allows to configure NMOS properties → see 8.10.4.34
- ConfigurationChange for ports now allows to set 'LongName', 'Label', 'Alias' and 'Subtitle' → See 8.10.4.5.1
- ConfigurationChange for panel-key now allows to set 'AutoSubTitle', 'SubTitle' and 'GroupColor' → see 8.10.4.33.1
- ConfigurationChange now allows to create and delete conferences → see 8.10.4.2

- ConfigurationChange for call-to-conference now supports AllowSelectingConf and AllowChangingConfDest → See 8.10.4.8.1
- Added GetAllActiveXpsCount
- ConfigurationChange for AES67Inputs now allows to set ‘PlayMode’ to switch between synton and synchron mode → See 8.10.4.5.1
- ConfigurationChange now allows to set ‘IGMPVersion‘, ‘PtpPriority2’ and ‘AnnounceInterval’ to AES67 Client Cards → see 8.10.4.34
- Introduced new feature ‘UpdateTrunkingLabels’ → See 8.16
- Bugfix: Label & Alias are available again on ConfigurationChange of a Port → see 8.10.4.5

Version 7.40.1 04/03/2019

- ConfigurationChange supports creating, editing and deleting PortRSP1232HL → See 8.10.4.5.1

Version 7.30.1 06/08/2018

- ConfigurationChange supports editing Madi-Client Card → See 8.10.4.34
- ConfigurationChange supports editing several port types → See 8.10.4.5.1
- ConfigurationChange supports editing several general settings, like Vox… → See 8.10.4.5.1
- ConfigurationChange supports creating and deleting virtual keys → See 8.10.4.35
- Added example how to create and delete call to trunking enbabled IFBs

Version 7.20.1 05/03/2018

- Support for dial functionality
    - Added DialNumber
    - Added HangUpCall
    - Added LineStatus

Version 7.10.2 18/09/2017

- Bugfix: Values of boolean type in XML-RPC example described in section 8.10.4.12 (call-to-port-cmd) are now excellent correctly

Version 7.10.1 07/06/2017

- ConfigurationChange supports editing properties of port type SIP Phone, RSP, DSP, Bolero Beltpack, AES67Input and AES67Output. -> See chapter 8.10.4.6
- ConfigurationChange supports editing AES67-Client Card → See chapter 8.10.4.34
- Bugfix: Wrong data type definitions in ClientCard VoIP properties
- Bugfix: VoIP port properties accidentally added to GetAllIFB instead of GetAllPort
- Bugfix: Wrong data type definition in SetXPVolume (bool vs boolean)

Version 6.90.5 18/09/2017

- Bugfix: Values of boolean type in XML-RPC example described in section 8.10.4.12 (call-to-port-cmd) are now excellent correctly

Version 6.90.4 07/06/2017

- Bugfix: RRCS Specification: Client card VoIP properties are not accessible
- Bugfix: RRCS Specification: Port VoIP properties property type column is messed up.

Version 6.90.3 10/11/2016

- Bugfix: RRCS Specification: TrunkPortAddr and TrunkNetAddr has to be TrunkingPortAddr and TrunkingNetAddr
- Added “GetTrunkIfbs”
- Bugfix: RRCS Specification: XpVolumeChangeAdd must be XpVolumeChangeRegistryAdd
- Pdf document now has bookmarks
- Bugfix: RRCS Specification: change boolean to boolean due to RRCS is case sensitive and copying from specification may not work directly.

Version 6.90.1 22/04/2016

- Compatibility with 6.90
- Support for VoIP parameters (client card & port)
- Added “GetTrunkPorts”Added “GetTrunklineSetup”
- Adedd“GetTrunklineActivities”

Version 6.80.1 27/04/2016

- Compatibility with Release 6.80
- Added “Set label“ to the features list
- Added “Port Label” section

Version 6.70.2 06/10/2014

- Added “Set alias / label transfer“ to the features list

Version 6.70.1 24/04/2014

- Compatibility with Relase 6.70

Version 6.60.2 11/03/2013

- Compatibility with Relase 6.60

Version 6.50.2 21/04/2012

- RRCS alarms supported according to the new Artist alarm system. Outgoing alarm notifications have not changed.

Version 6.40.5 05/04/2012

- New function GetAllLogicSources_v2

Version 6.30.3 14/04/2011

- IFB Volume

Version 6.30.1 08/03/2011

- Port Cloning

Version 6.20.3 21/09/2010

- The helper type TGroupPortAddress (see 6.2) may contain the member ‘IsInput’ to address both: matrix outputs and inputs (which is a new feature since artist v6.20)

Version 6.20.2 14/09/2010

- The IFB properties ‘Input’, ‘Output’ and ‘MixMinus’ can be groups (see 8.10.4.4)
- The command position can now also be a virtual key (see 6.6)
- The new object type ‘command-container’ allows to delete all commands from a given command position (see 8.10.4.7)

Version 6.20.1 30/07/2010

- possibility to create all available artist commands

Version 6.10.1 30/06/2010

- Added method documentation:
- IsRegisteredForAllEvents
- RegisterForAllEvents
- UnregisterForAllEvents
- ChangePanelSpy
- Added notifications:
- GetAlive
- PanelSpyStateChanged
- PanelSpyRotateEvent
- PanelSpyKeyEvent
- PanelSpyFuncKeyEvent
- PanelSpyNumKeyEvent
- Added error code 26 (limit exceeded)

Version 6.00 26/02/2010

- Added documention for
- ‘ApplyConfigurationChange’
- and ‘BufferConfigurationChange’

Version 2.0 18/10/2009

- Changed document layout
- Added logic sources
- Added object lists
- Added configuration changes
- Added Volume notifications
- Added XP change notifications
- Added GetAllIFBs (Artist version >= 5.91)
- Added GetAllPorts (Artist version >= 5.91)
- Added GetCommandList (Artist version >= 5.91)
- Added GetErrorCodeList (Artist version >= 5.91)
- Added IsConnectedToArtist (Artist version >= 5.91)
- Added IsRegisteredForEvents (Artist version >= 5.91)

Version 1.9 19/12/2008

- Added methods for Logic Sources

4 Glossary

8 char. Name	**8 character name, which is typically shown on a Riedel panel key**
Conference	A communication with n members. If one member is talking the other n-1 members are listening.
Longname	32 character name
Net	An Artist network containing several Nodes
Node	A single Artist Frame with at least one Node Controller and client cards
Object ID	32 Bit unique Identifier
Port	A 4-Wire/4-Wire-split audio port or a Panel
RRCS	Riedel Router Control Software
RPC	Remote Procedure Call
XML	Extended Mark-up Language

5 Interface Specification

5.1 Overview

The following figure illustrates the protocol structure of the interface. This chapter describes the individual layers in detail.

Ethernet
IP
TCP
HTTP
Messages in XML

5.2 Physical Layer and data link

Generally there aren't any limitations concerning physical network type or topology. However Ethernet is the favorite network type since the gateway applications are running on a standard PC.

5.3 Internet Protocol (IP)

RRCS and RBIS make use of IP. There are no further restrictions concerning the IP-addresses.

5.4 Transmission Control Protocol (TCP)

RRCS and RBIS make use of TCP. The TCP port settings are provisionable. RRCS will listen to TCP Port 8193 per default. RBIS will listen to TCP Port 8194 per default.

5.5 Hypertext Transfer Protocol (HTTP)

RRCS and RBIS make use of HTTP. All messages must be HTTP-POST requests. A User-Agent and Host must be specified. The Content-Type must be text/xml. The Content-Length must be specified and must be correct.

5.6 Extended Markup Language – Remote Procedure Call (XML-RPC)

All messages have to be encoded in XML and follow the XML-RPC specification. What is XML-RPC? It's a spec and a set of implementations that allow software running on disparate operating systems, running in different environments to make procedure calls over a network. It's remote procedure calling using HTTP as the transport and XML as the encoding. XML-RPC is designed to be as simple as possible, while allowing complex data structures to be transmitted, processed and returned

This chapter is a brief description of XML-RPC. Further information can be found on http://www.xmlrpc.com/

5.6.1 XML-RPC structure

Example:

```
<?xml version="1.0"?>
<methodCall>
   <methodName>examples.getStateName</methodName>
   <params>
      <param>
         <value><i4>41</i4></value>
         </param>
      </params>
   </methodCall>
```

The XML code has a single <methodCall> structure. The <methodCall> must contain a <methodName> sub-item, a string, containing the name of the method to be called. The string may only contain identifier characters, upper and lower-case A-Z, the numeric characters, 0-9, underscore, dot, colon and slash.

If the procedure call has parameters, the <methodCall> must contain a <params> sub-item. The <params> sub-item can contain any number of <param>s, each of which has a <value>.

5.6.2 Scalar <value>s

<value>s can be scalars, type is indicated by nesting the value inside one of the tags listed in this table:

Tag	Type	Example
<i4> or <int>	four-byte signed integer	-12
<boolean>	0 (false) or 1 (true)	1
<string>	string	hello world
<double>	double-precision signed floating point number	-12.214
<dateTime.iso8601>	date/time	19980717T14:08:55
<base64>	base64-encoded binary	eW91IGNhbid0IHJlYWQgdGhpcyE=

If no type is indicated, the type is string.

5.6.3 <struct>s

A value can also be of type <struct>.

A <struct> contains <member>s and each <member> contains a <name> and a <value>.

Here's an example of a two-element <struct>:

```
<struct>
   <member>
      <name>lowerBound</name>
      <value><i4>18</i4></value>
      </member>
   <member>
      <name>upperBound</name>
      <value><i4>139</i4></value>
      </member>
   </struct>
```

<struct>s can be recursive, any <value> may contain a <struct> or any other type, including an <array>, described below.

5.6.4 <array>s

A value can also be of type <array>.

An <array> contains a single <data> element, which can contain any number of <value>s.

Here's an example of a four-element array:

```
<array>
   <data>
      <value><i4>12</i4></value>
      <value><string>Egypt</string></value>
      <value><boolean>0</boolean></value>
      <value><i4>-31</i4></value>
      </data>
   </array>
```

<array> elements do not have names.

You can mix types as the example above illustrates.

<arrays>s can be recursive, any value may contain an <array> or any other type, including a <struct>, described above.

5.6.5 XML-RPC Response

Here's an example of a response to an XML-RPC request:

HTTP/1.1 200 OK

Connection: close

Content-Length: 158

Content-Type: text/xml

Date: Fri, 17 Jul 1998 19:55:08 GMT

Server: UserLand Frontier/5.1.2-WinNT

```
<?xml version="1.0"?>
<methodResponse>
   <params>
      <param>
         <value><string>South Dakota</string></value>
       </param>
   </params>
</methodResponse>
```

Response format:

Unless there's a lower-level error, always return 200 OK.

The Content-Type must be text/xml. Content-Length must be present and correct.

The body of the response is a single XML structure, a <methodResponse>, which can contain a single <params> which contains a single <param> which contains a single <value>.

The <methodResponse> could also contain a <fault> which contains a <value> which is a <struct> containing two elements, one named <faultCode>, an <int> and one named <faultString>, a <string>.

A <methodResponse> can not contain both a <fault> and a <params>.

Fault example:

HTTP/1.1 200 OK

Connection: close

Content-Length: 426

Content-Type: text/xml

Date: Fri, 17 Jul 1998 19:55:02 GMT

Server: UserLand Frontier/5.1.2-WinNT

```
<?xml version="1.0"?>
<methodResponse>
   <fault>
      <value>
         <struct>
            <member>
               <name>faultCode</name>
               <value><int>4</int></value>
               </member>
            <member>
               <name>faultString</name>
               <value><string>Too many parameters.</string></value>
               </member>
            </struct>
         </value>
      </fault>
   </methodResponse>
```

6 Helper types

Helper types are used in the following chapters as placeholders in order to reduce space in this document.

6.1 TPortAddress

The port-address type is used to address Artist ports. The format is as follows:

```
<value><struct>
   <member>
      <name>Node</name>
      <value><i4>{Node}</i4></value>
   </member>
   <member>
      <name>Port</name>
      <value><i4>{Port}</i4></value>
   </member>
   <member>
      <!-- optional, default is false/0-->
      <name>IsInput</name>
      <value><boolean>{0/1}</boolean></value>
   </member>
   <member>
      <!-- optional, default is -1 (Not defined) -->
      <name>PoolPort</name>
      <value><i4>{0..32}</ i4></value>
   </member>

</struct></value>
```

If a null-port is required (e.g.: to clear the mix-minus of an IFB), the Node and Port member must be set to 0.

6.2 TGroupPortAddress

The group-port-address type represents a member in an Artist-group.

The format is as follows:

```
<value><struct>
<!—note: since artist version 6.20 or later group members can be matrix inputs. Thereto it is necessary to be
able to separate input from outputs. To be backward compatible, RRCS adds the ‘IsInputOnly ‘member for pure
matrix inputs. -->
   <member>
      <name>IsInputOnly</name>
      <value><boolean>1</boolean></value>
   </member>
   <member>
   <member>
      <name>Node</name>
      <value><i4>{Node}</i4></value>
   </member>
   <member>
      <name>Port</name>
      <value><i4>{Port}</i4></value>
   </member>
   <member>
      <name>SecondChannel</name>
      <value><boolean>{0/1}</boolean></value>
   </member>
   <member>
      <!-- optional, default is -1 (Not defined) -->
      <name>PoolPort</name>
      <value><i4>{0..32}</ i4></value>
   </member>
</struct></value>
```

6.3 TConferencePortAddress

The conference-port-address type represents a member in an Artist-conference.

The format is as follows:

```
<value><struct>
   <member>
      <name>Node</name>
      <value><i4>{Node}</i4></value>
   </member>
   <member>
      <name>Port</name>
      <value><i4>{Port}</i4></value>
   </member>
   <member>
      <name>UseSecondChannel </name>
      <value><boolean>{0/1}</boolean></value>
   </member>
   <member>
      <!-- talk right -->
      <name>Talk</name>
      <value><boolean>{0/1}</boolean></value>
   </member>
   <member>
      <!-- listen right -->
      <name>Listen</name>
      <value><boolean>{0/1}</boolean></value>
   </member>
   <member>
      <!-- optional, default is -1 (Not defined) -->
      <name>PoolPort</name>
      <value><i4>{0..32}</ i4></value>
   </member>
</struct></value>
```

6.4 TMemberChangeList

This type is used to add and/or remove ports to/from a group. The format is as follows:

```
<value><array><data>
   <!-- 1st array element -->
   <value><struct>
      <!-- tells, whether this port should be added or removed from the group-->
      <member>
         <name>AddToGroup</name>
         <value><boolean>0</boolean></value>
      </member>

      <!-- the port address which should be added or removed from the Group (notice: this is the type
           TPortAddress -->
      <member>
         <name>PortAddress</name>
         <value><struct>
            <member>
               <name>IsInput</name>
               <value><boolean>0</boolean></value>
            </member>
            <member>
               <name>Node</name>
               <value><i4>{Node}</i4></value>
            </member>
            <member>
               <name>Port</name>
               <value><i4>{Port}</i4></value>
            </member>
            <member>
               <!-- optional, default is -1 (Not defined) -->
               <name>PoolPort</name>
               <value><i4>{0..32}</ i4></value>
            </member>
         </struct></value>
      </member>

      <!-- tells, whether the 1st or the 2nd channel of the port is used -->
      <member>
         <name>UseSecondChannel</name>
         <value><boolean>{0/1}</boolean></value>
      </member>
   </struct></value>

   <!-- additional array elements -->

</data></array></value>
```

6.5 TKeyPosition

TKeyPosition defines a key on a Panel / Expansion Panel or a virtual key of any port type. If you want to specify a key, you need to use the members ExpansionPanel, Page and KeyNumber.

The member 'PositionType' always exists and can currently have one of the following values:

- 'key'
- -'virtual-key'

RRCS ignores struct-members that are unnecessary.

Key position of a physical key

The member expansion panel only exist, if the is on an expansion panel.

```
<struct>
<member>
   <name>PositionType</name>
   <value><string>key</string></value>
</member>
<member>
   <name>Node</name>
   <value><int>{ ... }</int></value>
</member>
<member>
   <!– counting starts with 0 -->
   <name>Port</name>
   <value><int>{...}</int></value>
</member>
<member>
   <!– optional, counting starts with 0 -->
   <name>PoolPort</name>
   <value><int>{...}</int></value>
</member>
<member>
   <!– counting starts with 1 -->
   <name>ExpansionPanel</name>
   <value><int>{...}</int></value>
</member>
<member>
   <!– counting starts with 1 -->
   <name>Page</name>
   <value><int>{...}</int></value>
</member>
<member>
   <!– counting starts with 1-->
   <name>KeyNumber</name>
   <value><int>{...}</int></value>
</member>
<member>
   <!-- optional,  default is false -->
   <name>IsInput</name>
   <value><boolean>{…}</boolean></value>
</member>
</struct>
```

6.6 TCmdPosition

TCmdPosition defines the place of a command. Commands can be placed onto a key or a virtual Function of a port. If you want to specify a key, you need to use the members ExpansionPanel, Page and KeyNumber. The member VirtualFunction is nonexistent in this case. If you want to specify a virtual Function, you need to use the member virtual function. In this case ExpansionPanel, Page and KeyNumber are nonexistent.

The member 'PositionType' always exists and can currently have one of the following values:

- 'key'
- 'virtual-function'
- -'virtual-key'

The number of available position types can grow in the future, e.g. "virtual-key" or "gp-output"

RRCS ignores struct-members that are unnecessary.

Command position is on a key

The member expansion panel only exist, if there is an expansion panel.

```
<struct>
<member>
   <name>PositionType</name>
   <value><string>key</string></value>
</member>
<member>
   <name>Node</name>
   <value><int>{ ... }</int></value>
</member>
<member>
   <!– counting starts with 0 -->
   <name>Port</name>
   <value><int>{...}</int></value>
</member>
<member>
   <!– optional, counting starts with 0 -->
   <name>PoolPort</name>
   <value><int>{...}</int></value>
</member>
<member>
   <!– counting starts with 1 -->
   <name>ExpansionPanel</name>
   <value><int>{...}</int></value>
</member>
<member>
   <!– counting starts with 1 -->
   <name>Page</name>
   <value><int>{...}</int></value>
</member>
<member>
   <!– counting starts with 1-->
   <name>KeyNumber</name>
   <value><int>{...}</int></value>
</member>
<member>
   <!-- optional,  default is false -->
   <name>IsInput</name>
   <value><boolean>{…}</boolean></value>
</member>
</struct>
```

Command position is on a virtual function

The member 'VirtualFunctionType' can have one of the following values:

- 'vox'
- 'always'
- 'on-call'

```
<struct>
<member>
   <name>PositionType</name>
   <value><string>virtual-function</string></value>
</member>
<member>
   <name>Node</name>
   <value><int>{ ... }</int></value>
</member>
<member>
   <name>Port</name>
   <value><int>{...}</int></value>
</member>
<member>
   <name>VirtualFunctionType</name>
   <value><string>{...}</string></value>  <!– e.g. {vox} -->
</member>
<member>
   <!-- optional,  default is false -->
   <name>IsInput</name>
   <value><boolean>{…}</boolean></value>
</member>
</struct>
```

6.7 TGPIOAddress

The gpio-address type is used to address Artist GPIOs. GPIOs can be located on client cards, panels or SICs. All three variants are mapped using this type.

Please Note: You can dynamically assign up to 64 GPIOs per SIC. How those 64 GPIOs are distributed across the media interface get´s defined by the Allocated-GPIOs configuration you can set on AES67-SICs. To ensure a stable addressing scheme GPIOs located on a AES67-SIC gets addressed the same way Ports get addressed. Thus an addressing range from 0-63 is reserved for Media1-GPIOs, 64-127 is reserved for Media2-GPIOs and 128-191 is reserved for #7-GPIOs.

Addressing a GPIO located on a client-card

```
<member>
    <name>GpioAddress</name>
    <value><struct>
            <member>
                    <name>Index</name>
                    <value><i4>5</i4></value>
            </member>
            <member>
                    <name>IsInput</name>
                    <value><boolean>1</boolean></value>
            </member>
            <member>
                    <name>Node</name>
                    <value><i4>2</i4></value>
            </member>
            <member>
                    <name>Bay</name>
                    <value><i4>6</i4></value>
            </member>
    </struct></value>
</member>
```

Addressing a GPIO located on a Port

```
<member>
    <name>GpioAddress</name>
    <value><struct>
            <member>
                    <name>PortAddress</name>
                    <value><struct>
                            <member>
                                    <name>IsInput</name>
                                    <value><boolean>1</boolean></value>
                            </member>
                            <member>
                                    <name>Node</name>
                                    <value><i4>4</i4></value>
                            </member>
                            <member>
                                    <name>Port</name>
                                    <value><i4>9</i4></value>
                            </member>
                    </struct></value>
            </member>
            <member>
                    <name>Index</name>
                    <value><i4>0</i4></value>
            </member>
            <member>
                    <name>IsInput</name>
                    <value><boolean>1</boolean></value>
            </member>
    </struct></value>
</member>
```

Addressing a GPIO located on a SIC

```
<member>
    <name>GpioAddress</name>
    <value><struct>
            <member>
                    <name>Index</name>
                    <value><i4>5</i4></value>
            </member>
            <member>
                    <name>IsInput</name>
                    <value><boolean>1</boolean></value>
            </member>
            <member>
                    <name>Node</name>
                    <value><i4>4</i4></value>
            </member>
    </struct></value>
</member>
```

6.8 TTextColor

This type is used to define color of a key on panel (when custom key color is supported).

```
<member>
   <name>TextColor</name>
   <value><struct>
      <!—boolean swtich that defines if default color should be used or RGB values defined below--!>
      <member>
         <name>PanelDefault</name>
         <value><boolean>0</boolean></value>
      </member>
      <member>
         <name>Red</name>
         <value><i4>254</i4></value>
      </member>
      <member>
         <name>Green</name>
         <value><i4>0</i4></value>
      </member>
      <member>
         <name>Blue</name>
         <value><i4>0</i4></value>
      </member>
   </struct></value>
</member>
```

6.9 Miscellaneous

Expressions in curly brackets used in this chapter are placeholders. The following table is a brief description:

Expression	Description
{Net} {Source Net} {Dest Net} {Net of memb#n}	Artist Network Address. Range 1..255. In most cases there is only one network with address 1.
{Node} {Source Node} {Dest Node} {Node of memb#n}	Addresses Artist 32/64/128 nodes and Artist1024 SICs. Range 2..255
{Port} {Source Port} {Dest Port} {Port of memb#n}	Artist Port address. Range 0.. 1151. For Artist32/64/128 Port address = ((Slot no. – 1) * 8) + Port no. on Client Card -1 e.g. Port 5 on Slot 2 has the port address ((2 – 1) * 8 + 5 – 1) = 12 For Artist 1024 AES67 SIC: ports on media 1 are to be addressed from 0 to 127, ports on media 2 from 512 to 639, ports on -7 from 1024 to 1151. Madi SICs are to be addressed from 0 to 127 (0..63 = Interface1, 64-127 = Interface2).
{Slot-of-client-card}	Client Card slot number. For Artist S the range is 1..4, for Artist M it's 1..16
{Number of XPs}	Number of Crosspoints in the message.
{Priority}	Priority of a crosspoint. Range 0..4 0 = below standard 1 = standard 2 = high 3 = paging 4 = emergency
{Gain}	Input Gain: Range = -36,-35,..,0,..,35,36 gain [dB] = Gain / 2.0
{Status}	Status. Active = 1. Inactive = 0.
{Number of Ports}	Number of Ports in the message.
{IP-address}	IP address in syntax "xxx.xxx.xxx.xxx" or "localhost"
{TCP-Port}	TCP Port number.
{GatewayState}	Gateway status. "Working" or "Standby".
{Volume}	<= 0 → mute 1..255 → (volume-230)/2 dB > 255 → +12.5 dB
{Alias}	Alias name for the given port which is displayed on the panel keys. A string with maximum of 8 characters.
{Slot}	if Port = 128 (GPIO card): 0..15 → bay 1..16 16 → bay X 17 → bay Y 20 → bay B If Port = 0,1,..,127 (Panel GPIO) slot-value not relevant
{GPIO no.}	GPIO number 0..15 on this port / slot
{ErrorCode}	Error code. The error code table provides detailed information.

Expression	Description
{TransKey}	The transaction key is used in all requests. The response to a request must use the same transaction key. The String has a starting charcater followed by 10 digits. The starting character is "R" if RRCS sends a request. An external Control System can choose a different character. There are no limitations. The 10 digits are free of choice. It's up to the sender of the request how to choose the 10 digits.
{EthernetLinkMode}	Range 1..5 1 = AutoNegotiation 2 = 10 Mbit Half 3 = 10 Mbit Full 4 = 100 Mbit Half 5 = 100 Mbit Full
{TrunkCmdType}	1 = Call to Port 2 = Listen to Port 4 = Call to Conference 5 = Call to Group 48 = Call to IFB 201 = IFB Listen to port 202 = Listen to telex IFB Listen Source 203 = Listen to Artist IFB - MixMinus 204 = Listen to Artist IFB-Input 205 = Conference Call to Conference
{LineStatus}	0 – unknown 1 – Line is free 2 – Waiting for Connection 3 – Line is busy

7 Error codes

The following error codes apply to all APIs defined in this document.

Error code	Description
0	Success
1	Transaction key invalid
2	Net address invalid
3	Node address invalid
4	Port address invalid
5	Slot no. invalid
6	Input Gain invalid
7	IP-address invalid
8	TCP-Port invalid
9	Label invalid
10	Conference position invalid
11	Operation failed, because Artist-network not connected
12	Operation failed, because route does not exist
13	Operation not possible, because gateway is standby
14	XML-RPC parameters wrong for this request
15	Invalid conference or conference not found
16	Invalid conference member or conference member not found
17	Invalid priority
18	Invalid GPIO number
19	Invalid gain value
20	Timeout
21	No permission
22	Object does not exist
23	No USB-dongle available
24	Port is not online
25	Object property not supported
26	Limit exceeded
99	Generic error

With version 8.5 new ErrorCodes have been introduced that are only used on the APIs ConfigurationChangeEx and BufferConfigurationChangeEx. These errors are categorized in Application Erros, XML Errors and Cfg Errors.

Error code	Description
0	Success
1	Internal Application Error
2	Configuration unavailable / not (yet) loaded
3	Node address invalid
101	XML member required / mandatory
102	XML member ambiguous
103	XML member invalid
104	XML member unknown
105	XML member value type boolean required
106	XML member value type int required
107	XML member value type struct required
108	XML member value type array required

Error code	Description
109	XML member value type string required
110	XML member value invalid
111	XML member value out of range
201	Instruction unsupported with current configuration
202	Duplicate configuration object
203	Addressed configuration object invalid
204	Addressed configuration object unsupported
205	Address already in use in configuration
206	Configuration member unsupported
207	Configuration member ambiguous
208	Configuration value invalid
209	Configuration value already in use
401	Generic

8 Incoming Remote Procedures requests → RRCS

8.1 Crosspoints

Description	Remote Procedure	Parameters	Return Value on success
Activate XP source, destination. With standard priority	SetXp	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Source Net}</int></value></param> <param><value><int>{Source Node}</int></value></param> <param><value><int>{Source Port}</int></value></param> <param><value><int>{Dest Net}</int></value></param> <param><value><int>{Dest Node}</int></value></param> <param><value><int>{Dest Port}</int></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>
Activate XP source, destination, Priority	SetXpPrio	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Source Net}</int></value></param> <param><value><int>{Source Node}</int></value></param> <param><value><int>{Source Port}</int></value></param> <param><value><int>{Dest Net}</int></value></param> <param><value><int>{Dest Node}</int></value></param> <param><value><int>{Dest Port}</int></value></param> <param><value><int>{Priority}</int></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>
Activate XP source, destination with audio priority. An existing route to the given destination will be removed. The values Net = 0, Node = 0, port = 0 remove all existing routes for the given source or destination.	SetXpDestructive	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Source Net}</int></value></param> <param><value><int>{Source Node}</int></value></param> <param><value><int>{Source Port}</int></value></param> <param><value><int>{Dest Net}</int></value></param> <param><value><int>{Dest Node}</int></value></param> <param><value><int>{Dest Port}</int></value></param> <param><value><int>{Priority}</int></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>
Kill all crosspoints between the given source,destination Hint: The method deletes only those crosspoints, which have been activated by RRCS.	KillXp	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Source Net} </int></value></param> <param><value><int>{Source Node}</int></value></param> <param><value><int>{Source Port}</int></value></param> <param><value><int>{Dest Net}</int></value></param> <param><value><int>{Dest Node}</int></value></param> <param><value><int>{Dest Port}</int></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>

Description	Remote Procedure	Parameters	Return Value on success
Query status of XP source, destination Hint: The method can return true even if KillXp has been called beforehand.	GetXpStatus	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Source Net}</int></value></param> <param><value><int>{Source Node}</int></value></param> <param><value><int>{Source Port}</int></value></param> <param><value><int>{Dest Net}</int></value></param> <param><value><int>{Dest Node}</int></value></param> <param><value><int>{Dest Port}</int></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> <value><boolean>{Status}</boolean ></value> </data></array></value></param>
Dump active XPs	GetAllActiveXps	<param><value><string>{TransKey}</string></value></param>	<param><value><struct> <member><name>ErrorCode</name> <value><int>{ ErrorCode }</int></value> </member> <member><name>TransKey</name> <value><string>{TransKey}</string></value> </member> <member><name>XP Count</name> <value><int>{Number of XPs}</int></value> </member> <member><name>XP#1</name> <value><array><data> <value><int>{Source Net}</int></value> <value><int>{Source Node}</int></value> <value><int>{Source Port}</int></value> <value><int>{Dest Net}</int></value> <value><int>{Dest Node}</int></value> <value><int>{Dest Port}</int></value> </data></array></value> </member> <member><name>XP#2</name> ... </struct></value></param>
Dumps active crosspoints for the given range. A crosspoint is in range, if the source <u>or</u> (changed in specification version 1.4) the destination of the crosspoint is in range (Start Net <= xp-net <= Stop Net Start Node <= xp-node <= Stop Node Start Port <= xp-port <= Stop Port)	GetActiveXpsRange	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Start Net}</int></value></param> <param><value><int>{Start Node}</int></value></param> <param><value><int>{Start Port}</int></value></param> <param><value><int>{Stop Net}</int></value></param> <param><value><int>{Stop Node}</int></value></param> <param><value><int>{Stop Port}</int></value></param>	List of crosspoints. Please see “GetAllActiveXps” HINT: since RRCS version 5500, the return parameters contain an additional error code (see also “GetAllCaps”)

8.2 Volume

Description	Remote Procedure	Parameters	Return Value on success
Sets the single or/and conference volume. Note: This API does not work for XPs that´s destination port is connected to an Artist-1024	SetXpVolume	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Source Net}</int></value></param> <param><value><int>{Source Node}</int></value></param> <param><value><int>{Source Port}</int></value></param> <param><value><int>{Dest Net}</int></value></param> <param><value><int>{Dest Node}</int></value></param> <param><value><int>{Dest Port}</int></value></param> <param><value><boolean>{single}</bool></value></param> <param><value>< boolean >{conference}</ boolean ></value></param> <param><value><int>{volume}</int></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>
Queries the single and conference volume. Note: This API does not work for XPs that´s destination port is connected to an Artist-1024	GetXpVolume	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Source Net}</int></value></param> <param><value><int>{Source Node}</int></value></param> <param><value><int>{Source Port}</int></value></param> <param><value><int>{Dest Net}</int></value></param> <param><value><int>{Dest Node}</int></value></param> <param><value><int>{Dest Port}</int></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> <value><int>{single volume}</int></value> <value><int>{conference volume}</int></value> </data></array></value></param>

8.3 Port Alias

Description	Remote Procedure	Parameters	Return Value on success
Set the alias for the given port. Hint: To change the port alias it is also possible to use the method ‘ConfigurationChange’ with the object type ‘port’ (requires version 5.91 or later)	SetPortAlias	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Net}</int></value></param> <param><value><int>{Node}</int></value></param> <param><value><int>{Port}</int></value></param> <param><value><string>{Alias}</string></value></param> <param><value><boolean>{IsInput}</boolean></value></param> //optional; default = 0/false	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>
Queries the alias for the given port.	GetPortAlias	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Net}</int></value></param> <param><value><int>{Node}</int></value></param> <param><value><int>{Port}</int></value></param> <param><value><boolean>{IsInput}</boolean></value></param> //optional; default = 0/false	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> <value><string>{Alias}</string></value> </data></array></value></param>

8.4 Port Label

Description	Remote Procedure	Parameters	Return Value on success
Set the label for the given port. Hint: To change the port label it is also possible to use the method 'ConfigurationChange' with the object type 'port' (requires version 5.91 or later)	SetPortLabel	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Node}</int></value></param> <param><value><int>{Port}</int></value></param> <param><value><string>{Label}</string></value></param> <param><value><boolean>{IsInput}</boolean></value></param> //optional; default = 0/false	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>
Queries the label for the given port.	GetPortLabel	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Node}</int></value></param> <param><value><int>{Port}</int></value></param> <param><value><boolean>{IsInput}</boolean></value></param> //optional; default = 0/false	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> <value><string>{Label}</string></value> </data></array></value></param>

8.5 Input / Output gain

Description	Remote Procedure	Parameters	Return Value on success
Set input gain of port. Value range: -36,-35,..,0,..,35,36 gain [dB] = Gain / 2.0 Gain = -128 → mute	SetInputGain	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Net}</int></value></param> <param><value><int>{Node}</int></value></param> <param><value><int>{Port}</int></value></param> <param><value><int>{Gain}</int></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>
Range, see SetInputGain	GetInputGain	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Net}</int></value></param> <param><value><int>{Node}</int></value></param> <param><value><int>{Port}</int></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> <value><int>{Gain}</int></value> </data></array></value></param>
Set the output gain for the given port. Range, see SetInputGain	SetOutputGain	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Net}</int></value></param> <param><value><int>{Node}</int></value></param> <param><value><int>{Port}</int></value></param> <param><value><int>{Gain}</int></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>
Queries the output gain for the given port. Range, see SetInputGain	GetOutputGain	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Net}</int></value></param> <param><value><int>{Node}</int></value></param> <param><value><int>{Port}</int></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> <value><int>{Gain}</int></value> </data></array></value></param>

Description	Remote Procedure	Parameters	Return Value on success
Activates/Deactivates the given GP output. Hint: The method deactivates only GP outputs that have been activated by RRCS.	SetGpOutput	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Net}</int></value></param> <param><value><int>{Node}</int></value></param> <param><value><int>{Port}</int></value></param> <param><value><int>{Slot}</int></value></param> <param><value><int>{GPIO no.}</int></value></param> <param><value><int>{state}</int></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>
Queries the level meter values. Note: For Artist-1024 connected ports, RRCS must be connected directly to the same SIC. See also 8.21	GetLevelMeterValues	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Source Node}</int></value></param> <param><value><int>{Source Port}</int></value></param> <param><value><int>{Source PoolPort}</int></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{Level}</int></value> <value><int>{InputGain}</int></value> <value><boolean>{VoxOn}</boolean></value> <value><int>{PeakLevel}</int></value> </data></array></value></param>

8.6 GPIOs

Description	Remote Procedure	Parameters	Return Value on success
Activates/Deactivates the given GP output. Hint: The method deactivates only GP outputs that have been activated by RRCS.	SetGpOutput	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Net}</int></value></param> <param><value><int>{Node}</int></value></param> <param><value><int>{Port}</int></value></param> <param><value><int>{Slot}</int></value></param> <param><value><int>{GPIO no.}</int></value></param> <param><value><int>{state}</int></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>
Queries the given GP input state	GetGpInputState	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Net}</int></value></param> <param><value><int>{Node}</int></value></param> <param><value><int>{Port}</int></value></param> <param><value><int>{Slot}</int></value></param> <param><value><int>{GPIO no.}</int></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> <value><boolean>{state}</boolean></value> </data></array></value></param>
Queries the given GP output state Hint: The method can return true, even if the GP output has been deactivated by "SetGpOutput "	GetGpOutputState	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Net}</int></value></param> <param><value><int>{Node}</int></value></param> <param><value><int>{Port}</int></value></param> <param><value><int>{Slot}</int></value></param> <param><value><int>{GPIO no.}</int></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> <value><boolean>{state}</boolean></value> </data></array></value></param>

Description	Remote Procedure	Parameters	Return Value on success
Queries all configured GPOutputs. Introduced with version 8.8	GetAllGpOuts	<param><value><string>{TransKey}</string></value></param>	<param> <value><array><data> <value><string>{TransKey}</string></value> <value><array><data> <!—1st GPOutput information --> <value><struct> ... </value></struct> </data></array></value></param>
Queries all configured GPInputs. Introduced with version 8.8	GetAllGpIns	<param><value><string>{TransKey}</string></value></param>	<param> <value><array><data> <value><string>{TransKey}</string></value> <value><array><data> <!—1st GPInput information --> <value><struct> ... </value></struct> </data></array></value></param>

8.7 Logic Sources

Description	Remote Procedure	Parameters	Return Value on success
Activates/Deactivates the given logic source. Hint: The method deactivates only logic sources that have been activated by RRCS.	SetLogicSourceState	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Object ID}</int></value></param> <param><value><boolean>{state}</boolean></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>

Description	Remote Procedure	Parameters	Return Value on success
Queries all logic sources in the system	GetAllLogicSources	<param><value><string>{TransKey}</string></value></param>	<param><value><struct> <member><name>ErrorCode</name> <value><int>{ ErrorCode }</int></value> </member> <member><name>TransKey</name> <value><string>{TransKey}</string></value> </member> <member><name>LogicSourceCount</name> <value><int>{Number of LS}</int></value> </member> <member><name>LogicSource#1</name> <value><array><data> <value><string>{Longname}</string></value> <value><string>{8 char. label}</string ></value> <value><int>{Object ID}</int></value> </data></array></value> </member> <member><name> LogicSource#2</name> . . . </struct></value></param>
Queries all logic sources in the system including the current state	GetAllLogicSources_v2	<param><value><string>{TransKey}</string></value></param>	<param><value><struct> <member><name>ErrorCode</name> <value><int>{ ErrorCode }</int></value> </member> <member><name>TransKey</name> <value><string>{TransKey}</string></value> </member> <member><name>LogicSourceCount</name> <value><int>{Number of LS}</int></value> </member> <member><name>LogicSource#1</name> <value><array><data> <value><string>{Longname}</string></value> <value><string>{8 char. label}</string ></value> <value><int>{Object ID}</int></value> <value><boolean>{State}<boolean></value> </data></array></value> </member> <member><name> LogicSource#2</name> . . . </struct></value></param>

8.8 Status

Description	Remote Procedure	Parameters	Return Value on success
Queries all ports in the system	GetAllCaps	<param><value><string>{TransKey}</string></value></param>	<param><value><struct> <member><name>ErrorCode</name> <value><int>{ ErrorCode }</int></value> </member> <member><name>TransKey</name> <value><string>{TransKey}</string></value> </member> <member><name>port count</name> <value><int>{Number of Ports}</int></value> </member> <member><name>port#1</name> <value><array><data> <value><int>{Net}</int></value> <value><int>{Node}</int></value> <value><int>{Port}</int></value> <!-- -1 if Porttype does not support PoolPorts, else the amount of PoolPorts --/> <value><int>{PoolPortAmount}</int></value> </data></array></value> </member> <member><name>port#2</name> . . . </struct></value></param>
Queries all ports in the system with more detailed information than 'GetAllCaps'. Note: This version exists since RRCS version 5.91. The returned data can become very big (approximately 5 kByte per port) The corresponding "return value on success" is only a snippet of the full data per port and may vary from port type to port type as	GetAllPorts	<param><value><string>{TransKey}</string></value></param>	<param> <value><array><data> <value><string>{TransKey}</string></value> <value><array><data> <!—1st port information --> <value><struct> <member> <name>Input</name> <value><boolean>…</boolean></value> </member> <member> <name>KeyCount</name> <value><i4>…</i4></value> </member> <member> <name>Label</name> <value><string>…</string></value> </member>

Description	Remote Procedure	Parameters	Return Value on success
different properties are supported/not supported. The property names and data types correspond to the definitions of CfgChange->Port or rather PortEx			<!—only, if an Alias is defined --> <member> <name>Alias</name> <value><string>…</string></value> </member> <member> <name>Name</name> <value><string>…</string></value> </member> <member> <name>Node</name> <value><i4>…</i4></value> </member> <member> <name>ObjectID</name> <value><i4>…</i4></value> </member> <member> <name>Output</name> <value><boolean>…</boolean></value> </member> <member> <name>PageCount</name> <value><i4>…</i4></value> </member> <member> <name>Port</name> <value><i4>…</i4></value> </member> <member> <!-- only if port is a PoolPort --> <name>PoolPort</name> <value><i4>…</i4></value> </member> <member> <name>PortType</name> <value><string>…</string></value> </member> <member> <name>IsTrunkEnabled</name> <value><boolean>…</boolean></value> </member> <member> <name>IsTrunkLineEnabled</name> <value><boolean>…</boolean></value>

Description	Remote Procedure	Parameters	Return Value on success
			</member> <member> <!-- only if TrunkingPortAddr is defined --> <name>TrunkingNetAddr</name> <value><i4>…</i4></value> </member> <member> <!-- only if TrunkingPortAddr is defined --> <name>TrunkingPortAddr</name> <value><i4>…</i4></value> </member> <member> <!– only if expansion panel exists --> <name>ConnectedExpansions</name> <value>array of expansion panel object-IDs </value> </member> <member> <name>PortVoIP</name> <value><struct> <member> <name>AudioCodec</name> <value><i4>…</i4></value> </member> <member> <name>AudioPacketSize</name> <value><i4>…</i4></value> </member> <member> <name>DiffServiceCodePoint</name> <value><i4>…</i4></value> </member> <member> <name>LocalSipId</name> <value><string>…</string></value> </member> <member> <name>ReceiveBufferSize</name> <value><i4>…</i4></value> </member> <member> <name>RemoteHost</name> <value><string>{IP-address}</string></value> </member>

Description	Remote Procedure	Parameters	Return Value on success
			<member> <name>RemoteSipId</name> <value><string>…</string></value> </member> <member> <name>VoiceActDetection</name> <value><boolean>…</boolean></value> </member> </struct></value> </member> </struct></value> <!—further ports --> … </data></array></value> </data></array></value> </param> </params>
Queries all properties of the specified port. Note: This API exists since RRCS version 8.3. The returned data can become very big (approximately 5 kByte per port).	GetPort	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Net}</int></value></param> <param><value><int>{Node}</int></value></param> <param><value><int>{Port}</int></value></param> <param><value><boolean>{IsInput}</boolean></value></param> <param><value><int>{PoolPort}</int></value></param>	See GetAllPorts
Queries all supported port types that can be used on the CfgChange→PortEx API See 8.10.4.6	GetPortExTypeList	<param><value><string>{TransKey}</string></value></param>	<methodResponse> <params> <param> <value> <array> <data> <value> <string>C0123456789</string> </value> <value> <array> <data> <value> <struct> <member>

Description	Remote Procedure	Parameters	Return Value on success
			<name>PortExType</name> <value><i4>0</i4></value> </member> <member> <name>PortExTypeString</name> <value><string>TWO_WIRE_IN</string></value> </member> </struct> </value> … … </data> </array> </value> </data> </array> </value> </param> </params> </methodResponse>
Queries all IFB's in Artist. Hint: This version exists since RRCS version 5.91	GetAllIFBs	<param><value><string>{TransKey}</string></value></param>	<param> <value><array><data> <value><string>{TransKey}</string></value> <value><array><data> <!-- 1st IFB information --> <value><struct> <member> <name>DimLevel</name> <value><i4>5</i4></value> </member> <member> <name>Input</name> <value>{TPortAddress}</value> </member> <member> <name>IsTrunkEnabled</name> <value><boolean>1</boolean></value> </member> <member> <name>Label</name> <value><string>…</string></value> </member> <member> <name>LongName</name>

Description	Remote Procedure	Parameters	Return Value on success
			<value><string>…</string></value> </member> <member> <name>MixMinus</name> <value>{TPortAddress}</value> </member> <member> <name>Number</name> <value><i4>…</i4></value> </member> <member> <name>ObjectID</name> <value><i4>…</i4></value> </member> <member> <name>Output</name> <value>{TPortAddress}</value> </member> <member> <name>Owner</name> <value><i4>…</i4></value> </member> </struct></value> <!-- further IFB information --> … </data></array></value> </data></array></value> </param>
Get a list of possible error codes including their error descriptions	GetErrorCodeList	<param><value><string>{TransKey}</string></value></param>	<param> <value><array><data> <value><string>{TransKey}</string></value> <value><array><data> <value><struct> <member> <name>Code</name> <value><i4>…</i4></value> </member> <member> <name>Description</name> <value><string>…</string></value> </member> </struct></value> <!—further error information --> … </data></array></value>

Description	Remote Procedure	Parameters	Return Value on success
			</data></array></value> </param> </params> </methodResponse>
Query gateway state	GetState	<param><value><string>{TransKey}</string></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> <value><string>{GatewayState}</string></value> </data></array></value></param>
Set gatway to Working	SetStateWorking	<param><value><string>{TransKey}</string></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>
Set gateway to Standby	SetStateStandby	<param><value><string>{TransKey}</string></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>
Ping the gatway	GetAlive	<param><value><string>{TransKey}</string></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>
Retreive the RRCS version	GetVersion	<param><value><string>{TransKey}</string></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> <value><string>{Version}</string></value> </data></array></value></param>
Tests whether the RRCS sends notifications to the given XML-RPC-server or not	IsRegisteredForEvents	<param> <value><string>{TransKey}</string></value> </param> <param> <value><string>{IP-address}</string></value> </param> <param> <value><int>{TCP-port}</int></value> </param>	<param> <value><struct> <member> <name>IsRegistered</name> <value><boolean>…</boolean></value> </member> <member> <name>TransKey</name> <value><string>{TransKey}</string></value> </member> </struct></value> </param>

Description	Remote Procedure	Parameters	Return Value on success
Tests whether RRCS is connected to Artist or not Hint: The method returns 'not connected', if no network connection to Artist is available or the configuration format does not match with the version of RRCS	IsConnectedToArtist	<param><value><string>{TransKey}</string></value></param>	<param> <value><struct> <member> <name>IsConnected</name> <value><boolean>…</boolean></value> </member> <member> <name>TransKey</name> <value><string>{TransKey}</string></value> </member> </struct></value> </param>
Tests whether RRCS sends notifications to the given address or not. Internally the method tests whether RRCS sends notifications to the url ("http://" [ip-address of incoming RPC] ":" [TCPPort] [URLPath]) or not. The url path is optional and is treated as “/RPC2” if left blank	IsRegisteredForAllEvents	<param> <value><string>{TransKey}</string></value> </param> <param> <value><i4>{TCPPort}</i4></value> </param> <param> <value><string>{URLPath}</string></value> </param>	<param> <value><struct> <member> <name>IsRegistered</name> <value><boolean>…</boolean></value> </member> <member> <name>TransKey</name> <value><string>{TransKey}</string></value> </member> </struct></value> </param>
Returns a list of all configured conferences including all their property values. Hint: (requires version 8.5 or later)	GetAllConferences	<param><value><string>{TransKey}</string></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><array><data> <value><struct> <member> <name>Alias</name> <value><string></string></value> </member> <member> <name>IsTrunkEnabled</name> <value><boolean>0</boolean></value> </member> <member> <name>Label</name> <value><string>CNF_0</string></value> </member> <member> <name>LongName</name> <value><string>CNF_0</string></value> </member>

Description	Remote Procedure	Parameters	Return Value on success
			<member> <name>MemberList</name> <value><array><data> <value><struct> <member> <name>Listen</name> <value><boolean>1</boolean></value> </member> <member> <name>Node</name> <value><i4>6</i4></value> </member> <member> <name>Port</name> <value><i4>1057</i4></value> </member> <member> <name>Talk</name> <value><boolean>1</boolean></value> </member> <member> <name>UseSecondChannel</name> <value><boolean>0</boolean></value> </member> </struct></value> </data></array></value> </member> <member> <name>ObjectID</name> <value><i4>1752587182</i4></value> </member> </struct></value> </data></array></value> </data></array></value></param>
Returns a list of all configured groups including all their property values. Hint: (requires version 8.5 or later)	GetAllGroups	<param><value><string>{TransKey}</string></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><array><data> <value><struct> <member> <name>IsTrunkEnabled</name> <value><boolean>0</boolean></value> </member> <member> <name>Label</name> <value><string>Group1</string></value>

Description	Remote Procedure	Parameters	Return Value on success
			</member> <member> <name>LongName</name> <value><string>Group 1</string></value> </member> <member> <name>ObjectID</name> <value><i4>211298706</i4></value> </member> </struct></value> <value><struct> <member> <name>IsTrunkEnabled</name> <value><boolean>0</boolean></value> </member> <member> <name>Label</name> <value><string>GRP #002</string></value> </member> <member> <name>LongName</name> <value><string>Group #002</string></value> </member> <member> <name>ObjectID</name> <value><i4>504046310</i4></value> </member> </struct></value> </data></array></value> </data></array></value></param>
Returns a list of all key configurations of a Port, including all their property values. Also includes virtual-keys. Note: 'IsInput' and 'PoolPort' are optional parameters Hint: (requires version 8.5 or later)	GetAllKeyConfiguration	<param><value>{TransKey}</value></param> <param><value><i4>{Node]</i4></value></param> <param><value><i4>{Port}</i4></value></param> <param><value><boolean>{IsInput]</boolean></value></param> <param><value><i4>{PoolPort}</i4></value></param>	<param> <value><struct> <member> <name>KeyList</name> <value><array><data> <value><struct> <member> <name>KeyPosition</name> <value><struct> <member> <name>ExpansionPanel</name> <value><i4>0</i4></value> </member> <member> <name>KeyNumber</name>

Description	Remote Procedure	Parameters	Return Value on success
			<value><i4>1</i4></value> </member> <member> <name>Page</name> <value><i4>1</i4></value> </member> <member> <name>PositionType</name> <value><string>key</string></value> </member> </struct></value> </member> <member> <name>KeyProperties</name> <value><struct> <member> <name>ActionByKeyPressed</name> <value><i4>1</i4></value> </member> <member> <name>AutoLabelFlag</name> <value><boolean>1</boolean></value> </member> <member> <name>AutoSubtitle</name> <value><boolean>1</boolean></value> </member> <member> <name>ClipLabelTo6Chars</name> <value><boolean>0</boolean></value> </member> <member> <name>Dim</name> <value><boolean>0</boolean></value> </member> <member> <name>GroupColor</name> <value><i4>16</i4></value> </member> <member> <name>Icon</name> <value><i4>1</i4></value> </member> <member> <name>KeyMode</name>

Description	Remote Procedure	Parameters	Return Value on success
			<value><i4>2</i4></value> </member> <member> <name>LabelValue</name> <value><string>P.4.2</string></value> </member> <member> <name>LatchingTimeOut</name> <value><i4>0</i4></value> </member> <member> <name>MonitoringState</name> <value><i4>0</i4></value> </member> <member> <name>ObjectID</name> <value><i4>2062361842</i4></value> </member> <member> <name>RadioButton</name> <value><i4>0</i4></value> </member> <member> <name>RestartLatchingTimer</name> <value><boolean>0</boolean></value> </member> <member> <name>RestoreVolumeLevel</name> <value><boolean>1</boolean></value> </member> <member> <name>ShowSubtitle</name> <value><boolean>1</boolean></value> </member> <member> <name>Subtitle</name> <value><string></string></value> </member> <member> <name>UseScrollListEntryKeyMode</name> <value><boolean>1</boolean></value> </member> </struct></value> </member> </struct></value>

Description	Remote Procedure	Parameters	Return Value on success
			</data></array></value> </member> <member> <name>TransKey</name> <value><string>C0123456789</string></value> </member> </struct></value> </param>
Returns a list of all configured Devices	GetAllDevices	<param><value><string>{TransKey}</string></value></param>	<param> <value><array><data> <value><string>C0123456789</string></value> <value><array><data> <value><struct> <member> <name>AssignedNode</name> <value><i4>4</i4></value> </member> <member> <name>AssignedPorts</name> <value><array><data> <value><struct> <member> <name>LongName</name> <value><string>Port_18</string></value> </member> <member> <name>Node</name> <value><i4>4</i4></value> </member> <member> <name>ObjectID</name> <value><i4>1538671339</i4></value> </member> <member> <name>Port</name> <value><i4>18</i4></value> </member> <member> <name>ChannelSelection</name> <value><i4>3</i4></value> </member> </struct></value> </data></array></value> </member> <member>

Description	Remote Procedure	Parameters	Return Value on success
			<name>DeviceConnectVoIP</name> <value><struct> <member> <name>ChannelNumber</name> <value><i4>2</i4></value> </member> <member> <name>IpAddress</name> <value><string>0.0.0.0</string></value> </member> <member> <name>IpPort</name> <value><i4>0</i4></value> </member> <member> <name>MulticastIpIn</name> <value><string>0.0.0.0</string></value> </member> <member> <name>MulticastIpOut</name> <value><string>0.0.0.0</string></value> </member> <member> <name>MulticastPortIn</name> <value><i4>5004</i4></value> </member> <member> <name>MulticastPortOut</name> <value><i4>5004</i4></value> </member> </struct></value> </member> <member> <name>DeviceType</name> <value><i4>0</i4></value> </member> <member> <name>LongName</name> <value><string>TestDevice</string></value> </member> <member> <name>ObjectID</name> <value><i4>1008870053</i4></value> </member> </struct></value>

Description	Remote Procedure	Parameters	Return Value on success
			</data></array></value> </data></array></value> </param>
Returns the long name of the Net	GetNetName	<param><value><string>{TransKey}</string></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> <value><string>{LongName}</string></value> </data></array></value></param>
Sets the long name of the Net	SetNetName	<param><value><string>{TransKey}</string></value></param> <param><value><string>{LongName}</string></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>

8.9 Objects Lists

8.9.1 Generic

Description	Remote Procedure	Parameters	Return Value on success
Gets all existing objects in the system of a specific type Object-type can be: - “conference” - “group” - “port” - “ifb” - “logic-source” - “logic-destination” - “gp-input” - “gp-output” - “user” - “audiopatch” - “client-card” - “device” The object type is case-insensitive	GetObjectList	<param> <value> <string>{TransKey}</string> </value> </param> <param> <value> <string>{object-type}</string> </value> </param>	<param> <value> <struct> <member> <name>TransKey</name> <value>{TransKey}</value> </member> <member> <name>ObjectList</name> <value> <array> <data> <value> <struct> <member> <name>ObjectID</name> <value>{Object ID}</value> </member> <member> <name>LongName</name> <value>{LongName}</value> </member> </struct> </value> … </data> </array> </value> </member> </struct> </value> </param>
Gets an object property. Hint: 1)	GetObjectProperty	<param> <value> <string>{TransKey}</string> </value> </param> <param> <value><int>{Object ID}</int></value> </param> <param>	<param> <value> <struct> <member> <name>TransKey</name> <value>{TransKey}</value> </member> <member> <name>{Property name}</name>

Description	Remote Procedure	Parameters	Return Value on success
The method “ConfigurationChange” with the object type “any” allows to set any property for any object. 2) If the property name is empty, the method returns all available object properties		<value> <string>{Property name}</string> </value> </param>	<value> {Property Value} </value> </member> </struct> </value> </param>
Gets a list of supported object properties.	GetObjectPropertyNames	<param> <value> <string>{TransKey}</string> </value> </param> <param> <value><int>{Object ID}</int></value> </param>	<param> <value> <struct> <member> <name>TransKey</name> <value>{TransKey}</value> </member> <member> <name>PropertyNames</name> <value><array><data> <value>{property name 1}</value> <value>{property name 2}</value> … </data></array></value> </member> </struct> </value> </param>

Object properties can be retrieved by using GetObjectProperty. This method does not require an object type since the object ID already provides enough information. However, different object types support different properties. The return type {Property type} depends on the property. E.g. an alias would be a <string>

Object-type	Supported properties	Property-type
conference	Alias	<string>
	Label	<string> (8 characters max)
	LongName	<string> (32 unicode characters max)
	Owner	<int> (object-ID of user-object)
	MemberList	<array> of <{TConferencePortAddress}> (read-only, see 6.3)
group	Label	<string> (8 characters max)
	LongName	<string> (32 unicode characters max)
	MemberList	<array> of <{TGroupPortAddress}> (see 6.2)
	Owner	<int> (Object-ID of user-object)
ifb	Label	<string> (8 characters max)
	LongName	<string> (32 unicode characters max)
	Number	<int> (read-only)
	Input	<{TPortAddress> (see 6.1)
	Output	<{TPortAddress> (see 6.1)
	MixMinus	<{TPortAddress> (see 6.1)
	Owner	<int> (Object-ID of user-object)

8.9.2 Method GetCommandList

Description	Remote Procedure	Parameters	Return Value on success
Queries the list of configured command and all their propertiesof a given position. Position can be a key or virtual function or virtual key. Note: The corresponding “return value on success” is only a snippet of the full data. The property names and data types correspond to the definitions of the different CfgChange -> Command types	GetCommandList	<param> <value><string>{TransKey}</string></value> </param> <param> <{TCmdPosition}> <param>	<param> <value> <struct> <member> <name>CommandList</name> <value> <array> <data> <value> <struct> <member> <name>Description</name> <value><string>{command desciption}</string></value> </member> <member> <name>{command Object ID}</name> <value><int>{command object ID}</int></value> </member> </struct> </value> <value> <struct> <member> <name>Description</name> <value><string>{command description}</string></value> </member> <member> <name>ObjectID</name> <value><int>{command object ID}</int></value> </member> </struct> </value> </data> </array> </value> </member> <member> <name>TransKey</name> <value><string>{TransKey}</string></value> </member> </struct> </value> </param>

8.9.3 Method GetPortsCommandLists

Queries the list of configured command and all their properties of all positions of a Port. Position can be a key or virtual function or virtual key. Note: This return value is only a snippet and shows the basic structure.	GetPortsCommandLists	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Net}</int></value></param> <param><value><int>{Node}</int></value></param> <param><value><int>{Port}</int></value></param> <param><value><boolean>{IsInput}</boolean></value></param> <param><value><int>{PoolPort}</int></value></param>	<?xml version="1.0" encoding="UTF-8"?> <methodResponse> <params> <param> <value><struct> <member> <name>CommandLists</name> <value><array><data> <value><struct> <member> <name>CommandList</name> <value><array><data> <value><struct> </struct></value> </data></array></value> </member> <member> <name>CommandPosition</name> <value><struct> <member> <name>ExpansionPanel</name> <value><i4>0</i4></value> </member> <member> <name>IsInput</name> <value><boolean>0</boolean></value> </member> <member> <name>KeyNumber</name> <value><i4>0</i4></value> </member> <member> <name>Net</name> <value><i4>1</i4></value> </member> <member> <name>Node</name> <value><i4>2</i4></value> </member> <member> <name>Page</name> <value><i4>0</i4></value> </member> <member> <name>Port</name> <value><i4>24</i4></value>

			</member> <member> <name>PositionType</name> <value><string>virtual-function</string></value> </member> <member> <name>VirtualFunctionType</name> <value><string>always</string></value> </member> </struct></value> </member> </struct></value> … … </data></array></value> </member> <member> <name>TransKey</name> <value><string>C0123456789</string></value> </member> </struct></value> </param> </params> </methodResponse>

8.10 Configuration changes

All configuration changes (create, replace, edit and delete) will use the method ConfigurationChange. The format of the parameters is flexible enough to support all objects and their properties in the future. Objects have their specific parameters since the properties are very different. For the specific parameters please see 8.10.4.

With version 8.5 the API ConfigurationChangeEx has been introduced. Whilst with the ConfigurationChange API a whole request is always handled atomar (all or nothing) with ConfigurationChangeEx API the individual instructions of the request are handled atomar. Please note that with ConfigurationChangeEx API, you will receive different error codes than with ConfigurationChange API. See chapter 7 for more information.

8.10.1 Method ConfigurationChange / ConfigurationChangeEx

Description	Remote Procedure	Parameters	Return Value on success
Performs a configuration change. { ChangeType } can be: “create” “edit” “delete” “delete-children” “swap” { ObjectType } can be: “any” “port” “call-to-conference” “ifb” “conference” “group” “call-to-ifb” “call-to-group” “dim-level-command” "call-to-port-cmd" "listen-to-port-cmd" "route-cmd" "switch-gpio-out-cmd" "select-audiopatch-cmd"	ConfigurationChange / ConfigurationChangeEx	<param> <value><string>{TransKey}</string></value> </param> <param> <value> <array> <data> <!—1st configuration change --> <value> <struct> <member> <name>ChangeType</name> <value>{ ChangeType }</value> </member> <member> <name>ObjectType</name> <value> <string>{ ObjectType }</string> </value> </member> <member> <name>SpecificParams</name> <value>{ specific parameter struct }</value> </member> </struct> </value> <!—2nd configuration change --> <value> <struct> …. </struct> </value> ….	<param> <value> <string>{TransKey}</string> </value> </param>

Description	Remote Procedure	Parameters	Return Value on success
"control-audiopatch-cmd" "remote-key-cmd" "reply-cmd" "edit-conference-cmd" "edit-ifb-cmd" "dim-panel-speaker-cmd" "beep-panel-cmd" "telephone-dial-keypad-cmd" "logic-cmd" "kill-partyline-mic-cmd" "auto-listen-off-cmd" "set-input-output-gain-cmd" "sidetone-cmd" "send-string-cmd" “command-container” “panel-key” “client-card” “virtual-key” “device”		<!– last configuration change --> <value> <struct> …. </struct> </value> </data> </array></value></param>	

Object-types	ChangeType
call-to-conference	Create, Edit, Delete
conference	Edit
ifb	Edit
group	Create, Edit, Delete
call-to-ifb	Create, Edit, Delete
call-to-group	Create, Edit, Delete
dim-level-command	Create, Edit, Delete
call-to-port-cmd	Create, Edit, Delete
listen-to-port-cmd	Create, Edit, Delete
route-cmd	Create, Edit, Delete
switch-gpio-out-cmd	Create, Delete
select-audiopatch-cmd	Create, Delete
control-audiopatch-cmd	Create, Delete
remote-key-cmd	Create, Edit, Delete
reply-cmd	Create, Edit, Delete
edit-conference-cmd	Create, Delete
edit-ifb-cmd	Create, Delete
dim-panel-speaker-cmd	Create, Edit, Delete
beep-panel-cmd	Create, Delete
telephone-dial-keypad-cmd	Create, Edit, Delete
logic-cmd	Create, Delete
kill-partyline-mic-cmd	Create, Delete
auto-listen-off-cmd	Create, Delete
set-input-output-gain-cmd	Create, Delete
sidetone-cmd	Create, Edit, Delete
send-string-cmd	Create, Delete
hot-mic-cmd	Create, Delete
command-container	delete-children
client-card	Edit, Swap
virtual-key	Create, Delete
port	Create, Edit, Delete
panel-key	Edit

8.10.2 Method BufferConfigurationChange

Buffers configuration changes

The caller should call the method 'ApplyConfigurationChange' / 'ApplyConfigurationChangeEx' later to send the buffered configuration changes to Artist. Hint: If the caller does not call 'ApplyConfigurationChange', 'ApplyConfigurationChangeEx' or 'BufferConfigurationChange' within the next 60 seconds, the server deletes all previously buffer configurations changes. In this case the next call of 'ApplyConfigurationChange', 'ApplyConfigurationChangeEx' or 'BufferConfigurationChange' will return an error. Once the error has been returned back to the client it is possible to restart buffering.

Description	Remote Procedure	Parameters	Return Value on success
Buffers configuration changes. { ChangeType }can be: “create” “edit” “delete” { ObjectType } can be: “any” “port” “call-to-conference” “ifb” “conference” “group” “call-to-ifb” “call-to-group” “dim-level-command” "call-to-port-cmd" "listen-to-port-cmd" "route-cmd" "switch-gpio-out-cmd" "select-audiopatch-cmd" "control-audiopatch-cmd" "remote-key-cmd"	BufferConfigurationChange / BufferConfigurationChangeEx	<param> <value><string>{TransKey}</string></value> </param> <param> <value> <array> <data> <!—1st configuration change --> <value> <struct> <member> <name>ChangeType</name> <value>{ ChangeType }</value> </member> <member> <name>ObjectType</name> <value> <string>{ ObjectType }</string> </value> </member> <member> <name>SpecificParams</name> <value>{ specific parameter struct }</value> </member> </struct> </value> <!—2nd configuration change --> <value> <struct> …. </struct> </value> ….	<param> <value> <string>{TransKey}</string> </value> </param>

Description	Remote Procedure	Parameters	Return Value on success
"reply-cmd" "edit-conference-cmd" "edit-ifb-cmd" "dim-panel-speaker-cmd" "beep-panel-cmd" "telephone-dial-keypad-cmd" "logic-cmd" "kill-partyline-mic-cmd" "auto-listen-off-cmd" "set-input-output-gain-cmd" "sidetone-cmd" "send-string-cmd" “command-container” “client-card” “panel-key” “virtual-key” “device”		<!– last configuration change --> <value> <struct> …. </struct> </value> </data> </array></value></param>	

8.10.3 Method ApplyConfigurationChange / ApplyConfigurationChangeEx

Sends buffered configuration changes (via 'BufferConfigurationChange') to Artist

'ApplyConfigurationChange' / ApplyConfigurationChangeEx must be called at most 60 seconds after the last 'BufferConfigurationChange' call. Otherwise the server returns a time out error and ignores previously buffered configuration changes.

If the server buffers nothing the server returns success.

With version 8.5 the API 'ApplyConfigurationChangeEx' has been introduced. Whilst with the 'ApplyConfigurationChange' API a whole request is always handled atomar (all or nothing) with 'ApplyConfigurationChangeEx' API the individual instructions of the request are handled atomar. Please note that with 'ApplyConfigurationChangeEx' API, you will receive different error codes than with 'ApplyConfigurationChange' API. See chapter 7 for more information.

Description	Remote Procedure	Parameters	Return Value on success

Sends buffered configuration changes (via 'BufferConfigurationChange') to Artist	ApplyConfigurationChange	`<param>`
		
		`<value><string>{TransKey}</string></value>`
		`</param>`
			`<param>`
			`<value>`
			`<string>{TransKey}</string>`
			`</value>`
			`</param>`

8.10.4 Specific configuration parameters

8.10.4.1 Object type 'any'

The object-type "any" can be used to edit properties of any object. It implements the inverse operation of the method "GetObjectProperty". This object-type is here just for completeness.

8.10.4.1.1 Specific parameters for edit

```
<member>
   <name>ObjectID</name>
   <value><int>{ ObjectID }</int ></value>
</member>
<member>
   <name>Properties</name>
   <value><struct>
   <!- 1st property to edit -->
<member>
   <name>{Property name}</name>
   <value><{Property type}>{Property Value}</{Property type}</value>
</member>
   …
   </struct></value>
</member>
```

8.10.4.2 Object type ‘conference’

8.10.4.2.1 Specific parameters for creat/edit “conference”

The edit conference functionality requires an object-ID of the conference to modify. The properties of the conference are shown in the table below:

Supported properties	Property-type
Alias	<string> (8 characters max)
Label	<string> (8 characters max)
LongName	<string> (32 unicode characters max)
ObjectID	<int> (object-ID of conference object) (mandatory, when editing)
IsTrunkEnabled	<boolean>
TrunkingAddr	<int> (object-ID of user-object)

8.10.4.2.2 Specific parameters for deleting “conference”

The delete conference functionality requires object id of the conference to be deleted:

Supported properties	Property-type
ObjectID	<int> (object-ID of user-object)

8.10.4.2.3 Example

The following example changes the alias, the label and the long-name of a conference with the object-ID 39153498:

```
<?xml version="1.0""?>
<methodCall>
<methodName>ConfigurationChange</methodName>
<params>
<param>
   <value>C0000000005</value>
</param>
<param>
   <value><array><data>
      <value><struct>
      <member>
         <name>ChangeType</name>
         <value>edit</value>
      </member>
      <member>
         <name>ObjectType</name>
         <value>conference</value>
      </member>
      <member>
         <name>SpecificParams</name>
         <value><struct>
            <member>
               <name>ObjectID</name>
               <value><i4>39153498</i4></value>
            </member>
            <member>
               <name>Alias</name>
               <value>confAlia</value>
            </member>
            <member>
               <name>Label</name>
               <value>confLabl</value>
            </member>
            <member>
```

```
                <name>LongName</name>
                <value>New long-name</value>
            </member>
        </struct></value>
    </member>
    </struct></value>
</data></array></value>
</param>
</params>
</methodCall>
```

8.10.4.3 Object type ‘group’

8.10.4.3.1 Specific parameters for creating “group”

The create group functionality requires a name for the new group. The change properties of the group-object are shown in the table below:

Supported properties	Property-type
Name	<string> (32 unicode characters max)

8.10.4.3.2 Specific parameters for editing “group”

The edit group functionality requires an object-ID of the group to modify. The change properties of the group-object are shown in the table below:

Supported properties	Property-type
Label	<string> (8 characters max)
LongName	<string> (32 unicode characters max)
MemberList	<{TMemberChangeList}> (an array of ports to add or remove, see 6.4) Notice: This is not (!) the same type as used by the method “GetObjectProperty”
ObjectID	<int> (object-ID of user-object)
KeypadShortcut	<int> (0-9999)
IsTrunkEnabled	<boolean>
TrunkingAddr	<int> (object-ID of user-object)

8.10.4.3.3 Specific parameters for deleting “group”

The delete group functionality requires object id for the group to be deleted. The change properties of the group-object are shown in the table below:

Supported properties	Property-type
ObjectID	<int> (object-ID of user-object)

8.10.4.3.4 Example

This example changes the alias, the label, and the long-name of the group with object-ID 17251. Additionally it adds the port with the address (node=2 and port=0) and removes the port with the address (node=2 and port=1).

```
<?xml version="1.0"?>
<methodCall>
<methodName>ConfigurationChange</methodName>
<params>
<param>
  <value>C0000000005</value>
</param>
<param>
  <value><array><data>
    <value><struct>
      <member>
        <name>ChangeType</name>
        <value>edit</value>
        </member>
      <member>
        <name>ObjectType</name>
        <value>group</value>
        </member>
      <member>
        <name>SpecificParams</name>
        <value><struct>
```

```
<member>
  <name>ObjectID</name>
  <value><i4>17251</i4></value>
  </member>

<member>
  <name>Alias</name>
  <value>grpAlias</value>
  </member>
<member>
  <name>Label</name>
  <value>grpLabel</value>
  </member>
<member>
  <name>LongName</name>
  <value>New group long-name </value>
  </member>
<member>
  <name>MemberList</name>
  <value><array><data>
    <value><struct>
      <member>
        <name>AddToGroup</name>
        <value><boolean>1</boolean></value>
        </member>
      <member>
        <name>PortAddress</name>
        <value><struct>
          <member>
            <name>IsInput</name>
            <value><boolean>0</boolean></value>
            </member>
          <member>
            <name>Node</name>
            <value><i4>2</i4></value>
            </member>
          <member>
            <name>Port</name>
            <value><i4>0</i4></value>
            </member>
          </struct></value>
        </member>
      <member>
        <name>UseSecondChannel</name>
        <value><boolean>0</boolean></value>
        </member>
      </struct></value>
    <value><struct>
      <member>
        <name>AddToGroup</name>
        <value><boolean>0</boolean></value>
        </member>
      <member>
        <name>PortAddress</name>
        <value><struct>
          <member>
            <name>IsInput</name>
            <value><boolean>0</boolean></value>
            </member>
          <member>
            <name>Node</name>
            <value><i4>2</i4></value>
            </member>
          <member>
            <name>Port</name>
            <value><i4>1</i4></value>
            </member>
          </struct></value>
        </member>
      <member>
        <name>UseSecondChannel</name>
```

```
                  <value><boolean>0</boolean></value>
                  </member>
                </struct></value>
              </data></array></value>
            </member>
          </struct></value>
        </member>
      </struct></value>
    </data></array></value>
  </param>
</params>
</methodCall>
```

8.10.4.4 Object type ‘ifb’

8.10.4.4.1 Specific parameters for edit “ifb”

Supported properties	Property-type
LongName	<string> (32 unicode characters max)
Label	<string> (8 characters max)
Input	<{TPortAddress}> or <TIFBAudioEndpoint>
Output	<{TPortAddress}> or <TIFBAudioEndpoint>
MixMinus	<{TPortAddress}> or <TIFBAudioEndpoint>
Owner	<int> (object-ID of user-object)
DimLevel	<int> (optional) 0 = 0 dB 1 = -3 dB 2 = -6 dB 3 = -9 dB 4 = -12 dB 5 = -18 dB 6 = -24 dB 7 = mute
IsTrunkEnabled	<boolean> (optional)

Hint: If the ‘Node’ and the ‘Port’ members of a {TPortAddress} are set to 0, the properties ‘Input’, ‘Output’ and ‘MixMinus’ are set to nothing.

Since RRCS 6.20 the members ‘Input’, ‘Output’ and ‘MixMinus’ can also be artist-groups. Hence these members can now contain the new data type **TIFBAudioEndpoint** which is defined as follows:

TIFBAudioEndpoint

Member	Type
AudioEndpointType	<string> Can contain one of the following values: <table><tr><th>Value</th><th>Meaning</th></tr><tr><td>Nil</td><td>the audio endpoint is empty</td></tr><tr><td>Group</td><td>the endpoint defines an artist group (see member “ObjectID”</td></tr></table>
ObjectID	<int> (AudioEndpointType must be „Group“ (case-insensitive))

If an audio-endpoint is empty, the method GetObjectProperty returns a TPortAddress with the node and port member set to zero.

This is done for backward compatibility reasons.

8.10.4.4.2 Example

Example for editing the label to “Sport” on the IFB number 7

```
<?xml version="1.0"?>
<methodCall>
<methodName>ConfigurationChange</methodName>
<params>
<param>
  <value>C0000000001</value>
```

```
</param>
<param>
  <value><array><data>
    <value><struct>
      <member>
        <name>ChangeType</name>
        <value>edit</value>
        </member>
      <member>
        <name>ObjectType</name>
        <value>ifb</value>
        </member>
      <member>
        <name>SpecificParams</name>
        <value><struct>

          <!-- Hint: This member addresses the IFB only, it does not change the number -->
          <member>
            <name>IFBNumber</name>
            <value><i4>7</i4></value>
            </member>

          <member>
            <name>Label</name>
            <value>Sport</value>
            </member>
          </struct></value>
        </member>
      </struct></value>
    </data></array></value>
  </param>
</params>
</methodCall>
```

8.10.4.5 Object type ‘port’ *(deprecated)*

8.10.4.5.1 Specific parameters for create/edit

Parameters	Type
PortAddress	<{TPortAddress}>
LongName	<string> (optional, 32 unicode characters max)
Label	<string> (optional, 8 characters max)
Alias	<string> (optional, 8 characters max)
Subtitle	<string> (optional, 16 characters max)
PoolPort	<int> (optional, counting starts with 0) Deprecated → Please use property in PortAddress
PoolPortAmount	<int> (optional) Note: This Parameter is only applied in create and defines the amount of PoolPorts a user wants to create at the Port, specified in PortAddress. This Port must be of Type SipPhonePort or TelephoneCodec as only those 2 port types support PoolPorts at all. 1..32
IsTrunkEnabled	<boolean> (optional)
IsTrunkLineEnabled	<boolean> (optional)
TrunkingPortAddr	<int> (optional) Can only be set if IsTrunkEnabled or IsTrunkLineEnabled is set to true Gets resetted when IsTrunkEnabled is set to false
PortVoIP	<struct> (optional)
PortSip	<struct> (optional)
PortAes67	<struct> (optional) Applies to Aes67 panel port types
PortAes67Input	<struct> (optional)
PortAes67Output	<struct> (optional)
PortAes67Bolero	<struct> (optional)
PortDSP2312Basic	<struct> (optional)
PortDSP2312Plus	<struct> (optional)
PortRSP2318Basic	<struct> (optional)
PortRSP2318Plus	<struct> (optional)
PortRSP2318Pro	<struct> (optional)
PortRsp1216HL	<struct> (optional)
PortRSP1232HL	<struct> (optional)
PortAcrobatCC60	<struct> (optional)
PortC3BeltPack	<struct> (optional)
PortCCP1116	<struct> (optional)
PortDCP1116	<struct> (optional)
PortDCP2016DP4	<struct> (optional)
PortDCP2116P4	<struct> (optional)
PortDCP5008	<struct> (optional)
PortRCP1128	<struct> (optional)
PortRCP2016DP4	<struct> (optional)
PortRCP2116P4	<struct> (optional)
Port2WireIn	<struct> (optional)
Port2WireOut	<struct> (optional)

Parameters	Type
Port4Wire	<struct> (optional)
Port4WireSplit	<struct> (optional)
PortDanteIn	<struct> (optional)
PortDanteOut	<struct> (optional)
PortDanteInSplit	<struct> (optional)
PortDanteOutSplit	<struct> (optional)

PortVoIP (all members are optional)	Property-type
RemoteHost	<string> (optional)
RemoteSipId	<string> (optional)
LocalSipId	<string> (optional)
AudioCodec	<int> (optional) 0 = G.711 U-law 8k 8 = G.711 A-law 8k 9 = G.722 64kbps PLC 84 = G.711 U-law 16k 91 = G.711 A-law 16k 97 = PCM 8k 110 = RARe U-law 111 = RARe A-law 112 = G.722 48kbps PLC
AudioPacketSize	<int> (optional) 0 = 20 ms 1 = 40 ms 2 = 80 ms 3 = 160 ms
ReceiveBufferSize	<int> (optional) Depending on the AudioPacketSize, Max is 5120 ms 0 = 4 x AudioPacketSize 1 = 8 x AudioPacketSize 2 = 16 x AudioPacketSize 3 = 32 x AudioPacketSize 4 = 64 x AudioPacketSize 5 = 128 x AudioPacketSize 6 = 256 x AudioPacketSize
DiffServiceCodePoint	<int> (optional)
VoiceActDetection	<boolean> (optional)

PortSip (all members are optional)	Property-type
DomainServer	<string> (optional)
ProxyServer	<string> (optional)
UserName	<string> (optional)
DisplayName	<string> (optional)
AuthUserName	<string> (optional)
AuthPassword	<string> (optional)
Reregister	<int> (optional)
SipProtocol	<int> (optional) 0 = TCP 1 = UDP
TrustedDomain	<boolean> (optional)

PortAes67 (all members are optional)	Property-type
IpAddress	{IpAddress} (optional)
ListenPort	<int> (optional)
PacketTime	<int> (optional) 125 = 0.125 ms 250 = 0.250 ms 333 = 0.333 ms 1000 = 1.000 ms 1333 = 1.333 ms
ReceiveBuffer	<int> (optional, default = 3) Artist-32/64/128 n x PacketTime <= 18.000ms; where 3 <= n <= 20 Artist-1024 n x PacketTime <= 150.000ms; where 3 <= n <= 112 In synton mode n x PacketTime >= 3.000ms
PlayMode	<int> (optional) 0 = Synton 1 = Synchron

PortAes67Input (all members are optional)	Property-type
BitDepth	<int> (optional) 16 = L16 24 = L24
PacketTime	<int> (optional) Depends on the amount of configured channels that may reduces the max MTU size 125 = 0.125 ms 250 = 0.250 ms 333 = 0.333 ms 1000 = 1.000 ms 1333 = 1.333 ms
ReceiveBuffer	<int> (optional, default = 3) Artist-32/64/128 n x PacketTime <= 18.000ms; where 3 <= n <= 20 Artist-1024 n x PacketTime <= 150.000ms; where 3 <= n <= 112 In synton mode n x PacketTime >= 3.000ms
PayloadType	<int> (optional) 96 - 127
SSRC	<string> (optional)
TimeStampOffset	<string> (optional)
Channels	<int> (optional) 1..16
Selection	<int> (optional)

PortAes67Input (all members are optional)	**Property-type**
	Depending on the configured channels 1 .. max 16
Protocol	<int> (optional) 2 = Manual 3 = RTSP 5 = NMOS
Multicast	{IpAddress} (optional)
MulticastPort	<int> (optional)
RTSPUri	<string> (optional)
SourceIp	{IpAddress} (optional)
Multicast2	{IpAddress} (optional)
MulticastPort2	<int> (optional)
RTSPUri2	<string> (optional)
SourceIp2	{IpAddress} (optional)
Mode	<int> (optional) 0..1151 Define the Port ID you want to link to Note: The Port must be located on the same media interface and may not be linked to another port itself
PlayMode	<int> (optional) 0 = Synton 1 = Synchron

PortAes67Output (all members are optional)	**Property-type**
BitDepth	<int> (optional) 16 = L16 24 = L24
PacketTime	<int> (optional) Depends on the amount of configured channels that may reduces the max MTU size 125 = 0.125 ms 250 = 0.250 ms 333 = 0.333 ms 1000 = 1.000 ms 1333 = 1.333 ms
PayloadType	<int> (optional) 96 - 127
SSRC	<string> (optional)
TimeStampOffset	<string> (optional)
Channels	<int> (optional) 1..16
Selection	<int> (optional) Depending on the configured channels 1 ..max 16
Protocol	<int> (optional) 2 = Manual 3 = RTSP 5 = NMOS
Multicast	{IpAddress} (optional)
MulticastPort	<int> (optional)
Multicase2	{IpAddress} (optional)
MulticastPort2	<int> (optional)
Mode	<int> (optional)

PortAes67Output (all members are optional)	Property-type
	Define the Port ID you want to link to Note: The Port must be located on the same media interface and may not be linked to another port itself

PortAes67Bolero (all members are optional)	Property-type
BoleroUserId	<int> (optional)
Multicast	{IpAddress} (optional)
MulticastPort	<int> (optional)

The settings of the following table applies to the panel family:

All Panels (all members are optional)	Property-type
MinSpeakerVol	<int> (optional) 0 = -45 dB 1 = -42 dB ... 14 = -3 dB 15 = 0 dB
MinHeadSetVol	<int> (optional) 0 = -45 dB 1 = -42 dB ... 14 = -3 dB 15 = 0 dB
BeepVol	<int> (optional) 0 = -45 dB 1 = -42 dB ... 14 = -3 dB 15 = 0 dB
BeepDuration	<int> (optional) Any value in ms
SpeakerDimLevel	<int> (optional) 0 = 0 dB 1 = -3 dB 2 = -6 dB 3 = -9 dB 4 = -12 dB 5 = -18 dB 6 = -24 dB 7 = mute
VoxOnTh	<int> (optional) 0 = 12 dBu 1 = 10 dBu ... 23 = -34 dBu 24 = -36 dBu 25 = permanent
VoxOffTh	<int> (optional) Input values depend on set VoxOnTh, General boundaries 9 dbU to -53 dBu

All Panels (all members are optional)	Property-type
	In case VoxOnTh is set to permanent, setting VoxOffTh is not allowed 0 = 9 dBu 1 = 7 dBu 2 = 5 dBu … 30 = -51 dBu 31 = -53 dBu You can calculate lists of allowed input values with following formulas: Input value range: (VoxOnTh – 3 <= value <= (VoxOnTh – 3 + 53) / 2 In case VoxOnTh is – 36 dBu, allowed VoxOfTh -39 dBu <= v in dBu <= 7 dBu 7 (7 dBu)<= index <= 24 (-39 dBu)
VoxHoldTime	<int> (optional) -1 = no delay 0 = 50 ms 1 = 100 ms 2 = 200 ms 3 = 400 ms 4 = 800 ms 5 =1600 ms 6 = 3200 ms 7 = 6400 ms
InitialSingleVol	<int> (optional) 0 = 6 dB 1 = 3 dB … 6 = -12 dB 7 = -18 dB 8 = -24 dB 9 = mute
InitialConferenceVol	<int> (optional) 0 = 6 dB 1 = 3 dB … 6 = -12 dB 7 = -18 dB 8 = -24 dB 9 = mute
ShiftKeyMode	<int> (optional) 0 = Shift lock invalid 1 = Shift unlocked 2 = Locked on norm 3 = Locked on shift
HeadSetMode	<int> (optional) 0 = Headset lock invalid 1 = Headset unlocked 2 = Headset locked at speaker
LedBrightness	<int> (optional)

All Panels (all members are optional)	Property-type
	0 = 10% 1 = 20% … 8 = 90% 9 = 100%
DisplayBrightness	<int> (optional) 0 = 10% 1 = 20% … 8 = 90% 9 = 100%
FontRotation	NOTE: not all panel types support setting this <int> (optional) 0 = 0° 1 = 90° 2 = 180° 3 = 270°
IndicateInUseOnCallRecv	<boolean> (optional)
IndicateInUseOnCallSend	<boolean> (optional)
EnableMuteFunction	<boolean> (optional)
ShowOnReply	<boolean> (optional)
OperationMode	NOTE: only RSP12xxHL supports this setting <int> (optional) 0 = Talk Mute 1 = Talk Listen
InitialIfbVol (Introduced with 8.0.RR21)	<int> (optional) 0 = 6 dB 1 = 3 dB … 6 = -12 dB 7 = -18 dB 8 = -24 dB 9 = mute

PortRSP1232HL (special properites) PortRSP1216HL (special properites) Note: Members of all Panels apply here too	Property-type
OperationMode	<int> (optional) 0 = Talk Mute 1 = Talk Listen
GroupColorIndication	<int> (optional) 0 = Display 1 = Key Ring
UseGroupColorOnPanel	<boolean> (optional)

Port4Wire	Property-type
VoxOnTh	<int> (optional) 0 = 12 dBu 1 = 10 dBu …

Port4Wire	Property-type
	23 = -34 dBu 24 = -36 dBu 25 = permanent
VoxOffTh	<int> (optional) Input values depend on set VoxOnTh, General boundaries 9 dbU to -53 dBu In case VoxOnTh is set to permanent, setting VoxOffTh is not allowed You can calculate lists of allowed input values with following formulas: Input value range: (VoxOnTh – 3 <= value <= (VoxOnTh – 3 + 53) / 2 For each input value you can calculate the corresponding dB value like this (VoxOnTh – 3) – (2 * VoxOffTh input value) In case VoxOnTh is – 36 dBu 0 = -39 dBu 1 = -41 dBu 2 = -43 dBu … 6 = -51 dBu 7 = -53 dBu
VoxHoldTime	<int> (optional) 0 = no delay 1 = 50 ms 2 = 100 ms 3 = 200 ms 4 = 400 ms 5 = 800 ms 6 =1600 ms 7 = 3200 ms 8 = 6400 ms
InitialSingleVol	<int> (optional) 0 = 6 dB 1 = 3 dB … 6 = -12 dB 7 = -18 dB 8 = -24 dB 9 = mute
InitialConferenceVol	<int> (optional) 0 = 6 dB 1 = 3 dB … 6 = -12 dB 7 = -18 dB 8 = -24 dB 9 = mute
IndicateInUseOnCallRecv	<boolean> (optional)
IndicateBusyOnCallSend	<boolean> (optional)
ShowOnReply	<boolean> (optional)
Enable2nd	<boolean> (optional)
InputChannel	<int> (optional)

Port4Wire	Property-type
	Applies for Madi ports only The same input channel can be assigned to multiple ports
OutputChannel	<int> (optional) Applies for Madi ports only The same output channel cannot be applied to 2 different ports

Port2WireIn	Property-type
VoxOnTh	<int> (optional) 0 = 12 dBu 1 = 10 dBu ... 23 = -34 dBu 24 = -36 dBu 25 = permanent
VoxOffTh	<int> (optional) Input values depend on set VoxOnTh, General boundaries 9 dbU to -53 dBu In case VoxOnTh is set to permanent, setting VoxOffTh is not allowed You can calculate lists of allowed input values with following formulas: Input value range: (VoxOnTh – 3 <= value <= (VoxOnTh – 3 + 53) / 2 For each input value you can calculate the corresponding dB value like this (VoxOnTh – 3) – (2 * VoxOffTh input value) In case VoxOnTh is – 36 dBu 0 = -39 dBu 1 = -41 dBu 2 = -43 dBu ... 6 = -51 dBu 7 = -53 dBu
VoxHoldTime	<int> (optional) 0 = no delay 1 = 50 ms 2 = 100 ms 3 = 200 ms 4 = 400 ms 5 = 800 ms 6 =1600 ms 7 = 3200 ms 8 = 6400 ms
IndicateBusyOnCallSend	<boolean> (optional)
ShowOnReply	<boolean> (optional)
Enable2nd	<boolean> (optional)
InputChannel	<int> (optional) Applies for Madi ports only

Port2WireOut	Property-type
InitialSingleVol	<int> (optional) 0 = 6 dB 1 = 3 dB ... 6 = -12 dB 7 = -18 dB 8 = -24 dB 9 = mute
InitialConferenceVol	<int> (optional) 0 = 6 dB 1 = 3 dB ... 6 = -12 dB 7 = -18 dB 8 = -24 dB 9 = mute
IndicateInUseOnCallRecv	<boolean> (optional)
ShowOnReply	<boolean> (optional)
Enable2nd	<boolean> (optional)
OutputChannel	<int> (optional) Applies for Madi ports only

8.10.4.5.1 Specific parameters for delete

Parameters	Type
PortAddress	<{TPortAddress}>

8.10.4.5.2 Example

Example to change the alias on physical port 0, pool port 1, node 2:

```
<?xml version="1.0"?>
<methodCall>
<methodName>ConfigurationChange</methodName>
<params>
<param>
  <value>C0000000005</value>
  </param>
<param>
  <value><array><data>
    <value><struct>
      <member>
        <name>ChangeType</name>
        <value>edit</value>
        </member>
      <member>
        <name>ObjectType</name>
        <value>port</value>
        </member>
      <member>
        <name>SpecificParams</name>
        <value><struct>

          <member>
            <name>PortAddress</name>
            <value><struct>
            <member>
              <name>IsInput</name>
              <value><boolean>1</boolean></value>
              </member>
```

```
<member>
  <name>Node</name>
  <value><i4>2</i4></value>
  </member>
<member>
  <name>Port</name>
  <value><i4>0</i4></value>
  </member>
</struct></value>
</member>

<member>
  <name>Port4Wire</name>
  <value><struct></struct></value>
</member>

<member>
  <name>PoolPort</name>
  <value><int>0</int></value>
</member>
<member>
  <name>Alias</name>
  <value>NewAlias</value>
</member>

</struct></value>
</member>
</struct></value>
</data></array></value>
</param>
</params>
</methodCall>
```

Example how to create PoolPorts:

```
<?xml version="1.0"?>
<methodCall>
   <methodName>ConfigurationChange</methodName>
   <params>
   <param>
       <value>C0000000001</value>
    </param>
    <param>
       <value>
       <array><data>
          <value>
          <struct>
             <member>
                <name>ChangeType</name>
                <value>create</value>
             </member>
             <member>
                 <name>ObjectType</name>
                <value>port</value>
             </member>
             <member>
                <name>SpecificParams</name>
                <value>
                   <struct>
                   <member>
                      <name>PortAddress</name>
                      <value>
                         <struct>
                         <member>
                            <name>Node</name>
                            <value><i4>2</i4></value>
                         </member>
                         <member>
                            <name>Port</name>
                            <value><i4>8</i4></value>
                         </member>
                         <member>
                            <name>PoolPortAmount</name>
```

```
                    <value><i4>2</i4></value>
                  </member>
                  </struct>
                </value>
              </member>
              <member>
                <name>Port4Wire</name>
                <value><struct /></value>
              </member>
              </struct>
            </value>
          </member>
        </struct>
        </value>
      </data>
      </array>
      </value>
    </param>
    </params>
</methodCall>
```

8.10.4.6 Object type ‘portex’

8.10.4.6.1 Specific parameters for create/edit

Parameters	Type
PortAddress	<{TPortAddress}> (mandatory)
Dedicated creation properties	
PortType	<int> (mandatory) 0 = TWO_WIRE_IN 1 = TWO_WIRE_OUT 2 = FOUR_WIRE 3 = FOUR_WIRE_SPLIT 4 = RCP_1012E 5 = RCP_1028E 6 = DCP_1016E 7 = RCP_1112 8 = RCP_1128 9 = DCP_1116 10 = CCP_1116 11 = RSP_1232HL 12 = RSP_1216HL 13 = RCP_2016_DP4 14 = DCP_2016_DP4 15 = RCP_2116_P4 16 = DCP_2116_P4 17 = RSP_2318_PRO 18 = RSP_2318_PLUS 19 = RSP_2318_BASIC 20 = DSP_2312_PLUS 21 = DSP_2312_BASIC 22 = RCP_3016_P4 23 = DCP_3016_P4 24 = DCP_5008 25 = DCP_5108 26 = DBM_1004E 27 = TELEPHONE_CODEC 28 = RIF_1032 29 = RIF_2064

Parameters	Type
	30 = C3_BELTPACK 31 = WB_2_BP 32 = AURUS_PANEL 33 = BOLERO_BP 34 = TRUNKLINE 35 = SIP 36 = VCP_1004 37 = VCP_1012 38 = CODEC_CONNECTION 39 = Stage 40 = NSA_CONNECTION 41 = NSA_CONNECTION_SPLIT
PoolPortAmount	<int> (optional) Note: This Parameter defines the amount of PoolPorts a user wants to create at a Port, specified in PortAddress. This Port must be of Type SipPhonePort or TelephoneCodec as only those 2 port types support PoolPorts at all. 1..32
All port related properties	
LongName	<string> (optional, 32 unicode characters max)
Label	<string> (optional, 8 characters max)
Alias	<string> (optional, 8 characters max)
Subtitle	<string> (optional, 16 characters max)
KeypadShortcut	<int> (optional, 0-9999)
All Trunking related port properties	
IsTrunkEnabled	<boolean> (optional)
IsTrunkLineEnabled	<boolean> (optional)
TrunkingPortAddr	<int> (optional) Can only be set if IsTrunkEnabled or IsTrunkLineEnabled is set to true Gets resetted when IsTrunkEnabled is set to false
All port input related properties	
VoxOnTh	<int> (optional) 0 = 12 dBu 1 = 10 dBu ... 23 = -34 dBu 24 = -36 dBu 25 = permanent
VoxOffTh	<int> (optional) Input values depend on set VoxOnTh, General boundaries 9 dbU to -53 dBu In case VoxOnTh is set to permanent, setting VoxOffTh is not allowed 0 = 9 dBu 1 = 7 dBu 2 = 5 dBu ...

Parameters	Type
	30 = -51 dBu 31 = -53 dBu You can calculate lists of allowed input values with following formulas: Input value range: (VoxOnTh – 3 <= value <= (VoxOnTh – 3 + 53) / 2 In case VoxOnTh is – 36 dBu, allowed VoxOfTh -39 dBu <= v in dBu <= 7 dBu 7 (7 dBu)<= index <= 24 (-39 dBu)
VoxHoldTime	<int> (optional) -1 = no delay 0 = 50 ms 1 = 100 ms 2 = 200 ms 3 = 400 ms 4 = 800 ms 5 =1600 ms 6 = 3200 ms 7 = 6400 ms
IndicateBusyOnCallSend	<boolean> (optional)
All port output related properties	
InitialSingleVol	<int> (optional) 0 = 6 dB 1 = 3 dB … 6 = -12 dB 7 = -18 dB 8 = -24 dB 9 = mute
InitialConferenceVol	<int> (optional) 0 = 6 dB 1 = 3 dB … 6 = -12 dB 7 = -18 dB 8 = -24 dB 9 = mute
InitialIfbVol	<int> (optional) 0 = 6 dB 1 = 3 dB … 6 = -12 dB 7 = -18 dB 8 = -24 dB 9 = mute
IndicateBusyOnCallRecv	<boolean> (optional)
All generic Panel related properties *Note: Not all properties are supported by every panel type*	
MinSpeakerVol	<int>(optional) 0 = -45 dB 1 = -42 dB …

Parameters	Type
	14 = -3 dB 15 = 0 dB
MinHeadSetVol	<int>(optional) 0 = -45 dB 1 = -42 dB ... 14 = -3 dB 15 = 0 dB
BeepVol	<int>(optional) 0 = -45 dB 1 = -42 dB ... 14 = -3 dB 15 = 0 dB
BeepDuration	<int> (optional) Any value in ms
SpeakerDimLevel	<int> (optional) 0 = 0 dB 1 = -3 dB 2 = -6 dB 3 = -9 dB 4 = -12 dB 5 = -18 dB 6 = -24 dB 7 = mute
ShiftKeyMode	<int> (optional) 0 = Shift lock invalid 1 = Shift unlocked 2 = Locked on norm 3 = Locked on shift
HeadSetMode	<int> (optional) 0 = Headset lock invalid 1 = Headset unlocked 2 = Headset locked at speaker
LedBrightness	<int> (optional) 0 = 10% 1 = 20% ... 8 = 90% 9 = 100%
DisplayBrightness	<int> (optional) 0 = 10% 1 = 20% ... 8 = 90% 9 = 100%
FontRotation	<int> (optional) 0 = 0° 1 = 90° 2 = 180° 3 = 270°
FontRotationExp	<int> (optional) 0 = 0° 1 = 90° 2 = 180°

Parameters	Type
	3 = 270°
EnableMuteFunc	<boolean> (optional)
ShowOnReply	<boolean> (optional)
Enable2nd	<boolean> (optional)
OperationMode	<int> (optional) 0 = Talk Mute 1 = Talk Listen
UseGroupColorOnPanel	<boolean> (optional)
GroupColorIndication	<int> (optional) 0 = Talk Mute 1 = Talk Listen
VolumeBarIndication	<int> (optional) 0 = dynamic 1 = permanent
IncomingCallSignaling	<int> (optional) 0 = Blinking 1 = Permanent
FeedbackSuppression	<boolean> (optional)
All client card related port properties	
PortVoIP	<struct> (optional)
PortSip	<struct> (optional)
PortMadi	<struct> (optional)
PortBolero	<struct> (optional)
PortAes67	<struct> (optional)
PortAes67Input	<struct> (optional)
PortAes67Output	<struct> (optional)
PortAes67Trunkline	<struct> (optional)
PortCodec	<struct> (optional)
PortNsaConnection	<struct> (optional)
PortNsaConnectionIn	<struct> (optional)
PortNsaConnectionOut	<struct> (optional)

PortVoIP (all members are optional)	Property-type
RemoteHost	<string> (optional)
RemoteSipId	<string> (optional)
LocalSipId	<string> (optional)
AudioCodec	<int> (optional) 0 = G.711 U-law 8k 8 = G.711 A-law 8k 9 = G.722 64kbps PLC 84 = G.711 U-law 16k 91 = G.711 A-law 16k 97 = PCM 8k 110 = RARe U-law 111 = RARe A-law 112 = G.722 48kbps PLC
AudioPacketSize	<int> (optional) 0 = 20 ms 1 = 40 ms 2 = 80 ms

PortVoIP (all members are optional)	**Property-type**
	3 = 160 ms
ReceiveBufferSize	<int> (optional) Depending on the AudioPacketSize, Max is 5120 ms 0 = 4 x AudioPacketSize 1 = 8 x AudioPacketSize 2 = 16 x AudioPacketSize 3 = 32 x AudioPacketSize 4 = 64 x AudioPacketSize 5 = 128 x AudioPacketSize 6 = 256 x AudioPacketSize
DiffServiceCodePoint	<int> (optional)
VoiceActDetection	<boolean> (optional)

PortSip (all members are optional)	**Property-type**
DomainServer	<string> (optional)
ProxyServer	<string> (optional)
UserName	<string> (optional)
DisplayName	<string> (optional)
AuthUserName	<string> (optional)
AuthPassword	<string> (optional)
Reregister	<int> (optional)
SipProtocol	<int> (optional) 0 = TCP 1 = UDP
TrustedDomain	<boolean> (optional)
EnableAutoHangUp	<boolean> (optional)

PortMadi (all members are optional) Only supported on Artist-1024 Madi SIC	**Property-type**
InputChannel	<int> (optional)
OutputChannel	<int> (optional)

PortBolero (all members are optional)	**Property-type**
BoleroUserId	<int> (optional)
Multicast	{IpAddress} (optional)
MulticastPort	<int> (optional)
MulticastPortToBolero	<int> (optional)

PortAes67 (all members are optional)	**Property-type**
IpAddress	{IpAddress} (optional)
ListenPort	<int> (optional)
IpAddress2	{IpAddress} (optional)
ListenPort2	<int> (optional)
PacketTime	<int> (optional)

PortAes67 (all members are optional)	Property-type
	125 = 0.125 ms 250 = 0.250 ms 333 = 0.333 ms 1000 = 1.000 ms 1333 = 1.333 ms
ReceiveBuffer	<int> (optional, default = 3) Artist-32/64/128 n x PacketTime <= 18.000ms; where 3 <= n <= 20 Artist-1024 n x PacketTime <= 150.000ms; where 3 <= n <= 112 In synton mode n x PacketTime >= 3.000ms
PlayMode	<int> (optional) 0 = Synton 1 = Synchron

PortAes67Input (all members are optional)	Property-type
Multicast	{IpAddress} (optional)
MulticastPort	<int> (optional)
RTSPUri	<string> (optional)
SourceIp	{IpAddress} (optional)
Multicast2	{IpAddress} (optional)
MulicastPort2	<int> (optional)
RTSPUri2	<string> (optional)
SourceIp2	{IpAddress} (optional)
Mode	<int> (optional) 0..1151 Define the Port ID you want to link to Note: The Port must be located on the same media interface and may not be linked to another port itself
Protocol	<int> (optional) 2 = Manual 3 = RTSP 5 = NMOS
BitDepth	<int> (optional) 16 = L16 24 = L24
PacketTime	<int> (optional) Depends on the amount of configured channels that may reduces the max MTU size 125 = 0.125 ms 250 = 0.250 ms 333 = 0.333 ms 1000 = 1.000 ms 1333 = 1.333 ms
PayloadType	<int> (optional)

PortAes67Input (all members are optional)	**Property-type**
	96 - 127
SSRC	<string> (optional)
TimeStampOffset	<string> (optional)
Channels	<int> (optional) 1..16
Selection	<int> (optional) Depending on the configured channels 1 .. max 16
ReceiveBuffer	<int> (optional, default = 3) Artist-32/64/128 n x PacketTime <= 18.000ms; where 3 <= n <= 20 Artist-1024 n x PacketTime <= 150.000ms; where 3 <= n <= 112 In synton mode n x PacketTime >= 3.000ms
PlayMode	<int> (optional) 0 = Synton 1 = Synchron

PortAes67Output (all members are optional)	**Property-type**
Multicast	{IpAddress} (optional)
MulticastPort	<int> (optional)
RTSPUri	<string> (optional)
SourceIp	{IpAddress} (optional)
Multicast2	{IpAddress} (optional)
MulicastPort2	<int> (optional)
RTSPUri2	<string> (optional)
SourceIp2	{IpAddress} (optional)
Mode	<int> (optional) 0..1151 Define the Port ID you want to link to Note: The Port must be located on the same media interface and may not be linked to another port itself
Protocol	<int> (optional) 2 = Manual 3 = RTSP 5 = NMOS
BitDepth	<int> (optional) 16 = L16 24 = L24
PacketTime	<int> (optional) Depends on the amount of configured channels that may reduces the max MTU size 125 = 0.125 ms 250 = 0.250 ms

PortAes67Output (all members are optional)	Property-type
	333 = 0.333 ms 1000 = 1.000 ms 1333 = 1.333 ms
PayloadType	<int> (optional) 96 - 127
SSRC	<string> (optional)
TimeStampOffset	<string> (optional)
Channels	<int> (optional) 1..16
Selection	<int> (optional) Depending on the configured channels 1 .. max 16

PortAes67Trunkline (all members are optional)	Property-type
LocalSipId	<string> (optional)
SipCalleeMode	<boolean> (optional)
PlayMode	<int> (optional) 0 = synton 1 = synchron
PacketTime	<int> (optional) 125 250 333 1000 1333
ReceiveBuffer	<int> (optional)
RemoteHost	<string> (optional)
RemoteSipId	<string> (optional)
RemoteSipPort	<int> (optional)

PortCodec (all members are optional)	Property-type
DeviceSelection	<string> (optional) the longname of a device an empty string resets the device assignement
ChannelSelection	<int> (optional)
AutoAnswer	<boolean> (optional)
IsAutoDialEnabled	<boolean> (optional)
AutoDialNumber	<string> (optional)

PortNsaConnection (all members are optional)	Property-type
DeviceSelection	<string> (optional) the longname of a device

PortNsaConnection (all members are optional)	Property-type
	an empty string resets the device assignement
InputChannel	<int> (optional)
OutputChannel	<int> (optional)

PortNsaConnectionIn (all members are optional)	Property-type
DeviceSelection	<string> (optional) the longname of a device an empty string resets the device assignement
InputChannel	<int> (optional)

PortNsaConnectionOut (all members are optional)	Property-type
DeviceSelection	<string> (optional) the longname of a device an empty string resets the device assignement
OutputChannel	<int> (optional)

8.10.4.6.1 Specific parameters for delete

Parameters	Type
PortAddress	<{TPortAddress}>

8.10.4.6.2 Example

Example to change the alias on physical port 0, pool port 1, node 2:

```
<?xml version="1.0"?>
<methodCall>
<methodName>ConfigurationChange</methodName>
<params>
    <param>
        <value>C0000000005</value>
    </param>
    <param>
        <value>
            <array>
                <data>
                    <value>
                        <struct>
                            <member>
                                <name>ChangeType</name>
                                <value>create</value>
                            </member>
                            <member>
                                <name>ObjectType</name>
                                <value>portex</value>
                            </member>
                            <member>
                                <name>SpecificParams</name>
                                <value>
                                    <struct>
                                        <member>
```

```
<name>PortAddress</name>
<value>
    <struct>
        <!--
        <member>
            <name>IsInput</name>
            <value>
                <boolean>1</boolean>
            </value>
        </member>
        -->
        <member>
            <name>Node</name>
            <value>
                <i4>2</i4>
            </value>
        </member>
        <member>
            <name>Port</name>
            <value>
                <i4>29</i4>
            </value>
        </member>
        <!--
        <member>
            <name>PoolPort</name>
            <value>
                <int>1</int>
            </value>
        </member>
        -->
    </struct>
</value>
</member>

<!-- Dedicated creation properties -->
<member>
    <name>PortType</name>
    <value><i4>2</i4></value>
</member>
<!--
<member>
    <name>PoolPortAmount</name>
    <value><i4>2</i4></value>
</member>
-->

<!-- All generic port related properties -->
<member>
    <name>PortType</name>
    <value><i4>2</i4></value>
</member>
<member>
    <name>LongName</name>
    <value>Port 29</value>
</member>
<member>
    <name>Label</name>
    <value>Port 29</value>
</member>
<member>
    <name>Alias</name>
    <value>Port 29</value>
</member>
<member>
    <name>Subtitle</name>
    <value>Port 29</value>
</member>

<!-- All Trunking related properties -->
<!--
<member>
```

```
        <name>IsTrunkEnabled</name>
        <value><boolean>0</boolean>
        </value>
    </member>
    <member>
        <name>IsTrunkLineEnabled</name>
        <value><boolean>0</boolean>
        </value>
    </member>
    <member>
        <name>TrunkingPortAddr</name>
        <value><i4>112</i4></value>
    </member>
    -->

    <!-- All port input related properties -->
    <!--
    <member>
        <name>VoxOnTh</name>
        <value><i4>0</i4></value>
    </member>
    <member>
        <name>VoxOffTh</name>
        <value><i4>5</i4></value>
    </member>
    <member>
        <name>VoxHoldTime</name>
        <value><i4>4</i4></value>
    </member>
    <member>
        <name>IndicateBusyOnCallSend</name>
        <value><boolean>1</boolean></value>
    </member>
    -->

    <!-- All port output related properties -->
    <!--
    <member>
        <name>InitialSingleVol</name>
        <value><i4>1</i4></value>
    </member>
    <member>
        <name>InitialConferenceVol</name>
        <value><i4>1</i4></value>
    </member>
    <member>
        <name>InitialIfbVol</name>
        <value><i4>1</i4></value>
    </member>
    <member>
        <name>IndicateBusyOnCallRecv</name>
        <value><boolean>0</boolean></value>
    </member>
    -->

    <!-- All generic Panel related properties -->
    <!--
    <member>
        <name>MinSpeakerVol</name>
        <value><i4>1</i4></value>
    </member>
    <member>
        <name>MinHeadSetVol</name>
        <value><i4>1</i4></value>
    </member>
    <member>
        <name>BeepVol</name>
        <value><i4>1</i4></value>
    </member>
    <member>
        <name>BeepDuration</name>
        <value><i4>1</i4></value>
```

```
</member>
<member>
    <name>SpeakerDimLevel</name>
    <value><i4>1</i4></value>
</member>
<member>
    <name>ShiftKeyMode</name>
    <value><i4>1</i4></value>
</member>
<member>
    <name>HeadSetMode</name>
    <value><i4>1</i4></value>
</member>
<member>
    <name>LedBrightness</name>
    <value><i4>1</i4></value>
</member>
<member>
    <name>DisplayBrightness</name>
    <value><i4>1</i4></value>
</member>
<member>
    <name>FontRotation</name>
    <value><i4>1</i4></value>
</member>
<member>
    <name>FontRotationExp</name>
    <value><i4>1</i4></value>
</member>
<member>
    <name>EnableMuteFunc</name>
    <value><boolean>1</boolean></value>
</member>
<member>
    <name>ShowOnReply</name>
    <value><boolean>1</boolean></value>
</member>
<member>
    <name>Enable2nd</name>
    <value><boolean>1</boolean></value>
</member>
<member>
    <name>OperationMode</name>
    <value><i4>1</i4></value>
</member>
<member>
    <name>UseGroupColorOnPanel</name>
    <value><boolean>1</boolean></value>
</member>
<member>
    <name>GroupColorIndication</name>
    <value><i4>1</i4></value>
</member>
-->

<!-- All Port VoIP related properties -->
<!--
<member>
    <name>PortVoIP</name>
    <value>
        <struct>
            <member>
                <name>RemoteHost</name>
                <value>remotehost.riedel.net</value>
            </member>
            <member>
                <name>RemoteSipId</name>
                <value>remotesipid</value>
            </member>
            <member>
                <name>LocalSipId</name>
                <value>localsipid</value>
```

```
                </member>
                <member>
                    <name>AudioCodec</name>
                    <value><int>112</int></value>
                </member>
                <member>
                    <name>AudioPacketSize</name>
                    <value><int>1</int></value>
                </member>
                <member>
                    <name>ReceiveBufferSize</name>
                    <value><int>3</int></value>
                </member>
                <member>
                    <name>DiffServiceCodePoint</name>
                    <value><int>0</int></value>
                </member>
                <member>
                    <name>VoiceActDetection</name>
                    <value><boolean>1</boolean></value>
                </member>
            </struct>
        </value>
    </member>
    -->

    <!-- All port SIP related properties -->
    <!--
    <member>
        <name>PortSip</name>
        <value>
            <struct>
                <member>
                    <name>DomainServer</name>
                    <value>domainserver.riedel.net</value>
                </member>
                <member>
                    <name>ProxyServer</name>
                    <value>proxyserver.riedel.net</value>
                </member>
                <member>
                    <name>UserName</name>
                    <value>userName</value>
                </member>
                <member>
                    <name>DisplayName</name>
                    <value>DisplayName</value>
                </member>
                <member>
                    <name>AuthUserName</name>
                    <value>AuthUsername</value>
                </member>
                <member>
                    <name>AuthPassword</name>
                    <value>AuthPassword</value>
                </member>
                <member>
                    <name>Reregister</name>
                    <value><int>3</int></value>
                </member>
                <member>
                    <name>SipProtocol</name>
                    <value><int>0</int></value>
                </member>
                <member>
                    <name>TrustedDomain</name>
                    <value><boolean>1</boolean></value>
                </member>
                <member>
                    <name>EnableAutoHangUp</name>
                    <value><boolean>0</boolean></value>
                </member>
```

```
        </struct>
    </value>
</member>
-->

<!-- All port MADI related properties; Artist-1024 only -->
<!--
<member>
    <name>PortMadi</name>
    <value>
        <struct>
            <member>
                <name>InputChannel</name>
                <value><int>3</int></value>
            </member>
            <member>
                <name>OutputChannel</name>
                <value><int>5</int></value>
            </member>
        </struct>
    </value>
</member>
-->

<!-- All port Bolero related properties -->
<!--
<member>
    <name>PortBolero</name>
    <value>
        <struct>
            <member>
                <name>Multicast</name>
                <value>224.0.2.1</value>
            </member>
            <member>
                <name>MulticastPort</name>
                <value><int>5005</int></value>
            </member>
            <member>
                <name>BoleroUserId</name>
                <value><int>5</int></value>
            </member>
        </struct>
    </value>
</member>
-->

<!-- All port AES67 Panel related properties -->
<!--
<member>
    <name>PortAes67</name>
    <value>
        <struct>
            <member>
                <name>IpAddress</name>
                <value>192.168.1.1</value>
            </member>
            <member>
                <name>ListenPort</name>
                <value><int>5005</int></value>
            </member>
            <member>
                <name>PacketTime</name>
                <value><int>125</int></value>
            </member>
            <member>
                <name>ReceiveBuffer</name>
                <value><int>5</int></value>
            </member>
            <member>
                <name>PlayMode</name>
                <value><int>1</int></value>
```

```
                </member>
            </struct>
        </value>
    </member>
    -->

    <!-- All port AES67 Input related properties -->
    <!--
    <member>
        <name>PortAes67Input</name>
        <value>
            <struct>
                <member>
                    <name>Multicast</name>
                    <value>224.0.2.1</value>
                </member>
                <member>
                    <name>MulticastPort</name>
                    <value><int>5005</int></value>
                </member>
                <member>
                    <name>Multicast2</name>
                    <value>224.0.2.1</value>
                </member>
                <member>
                    <name>MulticastPort2</name>
                    <value><int>5005</int></value>
                </member>
                <member>
                    <name>Mode</name>
                    <value><int>125</int></value>
                </member>
                <member>
                    <name>Protocol</name>
                    <value><int>125</int></value>
                </member>
                <member>
                    <name>BitDepth</name>
                    <value><int>125</int></value>
                </member>
                <member>
                    <name>PacketTime</name>
                    <value><int>125</int></value>
                </member>
                <member>
                    <name>PayloadType</name>
                    <value><int>125</int></value>
                </member>
                <member>
                    <name>SSRC</name>
                    <value><int>125</int></value>
                </member>
                <member>
                    <name>TimeStampOffset</name>
                    <value><int>125</int></value>
                </member>
                <member>
                    <name>Channels</name>
                    <value><int>125</int></value>
                </member>
                <member>
                    <name>Selection</name>
                    <value><int>125</int></value>
                </member>
                <member>
                    <name>SourceIp</name>
                    <value><int>125</int></value>
                </member>
                <member>
                    <name>RTSPUri</name>
                    <value><int>125</int></value>
                </member>
```

```
                <member>
                    <name>SourceIp2</name>
                    <value><int>125</int></value>
                </member>
                <member>
                    <name>RTSPUri2</name>
                    <value><int>125</int></value>
                </member>
                <member>
                    <name>ReceiveBuffer</name>
                    <value><int>5</int></value>
                </member>
                <member>
                    <name>PlayMode</name>
                    <value><int>1</int></value>
                </member>
            </struct>
        </value>
    </member>
    -->

    <!-- All port AES67 Output related properties -->
    <!--
    <member>
        <name>PortAes67Output</name>
        <value>
            <struct>
                <member>
                    <name>Multicast</name>
                    <value>224.0.2.1</value>
                </member>
                <member>
                    <name>MulticastPort</name>
                    <value><int>5005</int></value>
                </member>
                <member>
                    <name>Multicast2</name>
                    <value>224.0.2.1</value>
                </member>
                <member>
                    <name>MulticastPort2</name>
                    <value><int>5005</int></value>
                </member>
                <member>
                    <name>Mode</name>
                    <value><int>125</int></value>
                </member>
                <member>
                    <name>Protocol</name>
                    <value><int>125</int></value>
                </member>
                <member>
                    <name>BitDepth</name>
                    <value><int>125</int></value>
                </member>
                <member>
                    <name>PacketTime</name>
                    <value><int>125</int></value>
                </member>
                <member>
                    <name>PayloadType</name>
                    <value><int>125</int></value>
                </member>
                <member>
                    <name>SSRC</name>
                    <value><int>125</int></value>
                </member>
                <member>
                    <name>TimeStampOffset</name>
                    <value><int>125</int></value>
                </member>
                <member>
```

```
                                                        <name>Channels</name>
                                                        <value><int>125</int></value>
                                                    </member>
                                                    <member>
                                                        <name>Selection</name>
                                                        <value><int>125</int></value>
                                                    </member>
                                                </struct>
                                            </value>
                                        </member>
                                        -->

                                    </struct>
                                </value>
                            </member>
                        </struct>
                    </value>
                </data>
            </array>
        </value>
    </param>
</params>
</methodCall>
```

8.10.4.7 Object type ‘command-container’

This object type allows to delete all commands from a given command position. There’s an optional parameter ‘RestrictToCreator” where the caller can select the commands which have to be deleted on a creator basis. At present the only filter is RRCS (RestrictToCreator=1) or no filter (RestrictToCreator=0). If the parameter is missing, RRCS will delete all commands.

Example:

```
<?xml version="1.0"?>
<methodCall>
<methodName>ConfigurationChange</methodName>
<params>
<param>
  <value><string>C0000000003</string></value>
  </param>
<param>
  <value><array><data>
    <value><struct>
      <member>
        <name>ChangeType</name>
        <value><string>delete-children</string></value>
        </member>
      <member>
        <name>ObjectType</name>
        <value><string>command-container</string></value>
        </member>
      <member>
        <name>SpecificParams</name>
        <value><struct>
          <member>
             <name>RestrictToCreator</name>
             <value><int>1</int></value>
          </member>
          <member>
            <name>CommandPosition</name>
            <value><struct>
              <member>
                <name>KeyNumber</name>
                <value><i4>1</i4></value>
                </member>
              <member>
                <name>Node</name>
                <value><i4>2</i4></value>
                </member>
```

```
              <member>
                <name>Page</name>
                <value><i4>1</i4></value>
                </member>
              <member>
                <name>Port</name>
                <value><i4>0</i4></value>
                </member>
              <member>
                <name>PositionType</name>
                <value><string>key</string></value>
                </member>
              </struct></value>
            </member>
          </struct></value>
        </member>
      </struct></value>
    </data></array></value>
  </param>
</params>
</methodCall>
```

This object type can also be used to delete a batch of commands. In this case you can pass a list of object IDs that should be deleted by means of the member "CommandsToDelete".

Example:

```
<methodCall>
<methodName>
  ConfigurationChange
</methodName>
<params>
<param>
  <value><string>C0000000003</string></value>
</param>
<param>
  <value><array><data>
    <value><struct>
      <member>
        <name>ChangeType</name>
        <value><string>delete-children</string></value>
      </member>
      <member>
        <name>ObjectType</name>
        <value><string>command-container</string></value>
      </member>
      <member>
        <name>SpecificParams</name>
        <value><struct>
          <member>
            <name>RestrictToCreator</name>
            <value><int>1</int></value>
          </member>
          <member>
            <name>CommandPosition</name>
            <value><struct>
              <member>
                <name>KeyNumber</name>
                <value><i4>1</i4></value>
              </member>
              <member>
                <name>Node</name>
                <value><i4>2</i4></value>
              </member>
              <member>
                <name>Page</name>
                <value><i4>1</i4></value>
              </member>
              <member>
                <name>Port</name>
                <value><i4>0</i4></value>
```

```
              </member>
              <member>
                <name>PositionType</name>
                <value><string>key</string></value>
              </member>
            </struct></value>
          </member>
          <member>
            <name>CommandsToDelete</name>
            <value><array><data>
              <value><int>1426414447</int></value>
              <value><int>314528588</int></value>
              <value><int>1976270605</int></value>
              <value><int>584610977</int></value>
            </data></array></value>
          </member>
        </struct></value>
      </member>
    </struct></value>
  </data></array></value>
</param>
</params>
</methodCall>
```

8.10.4.8 Object type ‘call-to-conference’

8.10.4.8.1 Specific parameters for creatíng “call-to-conference”

The call to conference command has the following command specific parameters. The members AudioPriority, UseSecondChannel, Talk and Listen are optional. If they do not exist, RRCS takes default values.

Parameters	**Type**
Conference	<int> (object-ID of conference-object)
CommandPosition	<{TCmdPosition}>
UseSecondChannel	<boolean> (optional, default=false)
Talk	<boolean> (optional, default=true)
Listen	<boolean> (optional, default=true)
AllowSelectingConf	<boolean> (optional, default=false)
AllowChangingDestConf	<boolean> (optional, default=false)
AudioPriority	<int> (optional, default = 1) - 0 = below standard - 1 = standard - 2 = high - 3 = paging - 4 = emergency
ForceFail	<boolean> (optional, default=true)

8.10.4.8.2 Specific parameters for editing “call-to-conference

Parameters	**Type**
Conference	<int> (object-ID of conference-object)
CommandPosition	<{TCmdPosition}>
UseSecondChannel	<boolean> (default=false)
NewProperties	<struct> (contains the new properties of the call-to-conference-object, see table below)

NewProperties **(all members are optional)**	**Property-type**
Conference	<int> (object-ID of conference-object)
UseSecondChannel	<boolean>
Talk	<boolean>
Listen	<boolean>
Owner	<int> (object-ID of user-object)
AudioPriority	<int> - 0 = below standard - 1 = standard - 2 = high - 3 = paging - 4 = emergency
ForceFail	<boolean> (optional, default=true)

8.10.4.8.3 Specific parameters for deleting “call-to-conference

Hint: if the command does not exist, the delete operation is empty (no error on return)

```
<member>
<name>CommandPosition</name>
```

```
<value><{TCmdPosition}</value>
</member>
<member>
<name>Conference</name>
<value><int>{Object ID}</int></value>
</member>
<member>
<name>UseSecondChannel</name>                <!- Default: 0 -->
<value><boolean>{...}</boolean ></value>
</member>
```

8.10.4.8.4 Example

Example for creating a call to conference (Object ID 38476590) on Key 7 on physical port 5, Expansion Panel 3, Shift Page on node 2

```
<?xml version="1.0"?>
<methodCall>
<methodName>ConfigurationChange</methodName>
<params>
<param>
  <value>C0000000001</value>
  </param>
<param>
  <value><array><data>
    <value><struct>
      <member>
        <name>ChangeType</name>
        <value>create</value>
        </member>
      <member>
        <name>ObjectType</name>
        <value>call-to-conference</value>
        </member>
      <member>
        <name>SpecificParams</name>
        <value><struct>
          <member>
            <name>CommandPosition</name>
            <value><struct>
              <member>
                <name>ExpansionPanel</name>
                <value><i4>3</i4></value>
                </member>
              <member>
                <name>KeyNumber</name>
                <value><i4>7</i4></value>
                </member>
              <member>
                <name>Node</name>
                <value><i4>2</i4></value>
                </member>
              <member>
                <name>Page</name>
                <value><i4>2</i4></value>
                </member>
              <member>
                <name>Port</name>
                <value><i4>5</i4></value>
                </member>
              <member>
                <name>PositionType</name>
                <value>key</value>
                </member>
              </struct></value>
            </member>
          <member>
            <name>Conference</name>
            <value><i4>38476590</i4></value>
            </member>
          </struct></value>
        </member>
      </struct></value>
```

```
    </data></array></value>
  </param>
</params>
</methodCall>
```

Example for disabling the talk-privilege of the conference with object-ID=994082254 :

```
<?xml version="1.0"?>
<methodCall>
<methodName>ConfigurationChange</methodName>
<params>
<param>
  <value>C0000000011</value>
  </param>
<param>
  <value><struct>
    <member>
      <name>ChangeType</name>
      <value>edit</value>
      </member>
    <member>
      <name>ObjectType</name>
      <value>call-to-conference</value>
      </member>
    <member>
      <name>SpecificParams</name>
      <value><struct>
        <member>
          <name>CommandPosition</name>
          <value><struct>
            <member>
              <name>KeyNumber</name>
              <value><i4>7</i4></value>
              </member>
            <member>
              <name>Node</name>
              <value><i4>2</i4></value>
              </member>
            <member>
              <name>Page</name>
              <value><i4>1</i4></value>
              </member>
            <member>
              <name>Port</name>
              <value><i4>0</i4></value>
              </member>
            <member>
              <name>PositionType</name>
              <value>key</value>
              </member>
            </struct></value>
          </member>
        <member>
          <name>Conference</name>
          <value><i4>994082254</i4></value>
          </member>
        <member>
          <name>NewProperties</name>
          <value><struct>
            <member>
              <name>Talk</name>
              <value><boolean>0</boolean></value>
              </member>
            </struct></value>
          </member>
        </struct></value>
      </member>
    </struct></value>
  </param>
</params>
```

```
</methodCall>
```

8.10.4.9 Object type ‘call-to-group’

8.10.4.9.1 Specific parameters for creating “call-to-group”

Parameters	Type
Group	<int> (object-ID of group-object)
CommandPosition	<{TCmdPosition}>
UseSecondChannel	<boolean> (optional, default=false)
ShowIncomingMarker	<boolean> (optional, default=false)
DisableDestinVolumeAdjust	<boolean> (optional, default=false)
AudioPriority	<int> (optional, default = 1) - 0 = below standard - 1 = standard - 2 = high - 3 = paging - 4 = emergency
ForceFail	<boolean> (optional, default=true)

8.10.4.9.2 Specific parameters for editing “call-to-group”

Parameters	Type
Group	<int> (object-ID of group-object)
CommandPosition	<{TCmdPosition}>
UseSecondChannel	<boolean> (default=false)
NewProperties	<struct> (contains the new properties of the call-to-group-object, see table below)

NewProperties (all members are optional)	Property-type
Group	<int> (object-ID of group-object)
UseSecondChannel	<boolean>
Owner	<int> (object-ID of user-object)
AudioPriority	<int> - 0 = below standard - 1 = standard - 2 = high - 3 = paging - 4 = emergency
ShowIncomingMarker	<boolean> (optional, default=false)
DisableDestinVolumeAdjust	<boolean> (optional, default=false)

8.10.4.9.3 Specific parameters for deleting "call-to-group"

Hint: if the command does not exist, the delete operation is empty (no error on return)

```
<member>
<name>CommandPosition</name>
<value><{TCmdPosition}</value>
</member>
<member>
<name>Group</name>
<value><int>{Object ID}</int></value>
</member>
<member>
<name>UseSecondChannel</name>                <!-- Default: 0 -->
<value><boolean>{...}</boolean ></value>
</member>
```

8.10.4.9.4 Example

Example for creating a call to group (object ID 521994316) on the vox function on physical port 0 on node 2:

```
<?xml version="1.0"?>
<methodCall>
<methodName>ConfigurationChange</methodName>
<params>
<param>
  <value>C0000000002</value>
  </param>
<param>
  <value><array><data>
    <value><struct>
      <member>
        <name>ChangeType</name>
        <value>create</value>
        </member>
      <member>
        <name>ObjectType</name>
        <value>call-to-group</value>
        </member>
      <member>
        <name>SpecificParams</name>
        <value><struct>
          <member>
            <name>CommandPosition</name>
            <value><struct>
              <member>
                <name>IsInput</name>
                <value><boolean>1</boolean></value>
                </member>
              <member>
                <name>Node</name>
                <value><i4>2</i4></value>
                </member>
              <member>
                <name>Port</name>
                <value><i4>0</i4></value>
                </member>
              <member>
                <name>PositionType</name>
                <value>virtual-function</value>
                </member>
              <member>
                <name>VirtualFunctionType</name>
                <value>vox</value>
                </member>
              </struct></value>
            </member>
          <member>
            <name>Group</name>
            <value><i4>521994316</i4></value>
            </member>
          </struct></value>
```

```
        </member>
      </struct></value>
    </data></array></value>
  </param>
</params>
</methodCall>
```

Example for enabling the incoming marker of the group with object-ID=521994316:

```
<?xml version="1.0"?>
<methodCall>
<methodName>ConfigurationChange</methodName>
<params>
<param>
  <value>C0000000008</value>
  </param>
<param>
  <value><struct>
    <member>
      <name>ChangeType</name>
      <value>edit</value>
      </member>
    <member>
      <name>ObjectType</name>
      <value>call-to-group</value>
      </member>
    <member>
      <name>SpecificParams</name>
      <value><struct>
        <member>
          <name>CommandPosition</name>
          <value><struct>
            <member>
              <name>KeyNumber</name>
              <value><i4>7</i4></value>
              </member>
            <member>
              <name>Node</name>
              <value><i4>2</i4></value>
              </member>
            <member>
              <name>Page</name>
              <value><i4>1</i4></value>
              </member>
            <member>
              <name>Port</name>
              <value><i4>0</i4></value>
              </member>
            <member>
              <name>PositionType</name>
              <value>key</value>
              </member>
            </struct></value>
          </member>
        <member>
          <name>Group</name>
          <value><i4>521994316</i4></value>
          </member>
        <member>
          <name>NewProperties</name>
          <value><struct>
            <member>
              <name>ShowIncomingMarker</name>
              <value><boolean>1</boolean></value>
              </member>
            </struct></value>
          </member>
        </struct></value>
      </member>
    </struct></value>
  </param>
</params></methodCall>
```

8.10.4.10 Object type 'call-to-ifb'

8.10.4.10.1 Specific parameters for creating and editing "call-to-ifb"

Parameters	Type
IFBObjectID	<int> (object-ID of IFB-object)
TrunkingNetAddr	<int>
TrunkingIFBNumber	<int>
CommandPosition	<{TCmdPosition}>
UseSecondChannel	<boolean> (optional, default=false)
Monitoring	<int> (optional, default = 0) 0 = Switchable 1 = Always On 2 = Always Off
ForceFail	<boolean> (optional, default=true)

8.10.4.10.2 Specific parameters for deleting "call-to-ifb"

Hint: if the command does not exist, the delete operation is empty (no error on return)

Parameters	Type
IFBObjectID	<int> (object-ID of IFB-object)
TrunkingNetAddr	<int>
TrunkingIFBNumber	<int>
CommandPosition	<{TCmdPosition}>
UseSecondChannel	<boolean> (optional, default=false)

8.10.4.10.3 Example

```
<?xml version="1.0"?>
<methodCall>
<methodName>ConfigurationChange</methodName>
<params>
<param>
  <value>C0000000003</value>
  </param>
<param>
  <value><array><data>
    <value><struct>
      <member>
        <name>ChangeType</name>
        <value>create</value>
        </member>
      <member>
        <name>ObjectType</name>
        <value>call-to-ifb</value>
        </member>
      <member>
        <name>SpecificParams</name>
        <value><struct>
          <member>
            <name>CommandPosition</name>
            <value><struct>
              <member>
                <name>KeyNumber</name>
                <value><i4>1</i4></value>
                </member>
              <member>
                <name>Node</name>
                <value><i4>2</i4></value>
                </member>
```

```
              <member>
                <name>Page</name>
                <value><i4>1</i4></value>
                </member>
              <member>
                <name>Port</name>
                <value><i4>0</i4></value>
                </member>
              <member>
                <name>PositionType</name>
                <value>key</value>
                </member>
              </struct></value>
            </member>
<!-- use either IFBObjectID or TrunkingNetAddr and TrunkingIFBNumber; delete appropriate member definitions
from request -->

          <member>
            <name>IFBObjectID</name>
            <value><i4>1932025380</i4></value>
          </member>
<member>
<name>TrunkingNetAddr</name>
  <value><i4>2</i4></value>
</member>
<member>
  <name>TrunkingIFBNumber</name>
  <value><i4>5</i4></value>
</member>
          <member>
            <name>UseSecondChannel</name>
            <value><boolean>0</boolean></value>
            </member>
          </struct></value>
        </member>
      </struct></value>
    </data></array></value>
  </param>
</params>
</methodCall>
```

8.10.4.11 Object type ‘dim-level-command’

8.10.4.11.1 Specific parameters for creating “dim-level-command”

Parameters	Type
CommandPosition	<{TCmdPosition}>
Source	<{TPortAddress}>
Destination	<{TPortAddress}>
SourceUsesSecondChannel	<boolean> (optional, default=false)
DestinationUsesSecondChannel	<boolean> (optional, default=false)
DimValue	<int> 0: 0 [dB] (attenuation) 1: 3 [dB] (..) 2: 6 [dB] (..) 3: 9 [dB] (..) 4: 12 [dB] (..) 5: 18 [dB] (..) 6: 24 [dB] (..) 7: Mute (..)
ForceFail	<boolean> (optional, default=true)

8.10.4.11.2 Specific parameters for editing “dim-level-command”

Parameters	Type
CommandPosition	<{TCmdPosition}>
Source	<{TPortAddress}>
Destination	<{TPortAddress}>
SourceUsesSecondChannel	<boolean> (optional, default=false)
DestinationUsesSecondChannel	<boolean> (optional, default=false)
NewProperties	<struct> (contains the new properties of the dim-level-command -object, see table below)

NewProperties (all members are optional)	**Property-type**
Source	<{TPortAddress}>
Destination	<{TPortAddress}>
SourceUsesSecondChannel	<boolean>
DestinationUsesSecondChannel	<boolean>
Source	<{TPortAddress}>
DimValue	<int> 0: 0 [dB] (attenuation) 1: 3 [dB] (..) 2: 6 [dB] (..) 3: 9 [dB] (..) 4: 12 [dB] (..) 5: 18 [dB] (..) 6: 24 [dB] (..) 7: Mute (..)

8.10.4.11.3 Specific parameters for deleting "dim-level-command"

Hint: if the command does not exist, the delete operation is empty (no error on return)

```
<member>
<name>CommandPosition</name>
<value><{TCmdPosition}</value>
</member>
<member>
<name>Source</name>
<value>{TPortAddress}</value>
</member>
<member>
<!- this member is optional, default value = false -->
<name>SourceUsesSecondChannel</name>
<value><boolean>..</boolean></value>
</member>
<member>
<name>Destination</name>
<value>{TPortAddress}</value>
</member>
<member>
<!- this member is optional, default value = false -->
<name>DestinationUsesSecondChannel</name>
<value><boolean>..</boolean></value>
</member>
```

8.10.4.11.4 Example

Example to create a dim-level command on the „On-Call" function on physical port 0, node 2. The command will dim/attenuate the crosspoint from port (node=2, port=26) to port (node=5, port=17) by 12 dB.

```
<?xml version="1.0"?>
<methodCall>
<methodName>ConfigurationChange</methodName>
<params>
<param>
  <value>C0000000002</value>
  </param>
<param>
  <value><array><data>
    <value><struct>
      <member>
        <name>ChangeType</name>
        <value>create</value>
        </member>
      <member>
        <name>ObjectType</name>
        <value>dim-level-command</value>
        </member>
      <member>
        <name>SpecificParams</name>
        <value><struct>
          <member>
            <name>CommandPosition</name>
            <value><struct>
              <member>
                <name>IsInput</name>
                <value><boolean>0</boolean></value>
                </member>
              <member>
                <name>Node</name>
                <value><i4>2</i4></value>
                </member>
              <member>
                <name>Port</name>
                <value><i4>0</i4></value>
                </member>
              <member>
                <name>PositionType</name>
                <value>virtual-function</value>
                </member>
```

```
              <member>
                <name>VirtualFunctionType</name>
                <value>on-call</value>
                </member>
              </struct></value>
            </member>
          <member>
            <name>Destination</name>
            <value><struct>
              <member>
                <name>IsInput</name>
                <value><boolean>0</boolean></value>
                </member>
              <member>
                <name>Node</name>
                <value><i4>5</i4></value>
                </member>
              <member>
                <name>Port</name>
                <value><i4>17</i4></value>
                </member>
              </struct></value>
            </member>
          <member>
            <name>DimValue</name>
            <value><i4>4</i4></value>
            </member>
          <member>
            <name>Source</name>
            <value><struct>
              <member>
                <name>IsInput</name>
                <value><boolean>1</boolean></value>
                </member>
              <member>
                <name>Node</name>
                <value><i4>2</i4></value>
                </member>
              <member>
                <name>Port</name>
                <value><i4>26</i4></value>
                </member>
              </struct></value>
            </member>
          </struct></value>
        </member>
      </struct></value>
    </data></array></value>
  </param>
</params>
</methodCall>
```

Example to change the dim-level 24 [dB] on the „On-Call“ function on physical port 0, node 2. The command will dim/attenuate the crosspoint from port (node=2, port=26) to port (node=5, port=17).

```
<?xml version="1.0"?>
<methodCall>
<methodName>ConfigurationChange</methodName>
<params>
<param>
  <value>C0000000002</value>
  </param>
<param>
  <value><array><data>
    <value><struct>
      <member>
        <name>ChangeType</name>
        <value>edit</value>
        </member>
      <member>
        <name>ObjectType</name>
        <value>dim-level-command</value>
```

```
    </member>
  <member>
    <name>SpecificParams</name>
    <value><struct>
      <member>
        <name>CommandPosition</name>
        <value><struct>
          <member>
            <name>IsInput</name>
            <value><boolean>0</boolean></value>
            </member>
          <member>
            <name>Node</name>
            <value><i4>2</i4></value>
            </member>
          <member>
            <name>Port</name>
            <value><i4>0</i4></value>
            </member>
          <member>
            <name>PositionType</name>
            <value>virtual-function</value>
            </member>
          <member>
            <name>VirtualFunctionType</name>
            <value>on-call</value>
            </member>
          </struct></value>
        </member>
      <member>
        <name>Source</name>
        <value><struct>
          <member>
            <name>IsInput</name>
            <value><boolean>1</boolean></value>
            </member>
          <member>
            <name>Node</name>
            <value><i4>2</i4></value>
            </member>
          <member>
            <name>Port</name>
            <value><i4>26</i4></value>
            </member>
          </struct></value>
        </member>
      <member>
        <name>Destination</name>
        <value><struct>
          <member>
            <name>IsInput</name>
            <value><boolean>0</boolean></value>
            </member>
          <member>
            <name>Node</name>
            <value><i4>5</i4></value>
            </member>
          <member>
            <name>Port</name>
            <value><i4>17</i4></value>
            </member>
          </struct></value>
        </member>
      <member>
        <name>NewProperties</name>
        <value><struct>
          <member>
            <name>DimValue</name>
            <value><i4>4</i4></value>
            </member>
          </struct></value>
        </member>
```

```
          </struct></value>
        </member>
      </struct></value>
    </data></array></value>
  </param>
</params>
</methodCall>
```

8.10.4.12 Object type 'call-to-port-cmd'

8.10.4.12.1 Specific parameters for creating/editing "call-to-port-cmd"

Parameters	Type
CommandPosition	<{TCmdPosition}>
SourceUsesSecondChannel	<boolean> (optional, default=false)
DestinationPortAddress	<{TPortAddress}>
DestinationUsesSecondChannel	<boolean> (optional, default = false)
TrunkingNetAddr	<int> (optional, for trunking)
TrunkingPortAddr	<int> (optional, for trunking)
NewProperties	<struct> (contains the new properties of the dim-level-command -object, see table below)
ForceFail	<boolean> (optional, default=true)

NewProperties (all members are optional)	Property-type
Priority	<int> (optional, default=1(standard))
TrunkcallPriority	<int> (optional, default=1(standard))
Isolate	<boolean> (optional, default=false)
IsolateSelf	<boolean> (optional, default = false)
DisableCrosspointVolume	<boolean> (optional, default = false)
BeepDestOnCall	<boolean> (optional, default = false)
AutolistenFromDest	<boolean> (optional, default = false)
AllowSetInOutGain	<boolean> (optional,default = true)
DuplexCall	<boolean> (optional, default = false)
Monitoring	<int> (optional, default = 0) 0 = Switchable 1 = Always On 2 = Always Off
AllowTelephoneCall	<boolean> (optional, default=false)
AllowFixedNumber	<boolean> (optional, default=false)
FixedNumber	<string> (optional)
AllowPhonebook	<boolean> (optional, default=false)
AllowDialpad	<boolean> (optional, default=false)

8.10.4.12.2 Specific parameters for deleting "call-to-port-cmd"

Hint: if the command does not exist, the delete operation is empty (no error on return)

Parameters	Type
CommandPosition	<{TCmdPosition}>
SourceUsesSecondChannel	<boolean> (optional, default=false)
DestinationPortAddress	<{TPortAddress}>
DestinationUsesSecondChannel	<boolean> (optional, default = false)
TrunkingNetAddr	<int> (optional, for trunking)
TrunkingPortAddr	<int> (optional, for trunking)

8.10.4.12.3 Example

Example for creating a call to port on Key 7 on physical port 5, Expansion Panel 3, Shift Page on node 2

```
<?xml version="1.0"?>
<methodCall>
<methodName>ConfigurationChange</methodName>
<params>
<param>
  <value>C0000000001</value>
  </param>
<param>
  <value><array><data>
    <value><struct>
      <member>
        <name>ChangeType</name>
        <value>create</value>
        </member>
      <member>
        <name>ObjectType</name>
        <value>call-to-port-cmd</value>
        </member>
      <member>
        <name>SpecificParams</name>
        <value><struct>
          <member>
            <name>CommandPosition</name>
            <value><struct>
              <member>
                <name>ExpansionPanel</name>
                <value><i4>3</i4></value>
                </member>
              <member>
                <name>KeyNumber</name>
                <value><i4>7</i4></value>
                </member>
              <member>
                <name>Node</name>
                <value><i4>2</i4></value>
                </member>
              <member>
                <name>Page</name>
                <value><i4>2</i4></value>
                </member>
              <member>
                <name>Port</name>
                <value><i4>5</i4></value>
                </member>
              <member>
                <name>PositionType</name>
                <value>key</value>
                </member>
              </struct></value>
            </member>
          <member>
            <name>SourceUsesSecondChannel</name>
            <value><boolean>0</boolean ></value>
          </member>
<!-- use either DestinationPortAddress and DestinationUseSecondChannel or TrunkingNetAddr and
TrunkingPortAddr; delete appropriate member definitions from request -->
          <member>
            <name>DestinationPortAddress</name>
            <value><{TportAdress}></value>
          </member>
          <member>
            <name>DestinationUsesSecondChannel</name>
            <value><boolean>0</boolean></value>
          </member>
          <member>
            <name>TrunkingNetAddr</name>
            <value><i4>0</i4></value>
          </member>
          <member>
            <name>TrunkingPortAddr</name>
            <value><i4>0</i4></value>
          </member>
```

```
          <member>
            <name>Priority</name>
            <value><i4>1</i4></value>
          </member>
          <member>
            <name>TrunkCallPriority</name>
            <value><i4>1</i4></value>
          </member>
          <member>
            <name>Isolate</name>
            <value><boolean>0</boolean></value>
          </member>
          <member>
            <name>IsolateSelf</name>
            <value><boolean>0</boolean></value>
          </member>
          <member>
            <name>DisableCrosspointVolume</name>
            <value><boolean>0</boolean></value>
          </member>
          <member>
            <name>BeepDestinationOnCall</name>
            <value><boolean>0</boolean></value>
          </member>
          <member>
            <name>AutolistenFromDest</name>
            <value><boolean>0</boolean></value>
          </member>
          <member>
            <name>AllowSetInOutGain</name>
            <value><boolean>0</boolean></value>
          </member>
        </struct></value>
      </member>
    </struct></value>
  </data></array></value>
</param>
</params>
</methodCall>
```

Example for deleting a call to port on Key 7 on physical port 5, Expansion Panel 3, Shift Page on node 2

```
<?xml version="1.0"?>
<methodCall>
<methodName>ConfigurationChange</methodName>
<params>
<param>
  <value>C0000000001</value>
  </param>
<param>
  <value><array><data>
    <value><struct>
      <member>
        <name>ChangeType</name>
        <value>delete</value>
        </member>
      <member>
        <name>ObjectType</name>
        <value>call-to-port-cmd</value>
        </member>
      <member>
        <name>SpecificParams</name>
        <value><struct>
          <member>
            <name>CommandPosition</name>
            <value><struct>
              <member>
                <name>ExpansionPanel</name>
                <value><i4>3</i4></value>
                </member>
              <member>
                <name>KeyNumber</name>
```

```
                <value><i4>7</i4></value>
                </member>
              <member>
                <name>Node</name>
                <value><i4>2</i4></value>
                </member>
              <member>
                <name>Page</name>
                <value><i4>2</i4></value>
                </member>
              <member>
                <name>Port</name>
                <value><i4>5</i4></value>
                </member>
              <member>
                <name>PositionType</name>
                <value>key</value>
                </member>
              </struct></value>
            </member>
          <member>
            <name>SourceUsesSecondChannel</name>
            <value><boolean>0</boolean ></value>
          </member>
<!-- use either DestinationPortAddress and DestinationUseSecondChannel or TrunkingNetAddr and
TrunkingPortAddr; delete appropriate member definitions from request -->
          <member>
            <name>DestinationPortAddress</name>
            <value><{TportAdress}></value>
          </member>
          <member>
            <name>DestinationUsesSecondChannel</name>
            <value><boolean>0</boolean></value>
          </member>
          <member>
            <name>TrunkingNetAddr</name>
            <value><i4>0</i4></value>
          </member>
          <member>
            <name>TrunkingPortAddr</name>
            <value><i4>0</i4></value>
          </member>
        </struct></value>
      </member>
    </struct></value>
  </data></array></value>
</param>
</params>
</methodCall>
```

8.10.4.13 Object type 'listen-to-port-cmd'

8.10.4.13.1 Specific parameters for creating/editing "listen-to-port-cmd"

Parameter	Type
CommandPosition	<{TCmdPosition}>
SourcePortAddress	<{TPortAddress}>
SourceUsesSecondChannel	<boolean> (optional, default=false)
LocalUsesSecondChannel	<boolean> (optional, default = false)
TrunkingNetAddr	<int> (optional, for trunking)
TrunkingPortAddr	<int> (optional, for trunking)
NewProperties	<struct> (contains the new properties of the dim-level-command -object, see table below)
ForceFail	<boolean> (optional, default=true)

NewProperties (all members are optional)	Property-type
Priority	<int> (optional, default=1(standard))
TrunkcallPriority	<int> (optional, default=1(standard))
DisableCrosspointVolume	<boolean> (optional, default=false)
AllowSetInOutGain	<boolean> (optional, default=false)

8.10.4.13.2 Specific parameters for deleting "listen-to-port-cmd"

Parameters	Type
CommandPosition	<{TCmdPosition}>
SourcePortAddress	<{TPortAddress}>
SourceUsesSecondChannel	<boolean> (optional, default=false)
LocalUsesSecondChannel	<boolean> (optional, default = false)
TrunkingNetAddr	<int> (optional, for trunking)
TrunkingPortAddr	<int> (optional, for trunking)

8.10.4.14 Object type “route-cmd”

8.10.4.14.1 Specific parameters for creating/editing “route-cmd”

Parameters	Type
CommandPosition	<{TCmdPosition}>
SourcePortAddress	<{TPortAddress}>
SourceUsesSecondChannel	<boolean> (optional, default = false)
DestinationPortAddress	<{TPortAddress}>
DestinationUsesSecondChannel	<boolean> (optional, default = false)
NewProperties	<struct> (contains the new properties of the dim-level-command -object, see table below)
ForceFail	<boolean> (optional, default=true)

NewProperties (all members are optional)	Property-type
Priority	<int> (optional, default=1(standard))
DisableCrosspointVolume	<boolean> (optional, default=false)

8.10.4.14.2 Specific parameters for deleting “route-cmd”

Parameters	Type
CommandPosition	<{TCmdPosition}>
SourcePortAddress	<{TPortAddress}>
SourceUsesSecondChannel	<boolean> (optional, default = false)
DestinationPortAddress	<{TPortAddress}>

8.10.4.15 Object type “switch-gpio-out-cmd”

8.10.4.15.1 Specific parameters for creating “switch-gpio-out-cmd”

Parameters	Type
CommandPosition	<{TCmdPosition}>
PortAddress	<{TPortAddress}>
Slot	<int>
GPIONumber	<int>
ForceFail	<boolean> (optional, default=true)

This object type hasn’t got any properties.

8.10.4.15.2 Specific parameters for deleting “switch-gpio-out-cmd”

Parameters	Type
CommandPosition	<{TCmdPosition}>
PortAddress	<{TPortAddress}>
Slot	<int>
GPIONumber	<int>

8.10.4.16 Object type “select-audiopatch-cmd”

8.10.4.16.1 Specific parameters for creating “select-audiopatch-cmd”

Parameters	Type
CommandPosition	<{TCmdPosition}>
PortAddress	<{TPortAddress}>
Audiopatch	<int> (object ID of the AudioPatch)
ForceFail	<boolean> (optional, default=true)

This object type hasn’t got any properties.

8.10.4.16.2 Specific parameters for deleting “select-audiopatch-cmd”

Parameters	Type
CommandPosition	<{TCmdPosition}>
PortAddress	<{TPortAddress}>
Audiopatch	<int> (object ID of the AudioPatch)

8.10.4.17 Object type “control-audiopatch-cmd”

8.10.4.17.1 Specific parameters for creating “control-audiopatch-cmd”

Parameters	Type
CommandPosition	<{TCmdPosition}>
PortAddress	<{TPortAddress}>
AudiopatchElement	<int>
ForceFail	<boolean> (optional, default=true)

This object type hasn’t got any properties.

8.10.4.17.2 Specific parameters for deleting “control-audiopatch-cmd”

Parameters	Type
CommandPosition	<{TCmdPosition}>
PortAddress	<{TPortAddress}>
AudiopatchElement	<int>

8.10.4.18 Object type “remote-key-cmd”

8.10.4.18.1 Specific parameters for creating “remote-key-cmd”

Parameters	Type
CommandPosition	<{TCmdPosition}>
DestinationPortAddress	<{TPortAddress}>
ExpansionPanel	<int> (optional)
KeyPage	<int> 0 – standard 1 – shift page 2 – virtual key
KeyNumber	<int>
NewProperties	<struct> (contains the new properties of the dim-level-command -object, see table below)
ForceFail	<boolean> (optional, default=true)

NewProperties (all members are optional)	Property-type
PressKey	<boolean> (default = false)
LockKey	<boolean> (default = false)
MarkerFlag	<boolean> (default = false)
MarkerValue	<int> (default=0)
TextFlag	<boolean> (default = false)
TextValue	<string> (default=””)
PressKeySecondary	<boolean> (default = false) Note: PressKey or PressKeySecondary cannot be activated at the same time.

8.10.4.18.2 Specific parameters for deleting “remote-key-cmd”

Parameters	Type
CommandPosition	<{TCmdPosition}>
DestinationPortAddress	<{TPortAddress}>
ExpansionPanel	<int> (optional)
KeyPage	<int> 0 – standard 1 – shift page 2 – virtual key
KeyNumber	<int>

8.10.4.19 Object type “reply-cmd”

8.10.4.19.1 Specific parameters for creating and editing“reply-cmd”

Parameters	Type
CommandPosition	<{TCmdPosition}>
NewProperties	<struct> (contains the new properties of the dim-level-command -object, see table below)
ForceFail	<boolean> (optional, default=true)

NewProperties (all members are optional)	Property-type
Priority	<int> (default=1) 0 = Below Standard 1= Standard 2 = High 3 = Paging 4 = Emergency
EnableForCallsFromConf	<boolean> (default = false)
DuplexEnabledForCall2Port	<boolean> (default = false) Not supported on Bolero Beltpacks and Menu Keys
ScrollEnabled	<boolean> (default = false)

8.10.4.19.2 Specific parameters for deleting “reply-cmd”

Parameters	Type
CommandPosition	<{TCmdPosition}>

8.10.4.20 Object type “edit-conference-cmd”

8.10.4.20.1 Specific parameters for creating “edit-conference-cmd”

Parameters	**Type**
CommandPosition	<{TCmdPosition}>
ForceFail	<boolean> (optional, default=true)

This object type hasn’t got any properties.

8.10.4.20.2 Specific parameters for deleting “edit-conference-cmd”

Parameters	**Type**
CommandPosition	<{TCmdPosition}>

8.10.4.21 Object type “edit-ifb-cmd”

8.10.4.21.1 Specific parameters for creating “edit-ifb-cmd”

Parameters	**Type**
CommandPosition	<{TCmdPosition}>
ForceFail	<boolean> (optional, default=true)

This object type hasn’t got any properties.

8.10.4.21.2 Specific parameters for deleting “edit-ifb-cmd”

Parameters	**Type**
CommandPosition	<{TCmdPosition}>

8.10.4.22 Object type “dim-panel-speaker-cmd”

8.10.4.22.1 Specific parameters for creating “dim-panel-speaker -cmd”

Parameters	Type
CommandPosition	<{TCmdPosition}>
DestinationPortAddress	<{TPortAddress}>
NewProperties	<struct> (contains the new properties of the dim-level-command -object, see table below)
ForceFail	<boolean> (optional, default=true)

NewProperties (all members are optional)	Property-type
DimSpeakerBy	<int> (default = 0)

8.10.4.22.2 Specific parameters for deleting “dim-panel-speaker-cmd”

Parameters	Type
CommandPosition	<{TCmdPosition}>
DestinationPortAddress	<{TPortAddress}>

8.10.4.23 Object type “beep-panel-cmd”

8.10.4.23.1 Specific parameters for creating “beep-panel-cmd”

Parameters	Type
CommandPosition	<{TCmdPosition}>
DestinationPortAddress	<{TPortAddress}>
ForceFail	<boolean> (optional, default=true)

This object type hasn’t got any properties.

8.10.4.23.2 Specific parameters for deleting “beep-panel-cmd”

Parameters	Type
CommandPosition	<{TCmdPosition}>
DestinationPortAddress	<{TPortAddress}>

8.10.4.24 Object type “telephone-dial-keypad-cmd”

8.10.4.24.1 Specific parameters for creating “telephone-dial-keypad-cmd”

Parameters	Type
CommandPosition	<{TCmdPosition}>
KeyFunction	<int>
ForceFail	<boolean> (optional, default=true)

This object type hasn’t got any properties.

8.10.4.24.2 Specific parameters for deleting “telephone-dial-keypad-cmd”

Parameters	Type
CommandPosition	<{TCmdPosition}>
KeyFunction	<int>

8.10.4.24.3 Specific parameters for editing “telephone-dial-keypad-cmd”

NewProperties	Property-type
KeyFunction	<int>
KeyPhoneNumber (Only available if KeyFunction = 17)	<string>

8.10.4.25 Object type “telephone-dial-hangup-cmd”

8.10.4.25.1 Specific parameters for creating “telephone-dial-hangup-cmd”

Parameters	Type
CommandPosition	<{TCmdPosition}>
Function	<int>
ForceFail	<boolean> (optional, default=true)

This object type hasn’t got any properties.

8.10.4.25.2 Specific parameters for deleting “telephone-dial-hangup-cmd”

Parameters	Type
CommandPosition	<{TCmdPosition}>
Function	<int>

8.10.4.26 Object type “logic-cmd”

8.10.4.26.1 Specific parameters for creating “logic-cmd”

Parameters	Type
CommandPosition	<{TCmdPosition}>
LogicSource	<int>
ForceFail	<boolean> (optional, default=true)

This object type hasn’t got any properties.

8.10.4.26.2 Specific parameters for deleting “logic-cmd”

Parameters	Type
CommandPosition	<{TCmdPosition}>
LogicSource	<int>

8.10.4.27 Object type “kill-partyline-mic-cmd”

8.10.4.27.1 Specific parameters for creating “kill-partyline-mic-cmd”

Parameters	Type
CommandPosition	<{TCmdPosition}>
ForceFail	<boolean> (optional, default=true)

This object type hasn’t got any properties.

8.10.4.27.2 Specific parameters for deleting “kill-partyline-mic-cmd”

Parameters	Type
CommandPosition	<{TCmdPosition}>

8.10.4.28 Object type “auto-listen-off-cmd”

8.10.4.28.1 Specific parameters for creating “auto-listen-off-cmd”

Parameters	Type
CommandPosition	<{TCmdPosition}>
ForceFail	<boolean> (optional, default=true)

This object type hasn’t got any properties.

8.10.4.28.2 Specific parameters for deleting “auto-listen-off-cmd”

Parameters	Type
CommandPosition	<{TCmdPosition}>

8.10.4.29 Object type “set-input-output-gain-cmd”

8.10.4.29.1 Specific parameters for creating “set-input-output-gain-cmd”

Parameters	Type
CommandPosition	<{TCmdPosition}>
DestinationPortAddress	<{TPortAddress}>
IsInput	<boolean>
ForceFail	<boolean> (optional, default=true)

This object type hasn’t got any properties.

8.10.4.29.2 Specific parameters for deleting “set-input-output-gain-cmd”

Parameters	Type
CommandPosition	<{TCmdPosition}>
DestinationPortAddress	<{TPortAddress}>
IsInput	<boolean>

8.10.4.30 Object type “sidetone-cmd”

8.10.4.30.1 Specific parameters for creating “sidetone-cmd”

Parameters	**Type**
CommandPosition	<{TCmdPosition}>
NewProperties	<struct> (contains the new properties of the dim-level-command -object, see table below)
ForceFail	<boolean> (optional, default=true)

NewProperties (all members are optional)	**Property-type**
EnableForSpeakerMode	<boolean> (default = false)
EnableForHeadsetMode	<boolean> (default = false)
NormSidetoneLevel	<int> (default = -6)

8.10.4.30.2 Specific parameters for deleting “sidetone-cmd”

Parameters	**Type**
CommandPosition	<{TCmdPosition}>

8.10.4.31 Object type “send-string-cmd”

8.10.4.31.1 Specific parameters for creating “send-string-cmd”

Parameters	Type
CommandPosition	<{TCmdPosition}>
SendString	<string>
ForceFail	<boolean> (optional, default=true)

This object type hasn’t got any properties.

8.10.4.31.2 Specific parameters for deleting “send-string-cmd”

Parameters	Type
CommandPosition	<{TCmdPosition}>
SendString	<string>

8.10.4.32 Object type “clone-output-port-cmd”

8.10.4.32.1 Specific parameters for creating “clone-output-port-cmd”

Parameters	Type
CommandPosition	<{TCmdPosition}>
ForceFail	<boolean> (optional, default=true)

NewProperties (all members are optional)	Property-type
OutputToCloneAddress	<{TPortAddress}>
OutputToCloneSecondChannel	<boolean> (optional, default=false)
ClonedOutputAddress	<{TPortAddress}>
ClonedOutputSecondChannel	<boolean> (optional, default=false)

8.10.4.32.2 Specific parameters for deleting “clone-output-port-cmd”

Parameters	Type
CommandPosition	<{TCmdPosition}>

8.10.4.33 Object type “panel-key”

8.10.4.33.1 Specific parameters for editing “panel-key”

Parameters	Type
Node	<int>
Port	<int>
ExpansionPanel	<int>
Page	<int>
KeyNumber	<int>
IsInput	<boolean>
NewProperties	<struct> (contains the new properties of the dim-level-command -object, see table below)

NewProperties (all members are optional)	Property-type
AutoLabelFlag	<boolean> (optional, default=true)
LabelValue	<string> (optional, default=””, 8 characters max)
KeyMode	<int> (optional, default=1) 0 – Auto 1 – Momentary(PTT) 2 - Latching
LatchingTimeout	<int> (optional, default = 0) 0 – Net Default 1 – Permanent 2 - 1 sec ……
RadioButton	<int> (optional, default = 0) 0 - <none> 1 – Group 1 2 – Group 2 3 – Group 3 4 - Group 4
Dim	<boolean> (optional, default = true)
ActionByKeyPressed	<int> (optional, default = 1) 0 – No Unmute when activated, no mute when deactivated 1 – Unmute when activated, no mute when deactivated 2 – Unmute when activated, mute again when deactivated
ScrollEnable	<boolean> (optional,default = false)
RestoreVolumeLevel	<boolean> (optional, default = false)
TextColor	TTextColor (optional, default = Panel-Default)
AutoSubtitle	<boolean> (optional, default = true) Supported only on RSP 1232 panels
Subtitle	<string> (optional, default = empty) Supported only on RSP 1232 panels
ShowSubtitle	<boolean> (optional, default=0)
GroupColor	<int> (optional, default = 0) 16 - <none> 0 – 15 Color values Supported only for RSP 12xx series

NewProperties (all members are optional)	Property-type
UseScrollListEntryKeyMode	<boolean> (optional)
MonitoringState	<int> (optional, default=0) 0 = initial off 1 = initial on

8.10.4.34 Object type “client-card”

8.10.4.34.1 Specific parameters for editing “client-card”

Parameters	Type
Node	<int>
Slot	<int>
LongName	<string> max 32 char Only supported on Artist-1024 SICs
VoIP	<struct> (contains the type-specific properties for VoIP Client-Card; see table below)
Aes67	<struct> (contains the type-specific properties for AES67 Client-Card and UIC; see table below)
Madi	<struct> (contains the type-specific properties for MADI Client-Card and UIC; see table below)
Nmos	<struct> (contains the NMOS-specific properties for AES67 Client-Card and UIC; see table below) Can be set in parallel to Aes67 struct
Dante	<struct> (contains the type-specific properties for Dante Client-Card and UIC; see table below)

VoIP (all members are optional)	Property-type
ObtainIpAddrAutomatic	<boolean> (optional)
ObtainDnsAddrAutomatic	<boolean> (optional)
IpAddress	{IP-address} (optional)
NetMask	<string> (optional)
DefaultGateway	{IP-address} (optional)
PrimaryDnsServer	{IP-address} (optional)
SecondaryDnsServer	{IP-address} (optional)
DnsHostName	<string> (optional)
TcpUdpPort	<int> (optional)
ServiceCodePoint	<int> (optional)
EthernetLinkMode	{EthernetLinkMode} (optional) 1 = Autonegotiation 2 = 10 Mbit Half 3 = 10 Mbit Full 4 = 100 Mbit Half 5 = 100 Mbit Full

Aes67 (all members are optional)	**Property-type**
MediaInterface	<int> (optional) 1 = Interface 1 -> default 2 = Interface 2
IpAddress	{IP-address} (optional)
Network	<string> (optional)
DefaultGateway	{IP-address} (optional)
TcpUdpPort	<int> (optional) 1024..65535
ServiceCodePoint	<int> (optional) 0..63
IGMPVersion	<int> (optional) 0..1
PTP	<int> (optional) 0..127
PtpPriority	<int> (optional) 0..255
PtpPriority2	<int> (optional) 0..255
PtpMode	<int> (optional) 0 = Multicast 1 = Hybrid
PtpAnnounceInterval	<int> (optional) 3..4
AllocatedPorts	<struct> see struct AllocatedPorts definition
NetworkSpeed	<int> (optional) 0 → Auto 1 → FullDuplex 1G
AllocatedGPIOs	<struct> see struct AllocatedGPIOs definition

Nmos (all members are optional)	Property-type
Enable	<boolean> (optional)
SupportedVersion04	<array><int> (optional) 1 = NMOS API Version 1.1 2 = NMOS API Version 1.2 3 = NMOS API Version 1.3
LocalPort	<int> (optional) 1024 – 65535
NetworkInterface	<int> (optional) 0 = Media_1 1 = Media_2 2 = Config
RegistrationApiVersion	<int> (optional) 1 = NMOS API Version 1.1 2 = NMOS API Version 1.2 3 = NMOS API Version 1.3
RegistrationIp	{IP-address} (optional) No multicast
RegistrationMode	<int> (optional) 0 = Automatic 1 = Peer2Peer 2 = Manual
RegistrationPort	<int> (optional) 1 – 65535

Madi (all members are optional)	Property-type
MadiInterface	<int> (optional) 0 = Interface 1 -> default 1 = Interface 2
FrameLength	<int> (optional) 0 = 56 Channels 1 = 64 Channels
Channels	<int> (optional) 0 = 01 - 08 1 = 09 - 16 2 = 17 - 24 3 = 25 - 32 4 = 33 - 40 5 = 41 - 48 6 = 49 - 56 7 = 57 -64 (need FrameLengt = 1)
UpInterface	<int> (optional) 0 = Optical 1 = Electrical
DownInterface	<int> (optional) 0 = Optical 1 = Electrical
AllocatedPorts	<struct> see struct AllocatedPorts definition
SyncSource	<int> (optional) 0 = Internal Clock 1 External Clock
Smux	<int> (optional)

Madi (all members are optional)	Property-type
	0 = Disabled 1 = Enabled

Dante (all members are optional)	Property-type
DanteDeviceName	<string> (optional)
AllocatedPorts	<struct> see struct AllocatedPorts definition

AllocatedPorts (all members are optional) Use as sub struct in Madi, Dante or AES67 UICs	Property-type
Media_1	Applies only to AES67 SIC <int> (optional) 0 ..128 in blocks of 8 Please note: A maximum of 128 ports are allocated across Media_1, Media_2 and 2022-7
Media_2	Applies only to AES67 SIC <int> (optional) 0 ..128 in blocks of 8 Please note: A maximum of 128 ports are allocated across Media_1, Media_2 and 2022-7
Dash7 2022_7 (deprecated)	Applies only to AES67 SIC <int> (optional) 0 ..128 in blocks of 8 Please note: A maximum of 128 ports are allocated across Media_1, Media_2 and 2022-7
Interface_1	Applies only to Madi and Dante SIC <int> (optional) 0 ..128 in blocks of 8
Interface_2	Applies only to Madi SIC <int> (optional) 0 ..128 in blocks of 8

AllocatedGPIOs (all members are optional) Use as sub struct in AES67 UICs	Property-type
Media_1	Applies only to AES67 SIC <int> (optional) 0 ..64 in blocks of 8 Please note: A maximum of 64 ports are allocated across Media_1, Media_2 and 2022-7
Media_2	Applies only to AES67 SIC <int> (optional) 0 ..64 in blocks of 8 Please note: A maximum of 64 ports are allocated across Media_1, Media_2 and 2022-7
Dash7 2022_7 (deprecated)	Applies only to AES67 SIC <int> (optional) 0 ..64 in blocks of 8 Please note: A maximum of 64 ports are allocated across Media_1, Media_2 and 2022-7

8.10.4.34.1 Specific parameters for swapping “client-card”

Parameters	Type
Node	<int>
Slot	<int>
NodeTarget	<int>
SlotTarget	<int>
SwapForced	<boolean> Note: If the target is not empty / has a configuration, with this flag you can force to swap. If in this case the flag is set to false, the configuration won´t get swapped.

8.10.4.34.1 Example

This example swaps 2 SIC configurations with each other. It ensures if also the target configuration exists, that it gets swaped.

```
<?xml version="1.0" encoding="UTF-8"?>
 <methodCall>
 <methodName>ConfigurationChange</methodName>
 <params>
     <param>
         <value>C0123456789</value>
     </param>
 <param>
    <value>
        <array>
            <data>
                <value>
                    <struct>
                        <member>
                            <name>ChangeType</name>
                            <value>swap</value>
                        </member>
                        <member>
                            <name>ObjectType</name>
                            <value>client-card</value>
                        </member>
                        <member>
                            <name>SpecificParams</name>
                            <value>
                                <struct>
                                    <member>
                                        <name>Node</name>
                                        <value><i4>84</i4></value>
                                    </member>
                                    <member>
                                        <name>Slot</name>
                                        <value><i4>1</i4></value>
                                    </member>
                                    <member>
                                        <name>NodeTarget</name>
                                        <value>
                                            <i4>84</i4>
                                        </value>
                                    </member>
                                    <member>
                                        <name>SlotTarget</name>
                                        <value>
                                            <i4>4</i4>
                                        </value>
                                    </member>
                                    <member>
                                        <name>SwapForced</name>
                                        <value><boolean>1</boolean></value>
                                    </member>
                                </struct>
                            </value>
```

```
                        </member>
                    </struct>
                </value>
            </data>
        </array>
    </value>
</param>
</params>
</methodCall>
```

8.10.4.35 Object type "virtual-key"

8.10.4.35.1 Specific parameters for creating "virtual-key"

Parameters	Type
Virtual Keys	<array><int> 1 .. 10
PortAddress	<{TPortAddress}>

8.10.4.35.2 Specific parameters for editing "virtual-key"

Parameters	Type
Virtual Key	<int> 1 - 10
PortAddress	<{TPortAddress}>
NewProperties	

NewProperties (all members are optional)	Property-type
AutoLabelFlag	<boolean> (optional)
KeyMode	<int> (optional) 0 Auto 1 Momentary 2 Latching
LatchingTimeOut	<int> (optional)
RadioButton	<int> (optional) 0 = None 1 = Group 1 2 = Group 2 3 = Group 3 4 = Group 4
DimSpeakerWhenActivated	<boolean> (optional)
RestartLatchingTimer	<boolean> (optional)

8.10.4.35.3 Specific parameters for deleting "virtual-key"

Parameters	Type
Virtual Keys	<array><int> 1 .. 10
PortAddress	<{TPortAddress}>

8.10.4.36 Object type ‘device

8.10.4.36.1 Specific parameters for creating / edit “device”

The create device functionality requires a name for the new device. The change properties of the device-object are shown in the table below:

Supported properties	Property-type
LongName	<string> (32 unicode characters max) (optional)
DeviceType	<int> (required) 0 = Connect VoIP 1 = NSA-003A 2 = NSA-004A 3 = NSA-005A 4 = NSA-006A 5 = NSA-007A 6 = NSA-010C
NewLongName	<string> (32 unicode characters max) (optional) Note: Only available when editing an existing device
AssignedPorts	<array> <struct>
AssignedGpios	<array> <struct>
DeviceConnectVoIP	<struct> (optional)
DeviceNsa	<struct> (optional)

DeviceConnectVoIP (all members are optional)	Property-type
IpAddress	<string> (optional)
IpPort	<int> (optional)
MulticastIpIn	<string> (optional)
MulticastIpOut	<string> (optional)
MulticastPortIn	<int> (optional)
MulticastPortOut	<int> (optional)
ChannelNumber	<int> (optional) 2 4 6 8 10 12 14 16

DeviceNsa (all members are optional)	Property-type
IpAddressMedia1	<string> (optional)
IpPortMedia1	<int> (optional)
IpAddressMedia2	<string> (optional)
IpPortMedia2	<int> (optional)
PlayMode	<int> (optional) 0 = Synton 1 = Synchron
ReceiveBuffer	<int> (optional)

DeviceNsa (all members are optional)	Property-type
PtpRole	<int> (optional)
WebUIIpAddress	<string> (optional)

AssignedPorts (all members are optional)	Property-type
AddToDevice	<boolean> (required)
PortAddress	{PortAddress} (required, when AddToDevice = true)
ChannelSelection	<int> (optional) Note: Must be used when adding VoIP Connections
InputChannel	<int> (optional) Note: Must be used when adding NSA Connections
OutputChannel	<int> (optional) Note: Must be used when adding NSA Connections

AssignedGPIOs (all members are optional)	Property-type
AddToDevice	<boolean> (required)
GPIOAddress	{GPIOAddress} (required, when AddToDevice = true)
InputChannel	<int> (optional) Note: Must be used when adding NSA Connections
OutputChannel	<int> (optional) Note: Must be used when adding NSA Connections

8.10.4.36.2 Specific parameters for deleting “device”

The delete device functionality requires the longname of the device to be deleted. All assigned ports will get unassigned.

Supported properties	Property-type
LongName	<string> (required) The name of the device to be deleted

8.10.4.36.3 Example

```
<?xml version="1.0"?>
<methodCall>
  <methodName>ConfigurationChangeEx</methodName>
  <params>
    <param>
      <value>C0000000005</value>
    </param>
    <param>
      <value>
        <array>
          <data>
            <value>
              <struct>
                <member>
                  <name>ChangeType</name>
                  <value>edit</value>
                </member>
                <member>
                  <name>ObjectType</name>
                  <value>device</value>
                </member>
                <member>
                  <name>SpecificParams</name>
```

```
<value>
  <struct>
    <member>
      <name>DeviceType</name>
      <value><i4>0</i4></value>
    </member>
    <member>
      <name>LongName</name>
      <value>TestDevice</value>
    </member>
    <member>
      <name>LongName</name>
      <value>TestDevice #1</value>
    </member>
    <member>
      <name>AssignedPorts</name>
      <value>
        <array>
          <data>
            <value>
              <struct>
                <member>
                  <name>AddToDevice</name>
                  <value><boolean>1</boolean></value>
                </member>
                <member>
                  <name>ChannelSelection</name>
                  <value><i4>3</i4></value>
                </member>
                <member>
                  <name>PortAddress</name>
                  <value>
                    <struct>
                      <member>
                        <name>Node</name>
                        <value><i4>4</i4></value>
                      </member>
                      <member>
                        <name>Port</name>
                        <value><i4>18</i4></value>
                      </member>
                    </struct>
                  </value>
                </member>
              </struct>
            </value>
            <value>
              <struct>
                <member>
                  <name>AddToDevice</name>
                  <value><boolean>0</boolean></value>
                </member>
                <member>
                  <name>PortAddress</name>
                  <value>
                    <struct>
                      <member>
                        <name>Node</name>
                        <value><i4>4</i4></value>
                      </member>
                      <member>
                        <name>Port</name>
                        <value><i4>17</i4></value>
                      </member>
                    </struct>
                  </value>
                </member>
              </struct>
            </value>
          </data>
        </array>
      </value>
```

```
                  </member>
                  <member>
                    <name>DeviceConnectVoIP</name>
                    <value>
                      <struct>
                        <value>
                          <struct>
                            <member>
                              <name>ChannelNumber</name>
                              <value><i4>12</i4></value>
                            </member>
                            <member>
                              <name>IpAddress</name>
                              <value><string>192.168.120.50</string></value>
                            </member>
                            <member>
                              <name>IpPort</name>
                              <value><i4>6061</i4></value>
                            </member>
                            <member>
                              <name>MulticastIpIn</name>
                              <value><string>0.100.239.239</string></value>
                            </member>
                            <member>
                              <name>MulticastIpOut</name>
                              <value><string>0.200.240.239</string></value>
                            </member>
                            <member>
                              <name>MulticastPortIn</name>
                              <value><i4>5005</i4></value>
                            </member>
                            <member>
                              <name>MulticastPortOut</name>
                              <value><i4>5005</i4></value>
                            </member>
                          </struct>
                        </value>
                      </member>
                    </struct>
                  </value>
                </member>
              </struct>
            </value>
          </data>
        </array>
      </value>
    </param>
  </params>
</methodCall>
```

8.10.4.37 Object type ‘gpio‘

8.10.4.37.1 Specific parameters for creating / edit “device”

The create and edit gpio functionality requires a GPIOAddress for identification. The IsInput flag within the TGPIOAddress type defines whether to create an input or output. The change properties of the gpio-object are shown in the table below:

Supported properties	Property-type
GpioAddress	<TGPIOAddress>
GpioInput	<struct> (optional)
GpioOutput	<struct> (optional)

GpioInput (all members are optional)	Property-type
LongName	<string> (32 unicode characters max) (optional)
GpioType	<int> (optional) 0 = Normal 1= Inverted
DeviceSelection	<string> (optional) The LongName of the device. Must be a NSA device
ChannelSelection	<int> (optional) Note: The value range is device specific

GpioOutput (all members are optional)	Property-type
LongName	<string> (32 unicode characters max) (optional)
GpioType	<int> (optional) 0 = Normal 1= Inverted
DeviceSelection	<string> (optional) The LongName of the device. Must be a NSA device
ChannelSelection	<int> (optional) Note: The value range is device specific
DisplayText	<string> (8 unicode characters max) (optional)
OffDelay	<int> (optional) 0 .. 10000

8.10.4.37.1 Specific parameters for deleting “gpio”

The delete device functionality requires the address of the GPIO to be deleted.

Supported properties	Property-type
GpioAddress	<TGPIOAddress>

8.11 Key and marker manipulation

Description	Remote Procedure	Parameters	Return Value on success (else XML-RPC-Fault-Response)
Clears a key label that has been set before	ClearKeyLabel	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Node}</int></value></param> <param><value><int>{Port}</int></value></param> <param><value><boolean>{IsInput}</boolean></value></param> <param><value><int>{page}</int></value></param> <param><value><int>{ExpansionPanel}</int></value></param> <param><value><int>{KeyNumber}</int></value></param> <param><value><boolean>{IsVirtKey}</boolean></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>
Clears a key label and marker that has been set before	ClearKeyLabelAndMarker	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Node}</int></value></param> <param><value><int>{Port}</int></value></param> <param><value><boolean>{IsInput}</boolean></value></param> <param><value><int>{page}</int></value></param> <param><value><int>{ExpansionPanel}</int></value></param> <param><value><int>{KeyNumber}</int></value></param> <param><value><boolean>{IsVirtKey}</boolean></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>
Clears a marker that has been set before	ClearKeyMarker	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Node}</int></value></param> <param><value><int>{Port}</int></value></param> <param><value><boolean>{IsInput}</boolean></value></param> <param><value><int>{page}</int></value></param> <param><value><int>{ExpansionPanel}</int></value></param> <param><value><int>{KeyNumber}</int></value></param> <param><value><boolean>{IsVirtKey}</boolean></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>
Queries all key manipulations that have been sent via RRCS with one the functions in this section	GetAllRemoteKeys	<param><value><string>{TransKey}</string></value></param>	<param><value><struct> <member> <name>RemoteKey</name> <value><array><data> <value><struct> <member> <name>EventId</name> <value><i4>{ EventId }</i4></value> </member> <member> <name>KeyNumer</name> <value><i4>{KeyNumer}</i4></value> </member> <member> <name>Label</name> <value><string></string></value>

Description	Remote Procedure	Parameters	Return Value on success (else XML-RPC-Fault-Response)
			</member> <member> <name>LabelFlag</name> <value> <boolean>{ LabelFlag }</boolean> </value> </member> <member> <name>LockKeyFlag</name> <value> <boolean>{ LockFlag }</boolean> </value> </member> <member> <name>Marker</name> <value><i4>{ Marker }</i4></value> </member> <member> <name>MarkerFlag</name> <value> <boolean>{Marker}</boolean> </value> </member> <member> <name>Node</name> <value><i4>{ Node }</i4></value> </member> <member> <name>PanelId</name> <value><i4>{ PanelId }</i4></value> </member> <member> <name>PressKeyFlag</name> <value> <boolean>{ PressFlag }</boolean> </value> </member> <member> <name>SubPanel</name> <value> <i4>{SubPanel}</i4> </value> </member> </struct></value>

Description	Remote Procedure	Parameters	Return Value on success (else XML-RPC-Fault-Response)
			… </data></array></value> </member> <member> <name>Transkey</name> <value> <string>{Transkey}</string> </value> </member> </struct></value></param>
Queries key manipulations for the given key	GetRemoteKey	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Node}</int></value></param> <param><value><int>{Port}</int></value></param> <param><value><boolean>{IsInput}</boolean></value></param> <param><value><int>{page}</int></value></param> <param><value><int>{ExpansionPanel}</int></value></param> <param><value><int>{KeyNumber}</int></value></param> <param><value><boolean>{IsVirtKey}</boolean></value></param>	Same as GetAllRemoteKeys
Locks a key, so the user cannot press it.	LockKey	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Node}</int></value></param> <param><value><int>{Port}</int></value></param> <param><value><boolean>{IsInput}</boolean></value></param> <param><value><int>{page}</int></value></param> <param><value><int>{ExpansionPanel}</int></value></param> <param><value><int>{KeyNumber}</int></value></param> <param><value><boolean>{IsVirtKey}</boolean></value></param> <param><value><boolean>{Lock}</boolean></value></param> <param><value><int>{PoolPort}</int></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>
Presses a key	PressKey	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Node}</int></value></param> <param><value><int>{Port}</int></value></param> <param><value><boolean>{IsInput}</boolean></value></param> <param><value><int>{page}</int></value></param> <param><value><int>{ExpansionPanel}</int></value></param> <param><value><int>{KeyNumber}</int></value></param> <param><value><boolean>{IsVirtKey}</boolean></value></param> <param><value><boolean>{Press}</boolean></value></param> <param><value><int>{PoolPort}</int></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>

Description	Remote Procedure	Parameters	Return Value on success (else XML-RPC-Fault-Response)
Presses a key Please note: The parameter Trigger can have the following values: 1 = Primary Trigger 2 = Secondary Trigger On 1200 series panels Primary Trigger is the Lever-Down, Secondary Lever-Up, whilst on a 1000/1100 series Panel it is Left- or rather Right-Key. On all other panel types only the Primary Trigger is supported.	PressKeyEx	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Node}</int></value></param> <param><value><int>{Port}</int></value></param> <param><value><boolean>{IsInput}</boolean></value></param> <param><value><int>{page}</int></value></param> <param><value><int>{ExpansionPanel}</int></value></param> <param><value><int>{KeyNumber}</int></value></param> <param><value><boolean>{IsVirtKey}</boolean></value></param> <param><value><boolean>{Press}</boolean></value></param> <param><value><int>{Trigger}</int></value></param> <param><value><int>{PoolPort}</int></value></param>	
Sets a key label. It overwrites any other key label of the configuration. {Label} is a 8 character string.	SetKeyLabel	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Node}</int></value></param> <param><value><int>{Port}</int></value></param> <param><value><boolean>{IsInput}</boolean></value></param> <param><value><int>{page}</int></value></param> <param><value><int>{ExpansionPanel}</int></value></param> <param><value><int>{KeyNumber}</int></value></param> <param><value><boolean>{IsVirtKey}</boolean></value></param> <param><value><string>{Label}</string></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>
Sets a key label and a marker . {Label} is a 8 character string. {Marker} is the index of the marker in the configuration. See the net properties dialog in Director.	SetKeyLabelAndMarker	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Node}</int></value></param> <param><value><int>{Port}</int></value></param> <param><value><boolean>{IsInput}</boolean></value></param> <param><value><int>{page}</int></value></param> <param><value><int>{ExpansionPanel}</int></value></param> <param><value><int>{KeyNumber}</int></value></param> <param><value><boolean>{IsVirtKey}</boolean></value></param> <param><value><string>{Label}</string></value></param> <param><value><int>{Marker}</int></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>
Sets a marker above the key. {Marker} is the index of the marker in the configuration. See the net properties dialog in Director.	SetKeyMarker	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Node}</int></value></param> <param><value><int>{Port}</int></value></param> <param><value><boolean>{IsInput}</boolean></value></param> <param><value><int>{page}</int></value></param> <param><value><int>{ExpansionPanel}</int></value></param> <param><value><int>{KeyNumber}</int></value></param> <param><value><boolean>{IsVirtKey}</boolean></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>

Description	Remote Procedure	Parameters	Return Value on success (else XML-RPC-Fault-Response)
		<param><value><int>{Marker}</int></value></param>	

8.12 Panel-Spy

Description	Remote Procedure	Parameters	Return Value on success
Starts/Stops panel spy notifications. Hint: The method fails, if RegisterForAllEvents has not been called.	ChangePanelSpyRegistry	<param><value><string>{TransKey}</string></value></param> <param><value><int>{TCPPort}</int></value></param> <param><value><string>{URLPath}</string></value></param> <param><value><int>{Node}</int></value></param> <param><value><int>{Port}</int></value></param> <param><value><struct>{EventInfo}</struct></value></param>	<param><value><string>{TransKey}</string></value>

Input parameters

Name	Type	Parameter description
TransKey	string	• Transaction Key
TCPPort	int	• TCP Port
URLPath	string	Defines the URL-path of the receiving computer. E.g. if path is 'notification' RRCS sends the events to the URL: "http://[ip-address of calling computer]:[TCPPort]/notification". String conditions: • All characters must be ASCII characters excluding the control characters ([0..31,127]) and the space character. • The leading character can be a slash however it needs not default: <value>/RPC2</value>
Node	int	• Node address
Port	int	• Port address
EventInfo	struct	The struct can contain the following members: • RotateEventsOn (type bool) • KeyEventsOn (type bool) • FuncKeyEventsOn (type bool) • NumKeyEventsOn (type bool) – if not present, the notification feature is nor turned on neither turned off. – if present, the notification feature is turned on (struct member = true) or turned off (struct member = false).

8.13 Port Cloning

Description	Remote Procedure	Parameters	Return Value on success (else XML-RPC-Fault-Response)
Clones the output signal of a port and routes all audios to a monitoring port.	StartPortCloning	<param><value><string>{TransKey}</string></value></param> <param><value>{Port to monitor}</value></param> <param><value>{Clone Port}</value></param>	<param> <value> <string>{TransKey}</string> </value> </param>
Stops cloneing. Use Port 0.0 as a wildcard in order to stop a batch of port clones.	StopPortCloning	<param><value><string>{TransKey}</string></value></param> <param><value>{Port to monitor}</value></param> <param><value>{Clone Port}</value></param>	<param> <value> <string>{TransKey}</string> </value> </param>
Returns an array of all active port clones.	GetAllActivePortClones	<param><value><string>{TransKey}</string></value></param>	<param> <value><array><data> <value><string>{TransKey}</string></value> <value><array><data> <!–1st port clone information --> <value><struct> <member> <name>PortToMonitor</name> <value> <struct>{Port to monitor}</struct> </value> </member> <member> <name>ClonePort</name> <value> <struct>{Clone Port}<struct> </value> </member> … </struct></value> <!–further port clones --> … </data></array></value> </data></array></value> </param> </params>

{Port to monitor} and {Clone Port} are using the existing helper type <TPortAddress>.

8.14 IFB Volume

Description	Remote Procedure	Parameters	Return Value on success
Sets the mix minus volume for an IFB. It affects the single volume. If the mix minus and the output is assigned when SetIFBVolumeMixMinus is called, the volume for the crosspoint MixMinus->Output is set. If SetIFBVolumeMixMinus is called and a mix minus is not assigned, it will return an error. Presetting the volume is therefor not possible. If SetIFBVolumeMixMinus is called and an output is not assigned, the value is stored. Later when the output is assigned the crosspoint volume is set to the IFB volume. If an output is reassigned, it does not affect the IFB volume. When the mix minus is assigned the initial volume is always unity gain. If a mix minus is a group, the volumes for the group members can be set seperately and do not interfere with each other.	SetIFBVolumeMixMinus	<param><value> <string>{TransKey}</string> </value></param> <param><value> <int>{Node}</int> </value></param> <param><value> <int>{Port}</int> </value></param> <param><value> <boolean>{IsInput}</boolean> </value></param> <param><value> <int>{IfbNumber}</int> </value></param> <param><value> <int>{Volume}</int> </value></param>	<param> <value> <string>{TransKey}</string> </value> </param>
Returns the the mix minus volume for an IFB If the volume was never set by SetIFBVolumeMixMinus, the return value is unity gain.	GetIFBVolumeMixMinus	<param><value> <string>{TransKey}</string> </value></param> <param><value> <int>{Node}</int> </value></param> <param><value> <int>{Port}</int> </value></param> <param><value> <boolean>{IsInput}</boolean> </value></param> <param><value> <int>{IfbNumber}</int> </value></param>	<param><value> <array><data> <value><string>{TransKey}</string></value> <value><array><data> <value><struct> <member> <name>IfbNumber</name> <value><int>{IfbNumber}</int></value> </member> <member> <name>Node</name> <value><int>{Node}</int></value> </member> <member> <name>Port</name> <value><int>{Port}</int></value> </member> <member> <name>Volume</name> <value><int>{Volume}</int></value> </member> </struct></value> </data></array></value> </data></array> </value></param>

Description	Remote Procedure	Parameters	Return Value on success

Removes the mix minus volume for an IFB. The volume returns to default (untiy gain)	RemoveIFBVolumeMixMinus	`<param><value>`
		`<string>{TransKey}</string>`
		`</value></param>`
		`<param><value>`
		`<int>{Node}</int>`
		`</value></param>`
		`<param><value>`
		`<int>{Port}</int>`
		`</value></param>`
		`<param><value>`
		`<boolean>{IsInput}</boolean>`
		`</value></param>`
		`<param><value>`
		`<int>{IfbNumber}</int>`
		`</value></param>`
		`<param><value>`
		`<int>{Volume}</int>`
		`</value></param>`
			`<param>`
			`<value>`
			`<string>{TransKey}</string>`
			`</value>`
			`</param>`

8.15 Registration for notifications

Description	Remote Procedure	Parameters	Return Value on success
Registers volume change notifications for the given crosspoints. Hint: - RRCS will send initial volume values for all crosspoints. - The method fails, if RegisterForEventsEx has not been called with the member “XpVolumeChange” set to true - The method fails, if the total number of registered crosspoints would exceed the limit of 8192 crosspoints. - If the method fails, the registry keeps unchanged Note: This API does not work for XPs that´s destination port is connected to an Artist-1024	XpVolumeChangeRegistryReset	<param> <value><string>{TransKey}</string></value> </param> <param> <value><string>{IP-address}</string></value> </param> <param> <value><i4>{TCP-port}</i4></value> </param> <param> <value><array><data> <!–1st crosspoint --> <value><struct> <member> <name>Destination</name> <value><struct> <member> <name>IsInput</name> <value><boolean>0</boolean></value> </member> <member> <name>Node</name> <value><i4>{Node}</i4></value> </member> <member> <name>Port</name> <value><i4>{Port}</i4></value> </member> </struct></value> </member> <member> <name>Source</name> <value><struct> <member> <name>IsInput</name> <value><boolean>1</boolean></value> </member> <member> <name>Node</name> <value><i4>{Node}</i4></value> </member> <member> <name>Port</name>	<param><value><string>{TransKey}</string></value>

Description	Remote Procedure	Parameters	Return Value on success
		<value><i4>{Port}</i4></value> </member> </struct></value> </member> </struct></value> <!-- further crosspoints --> … </param>	
Adds crosspoints to the volume change notification registry. Hint: - RRCS will send initial volume values for all newly registered crosspoints. - The method fails, if RegisterForEventsEx has not been called with the member "XpVolumeChange" set to true - The method fails, if the total number of registered crosspoints would exceed the limit of 8192 crosspoints. - If the method fails, the registry keeps unchanged Note: This API does not work for XPs that´s destination port is connected to an Artist-1024	XpVolumeChangeRegistryAdd	See method "XpVolumeChangeRegistryReset"	<param><value><string>{TransKey}</string></value>

Description	Remote Procedure	Parameters	Return Value on success
Removes a set of crosspoints from the volume change registry - The method fails, if RegisterForEventsEx has not been called with the member “XpVolumeChange” set to true. In this case nothing will be removed from the registry. Note: This API does not work for XPs that´s destination port is connected to an Artist-1024	XpVolumeChangeRegistryRemove	See method “XpVolumeChangeRegistryReset”	<param><value><string>{TransKey}</string></value>
Register Control System client for events	RegisterForEvents	<param><value><string>{TransKey}</string></value></param> <param><value><string>{IP-address}</string></value></param> <param><value><int>{TCP-Port}</int></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>
Unregister Control System client for events	UnregisterForEvents	<param><value><string>{TransKey}</string></value></param> <param><value><string>{IP-address}</string></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>
Registration for GP input change notifications	RegisterForGpInputChange	<param><value><string>{TransKey}</string></value></param> <param><value><string>{IP-address}</string></value></param> <param><value><int>{TCP-Port}</int></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>
Unregistration for GP input change notifications	UnregisterForGpInputChange	<param><value><string>{TransKey}</string></value></param> <param><value><string>{IP-address}</string></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>
Registration for GP output change notifications	RegisterForGpOutputChange	<param><value><string>{TransKey}</string></value></param> <param><value><string>{IP-address}</string></value></param> <param><value><int>{TCP-Port}</int></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>
Unregistration for GP output change notifications	UnregisterForGpOutputChange	<param><value><string>{TransKey}</string></value></param> <param><value><string>{IP-address}</string></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>

Description	Remote Procedure	Parameters	Return Value on success
Registration for change notifications Hint: The struct-members: General GpInChange GpOutChange XpChange LogicSourceChange XpVolumeChange ConfigurationChange are optional. If they are not given, no notifications will be sent.	RegisterForEventsEx	<param><value><string>{TransKey}</string></value></param> <param><value><string>{IP-address}</string></value></param> <param><value><int>{TCP-Port}</int></value></param> <param><value><struct> <member> <name>General</name> <value><boolean>{On/Off}</boolean></value> <member> <name>GpInChange</name> <value><boolean>{On/Off}</boolean></value> </member> <member> <name> GpOutChange</name> <value><boolean>{On/Off}</boolean></value> </member> </member> <member> <name>XpChange</name> <value><boolean>{On/Off}</boolean></value> </member> </member> <member> <!–requires RRCS v5.91 or later --> <name>LogicSourceChange</name> <value><boolean>{On/Off}</boolean></value> </member> <member> <!–requires RRCS v5.91 or later --> <!-- see methods “XpVolumeChangeRegistryReset”, “XpVolumeChangeRegistryAdd” and “XpVolumeChangeRegistryRemove” --> <name>XpVolumeChange</name> <value><boolean>{On/Off}</boolean></value> </member> </struct></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>
Unregistration for change notifications	UnregisterForEventsEx	<param><value><string>{TransKey}</string></value></param> <param><value><string>{IP-address}</string></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>

8.15.1 RegisterForAllEvents

Input parameters

Name	Type	Parameter description
TransKey	string	• Transaction Key default: <value>C0123456789</value>
TCPPort	int	• TCP-Port default: <value><i4>80</i4></value>
URLPath	string	Defines the URL-path of the receiving computer. E.g. if path is 'notification' RRCS sends the events to the URL: "http://[ip-address of calling computer]:[TCPPort]/notification". String conditions: • All characters must be ASCII characters excluding the control characters ([0..31,127]) and the space character. • The leading character can be a slash however it needs not default: <value>/RPC2</value>
AllowHttpPipelining	boolean	• 0 (==false): Wait for acknowlegde between outgoing notifications (slow, extra network round trips) • 1 (==true): RRCS sends multiple HTTP-packets without waiting for each response default: <value><boolean>0</boolean></value>
AllowSystemMulticall	boolean	NOTE: RRCS does not support the 'system.multicall' yet. However it may support it in future releases. • 0 (==false): Forbids RRCS to boxcar multiple RPC notifications in one request. • 1 (==true): Allows RRCS to boxcar multiple RPC notifications in one request. default: <value><boolean>0</boolean></value>

Description	Remote Procedure	Parameters	Return Value on success
Allows to receive any RRCS event. RRCS will send events to calling the computer with the given TCP-port.	RegisterForAllEvents	<param><value><string>{TransKey}</string></value></param> <param><value><int>{TCPPort}</int></value></param> <param><value><string>{URLPath}</string></value></param> <param><value><boolean>{AllowHttpPipelining}</boolean></value></param> <param><value><boolean>{AllowSystemMulticall}</boolean></value></param>	<param><value><string>{TransKey}</string></value>

8.15.2 UnregisterForAllEvents

Input parameters

Name	Type	Parameter description
TransKey	string	• Transaction Key —— default: <value>C0123456789</value>
TCPPort	int	• TCP-Port —— default: <value><i4>80</i4></value>
URLPath	string	Defines the URL-path of the receiving computer. E.g. if path is 'notification' RRCS sends the events to the URL: "http://[ip-address of calling computer]:[TCPPort]/notification". String conditions: • All characters must be ASCII characters excluding the control characters ([0..31,127]) and the space character. • The leading character can be a slash however it needs not —— default: <value>/RPC2</value>

Description	Remote Procedure	Parameters	Return Value on success

Unregisters a single event receiver for all change notifications	UnregisterForAllEvents	`<param><value><string>{TransKey}</string></value></param>`
		`<param><value><int>{TCPPort}</int></value></param>`
		`<param><value><string>{URLPath}</string></value></param>`
			`<param><value><string>{TransKey}</string></value>`

8.16 Trunking

Description	Remote Procedure	Parameters	Return Value on success
Queries all trunking ports . The function can be used to retrieve information about which port s are used as trunkline and which ports are enabled for trunking. Hint: (requires version 6.90 or later)	GetTrunkPorts	<param><value><string>{TransKey}</string></value></param>	<param> <value><array><data> <value><string>{TransKey}</string></value> <value><array><data><value><struct> <member> <name>EnabledForTrunking</name> <value><boolean>…</boolean></value> </member> <member> <name>NetName</name> <value><string>...</string></value> </member> <member> <name>NetTrAddr</name> <value><i4>…</i4></value> </member> <member> <name>PortDispName8</name> <value><string>…</string></value> </member> <member> <name>PortTrAddr</name> <value><i4>…</i4></value> </member> <member> <name>UsedAsTrunkLine</name> <value><boolean>…</boolean></value> </member> </struct></value></data></array></value> </data></array></value></param>
Queries setup information about Trunklines exist between the Riedel rings. Hint: (requires version 6.90 or later)	GetTrunklineSetup	<param><value><string>{TransKey}</string></value></param>	<param> <value><array><data> <value><string>{TransKey}</string></value> <value><i4>{ErrorCode}</i4></value> <value><array><data><value><struct> <member> <name>DestNetName</name> <value><string>…</string></value> </member> <member> <name>DestNetTrAddr</name> <value><i4>…</i4></value> </member> <member>

Description	Remote Procedure	Parameters	Return Value on success
			<name>DestPortName</name> <value><string>…</string></value> </member> <member> <name>DestTrAddr</name> <value><i4>…</i4></value> </member> <member> <name>SrcNetName</name> <value><string>,,,</string></value> </member> <member> <name>SrcNetTrAddr</name> <value><i4>…</i4></value> </member> <member> <name>SrcPortName</name> <value><string>…</string></value> </member> <member> <name>SrcPortTrAddr</name> <value><i4>…</i4></value> </member> </struct></value></data></array></value> </data></array></value></param>
Queries activity information of trunklines. The function can be used to retrieve information about trunkline activity including the command type a trunkline is currently used for. Hint: (requires version 6.90 or later)	GetTrunklineActivities	<param><value><string>{TransKey}</string></value></param>	<param> <value><array><data> <value><string>{TransKey}</string></value> <value><i4>{ErrorCode}</i4></value> <value><array><data> <value><struct> <member> <name>SrcTrAddr</name> <value> <struct> <member> <name>NetTrAddr</name> <value><i4>…</i4></value> </member> <member> <name>SrcObjAddr</name> <value><i4>…</i4></value> </member> </struct> </value> </member>

Description	Remote Procedure	Parameters	Return Value on success
			<member> <name>DestTrAddr</name> <value> <struct> <member> <name>NetTrAddr</name> <value><i4>…</i4></value> </member> <member> <name>DestObjAddr</name> <value><i4></i4></value> </member> </struct> </value> </member> <member> <name>Priority</name> <value><i4>1</i4></value> </member> <member> <name>TrunkCmdType</name> <value>{TrunkCmdType}</value> </member> <member> <name>TrunklinePath</name> <value> <array> <data> <value> <struct> <member> <name>sNetTrAddr</name> <value><i4>…</i4></value> </member> <member> <name>sPortTrAddr</name> <value><i4>…</i4></value> </member> <member> <name>dNetTrAddr</name> <value><i4>…</i4></value> </member> <member> <name>dPortTrAddr</name> <value><i4>…</i4></value>

Description	Remote Procedure	Parameters	Return Value on success
			</member> </struct> </value> </data> </array> </value></member></struct> </value></data></array></value> </param>
Queries all trunking ifbs . The function can be used to retrieve information about trunking ifbs. Hint: (requires version 6.90.RR5 or later)	GetTrunkIfbs	<param><value><string>{TransKey}</string></value></param>	<param> <value><array><data> <value><string>{TransKey}</string></value> <value><array><data><value><struct> <member> <name>NetTrAddr</name> <value><int>…</int></value> </member> <member> <name>IfbNumber</name> <value><int>...</int></value> </member> <member> <name>EnabledForTrunking</name> <value><boolean>…</boolean></value> </member> <member> <name>DispName8</name> <value><string>…</string></value> </member> <member> <name>LocalNetName</name> <value><string>…</string></value> </member> <member> <name>LocalNetTrAddr</name> <value><int>…</int></value> </member> <member> <name>ObjId</name> <value><int>…</int></value> </member> </struct></value></data></array></value> </data></array></value></param>
Updates the labels of trunked ports. Hint: (requires version 7.40.RR2 or later)	UpdateTrunkingLabels	<param><value><string>{TransKey}</string></value></param>	<param><value><string>{TransKey}</string></value></param>

Description	Remote Procedure	Parameters	Return Value on success
Gets the Trunking Net address of the ring	GetTrunkingNetAddr	<param><value><string>{TransKey}</string></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> <value><int>{TrunkingNetAddr}</int></value> </data></array></value></param>
Sets the Trunking Net address of the ring	SetTrunkingNetAddr	<param><value><string>{TransKey}</string></value></param> <param><value><int>{TrunkingNetAddr}</int></value></param>	<param><value><string>{TransKey}</string></value></param> <param><value><int>{ErrorCode}</int></value></param>
Gets the current configured local trunk controller.	GetLTC	<param><value><string>{TransKey}</string></value></param>	<param><value><string>{TransKey}</string></value></param> <param><value><int>{ErrorCode}</int></value></param> <param><value><int>{Node}</int></value></param>
Sets the local trunk controller Hint: This automatically moves the LTC role if the current LTC of a ring is online.	SetLTC	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Node}</int></value></param>	<param><value><string>{TransKey}</string></value></param> <param><value><int>{ErrorCode}</int></value></param>

8.17 Dial / HangUp of Call

Works only if one PoolPort is configured. Configuring a number on a PoolPort can only be done via Director and is a prerequisite for these functions.

Description	Remote Procedure	Parameters	Return Value on success
Instructs the system to dial the specified number. typically the returned LineState will be 2 (Waiting for Connection). Hint: (requires version 7.20 or later)	DialNumber	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Node}</int></value></param> <param><value><int>{Port}</int></value></param> <param><value><string>Number</string></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> <value><int>{LineStatus}</int></value> </data></array></value></param>
Instructs the system to draw the call. Hint: (requires version 7.20 or later)	HangUpCall	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Node}</int></value></param> <param><value><int>{Port}</int></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>
Queries information about the status of line Hint: (requires version 7.20 or later)	LineStatus	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Node}</int></value></param> <param><value><int>{Port}</int></value></param>	<param><value><struct> <member><name>ErrorCode</name> <value><int>{ ErrorCode }</int></value> </member> <member><name>Status</name> <value><int>{LineStatus}</int></value> </member> </struct></value></param>

8.18 Licensing

Works for all types of nodes, but only for configured Artist1024 detailed information will be returned.

Description	Remote Procedure	Parameters	Return Value on success

Returns the license information.
 	
Hint: (requires version 8.0.11 or later)
	GetLicenseInfo	`<param><value><string>{TransKey}</string></value></param>`
		`<param><value><int>{Node}</int></value></param>`
			`<param><value><array><data>`
			`<value><string>{TransKey}</string></value>`
			`<value><struct>`
			`<member>{PortsAllocated}</member>`
			`<member>{PortsAvailable_FVAE}</member>`
			`<member>{PortsConfigured}</member>`
			`<member>{PortsLicensed}</member>`
			`<member>{PortsLicensed_VAM}</member>`
			`</struct></value>`
			`</data></array></value></param>`

8.19 PoolPort

Works for all types of nodes, but only for configured Artist1024 detailed information will be returned.

Description	Remote Procedure	Parameters	Return Value on success

Returns PoolPort information regarding a specific port.
 	
Hint: (requires version 8.0.21 or later)
	GetPoolPortInfo	`<param><value><string>{TransKey}</string></value></param>`
		`<param><value><int>{Node}</int></value></param>`
		`<param><value><int>{Port}</int></value></param>`
			`<param><value><array><data>`
			`<value><string>{TransKey}</string></value>`
			`<value><boolean>{IsPoolPortSupported}</boolean></value>`
			`<value><int>{PoolPortAmount}</int></value>`
			`</data></array></value></param>`

8.20 System Time

Works for all types of nodes.

Description	Remote Procedure	Parameters	Return Value on success

Returns the amount of currently online nodes and the amount of updated nodes
 	
Hint: (requires version 8.1 or later)
	SetSystemTimeOnAllNodes	`<param><value><string>{TransKey}</string></value></param>`	`<param><value><array><data>`
			`<value><string>{TransKey}</string></value>`
			`<value><int>{AmountOfOnlineNodes}</int></value>`
			`<value><int>{UpdatedNodes}</int></value>`
			`</data></array></value></param>`

8.21 Connection Management

Works for all types of nodes.

Description	Remote Procedure	Parameters	Return Value on success
Disconnects from current connected Artist System, if necessary, and connects to the specified IPAddress. Hint: (requires version 8.4 or later)	ConnectToArtist	<param><value><string>{TransKey}</string></value></param> <param><value><string>{IpAddress}</string></value></param> <param><value><boolean>{IsRedundant}</boolean></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>
Disconnects from current connected Artist System. Hint: (requires version 8.4 or later)	DisconnectFromArtist	<param><value><string>{TransKey}</string></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>

8.22 Reset All Nodes

Works for all types of nodes.

Description	Remote Procedure	Parameters	Return Value on success
Resets all nodes that are online within the ring. Hint: (requires version 8.5 or later)	ResetAllNodes	<param><value><string>{TransKey}</string></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>

8.23 DeletePortCommands

Description	Remote Procedure	Parameters	Return Value on success

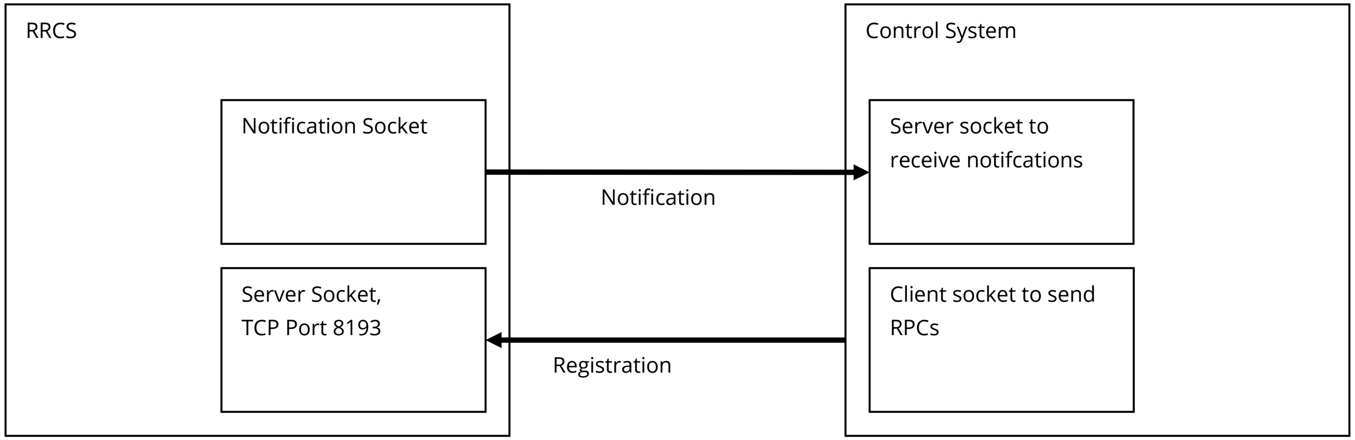
Deletes all commands of the specified command position.
 	
Note: When the member ‘PositionType’ in TCmdPosition is not defined, all commands of all command positions will be deleted, otherwise only the commands that are defined at the specified command position.
 	
Hint: (requires version 8.5 or later)
	DeletePortCommands	`<param><value><string>{TransKey}</string></value></param>`
		`<param><{TCmdPosition}><param>`
			`<param><value><array><data>`
			`<value><string>{TransKey}</string></value>`
			`<value><int>{ErrorCode}</int></value>`
			`</data></array></value></param>`

8.24 Stage Commands

Description	Remote Procedure	Parameters	Return Value on success
Gets the current configured Stage Net Address Hint: (requires version 8.7 or later)	GetStageNetAddr	<param><value><string>{TransKey}</string></value></param>	<params> <param> <value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> <value><i4>{StageNetAddr}</i4></value> </data></array></value> </param> </params>
Sets the Stage Net Address Hint: (requires version 8.7 or later)	SetStageNetAddr	<param><value><string>{TransKey}</string></value></param> <param><value><i4>{StageNetAddr}</i4></value></param>	<params> <param> <value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value> </param> </params>
Gets the current configured Stage Registry URL Hint: (requires version 8.7 or later)	GetStageRegistryUrl	<param><value><string>{TransKey}</string></value></param>	<params> <param> <value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> <value><string>{StageRegistryUrl}</string></value> </data></array></value> </param> </params>
Sets the Stage Registry URL Hint: (requires version 8.7 or later)	SetStageRegistryUrl	<param><value><string>{TransKey}</string></value></param> <param><value><string>{StageRegistryUrl}</string></value></param>	<params> <param> <value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value> </param> </params>

9 Notifications

The notifictions described in this chapter are used by RRCS in order to signal status changes of the Artist system. External control systems need to register for notifications beforehand, please see chapter 8.15. After registration RRCS will send XML-RPCs to the given IP- and TCP address, so RRCS acts as a client and the external control system as a server.



9.1 Send String

Description	Remote Procedure	Parameters	Return Value on success
Dump/send string notification. It is sent when the Send String command is executed in the system.	SendString	<param><value><string>{TransKey}</string></value></param> <param><value><string>{String}</string></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>
Dump/send string off notification. It is sent when the Send String command is executed in the system.	SendStringOff	<param><value><string>{TransKey}</string></value></param> <param><value><string>{String}</string></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>

9.2 GPIOs

Description	Remote Procedure	Parameters	Return Value on success
Notification for a GP input change.	GpInputChange	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Net}</int></value></param> <param><value><int>{Node}</int></value></param> <param><value><int>{Port}</int></value></param> <param><value><int>{Slot}</int></value></param> <param><value><int>{GPIO no.}</int></value></param> <param><value><boolean>{state}</boolean></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>
Notification for a GP output change.	GpOutputChange	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Net}</int></value></param> <param><value><int>{Node}</int></value></param> <param><value><int>{Port}</int></value></param> <param><value><int>{Slot}</int></value></param> <param><value><int>{GPIO no.}</int></value></param> <param><value><boolean>{state}</boolean></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>

9.3 Logic Source

Description	Remote Procedure	Parameters	Return Value on success
Notification for a Logic Source state change.	LogicSourceChange	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Object ID}</int></value></param> <param><value><boolean>{state}</boolean></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>

9.4 Volume change

<table>
<tr><th>Description</th><th>Remote Procedure</th><th>Parameters</th><th>Return Value on success</th></tr>
<tr><td>Notification for crosspoint volume change</td><td>XpVolumeChange</td><td>

```
<param><value><string>{TransKey}</string></value></param>
<param><value><array><data>
            <!- 1st volume change information -->
            <value><struct>
                  <member>
                               <name>Source</name>
                               <value>{TPortAddress}</value>
                  </member>
                  <member>
                               <name>Destination</name>
                               <value>{TPortAddress}</value>
                  </member>
                  <!-- this member only exists, if the single volume has changed. If this
member does not exist, the member ConferenceVolume must exist. -->
                  <member>
                               <name>SingleVolume</name>
                                <!–
                               -1: volume unavailable
                                0: volume mute
                                1.255: volume [dB] = (x - 230.0) / 2.0
                               -->
                               <value>{x}</value>
                  </member>
                  <!-- this member only exists, if the conference volume has changed. If
this member does not exist, the member SingleVolume must exist. -->
                  <member>
                               <name>ConferenceVolume</name>
                                <!–
                               -1: volume unavailable
                                0: volume mute
                                1.255: volume [dB] = (x - 230.0) / 2.0
                               -->
                               <value>{x}</value>
                  </member>
           </struct></value>
            <!-- further volume change information -->
            …

    </data></array></value>
</param>
```

</td><td>

```
No return value (ignored by RRCS)
```

</td></tr>
</table>

9.5 Configuration change

Description	Remote Procedure	Parameters	Return Value on success
Notification for a configuration change.	ConfigurationChange	<param><value><string>{TransKey}</string></value></param>	No return value (ignored by RRCS)

9.6 Crosspoint change

Description	Remote Procedure	Parameters	Return Value on success
Notification for a crosspoint change.	CrosspointChange	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Number of XPs}</int></value></param> <param><value><struct> <member><name>XP#1</name> <value><array><data> <value><int>{Source Net}</int></value> <value><int>{Source Node}</int></value> <value><int>{Source Port}</int></value> <value><int>{Dest Net}</int></value> <value><int>{Dest Node}</int></value> <value><int>{Dest Port}</int></value> <value><boolean>{State}</boolean></value> </data></array></value> </member> <member><name>XP#2</name> . . . </struct></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>

9.7 Alarms

Description	Remote Procedure	Parameters	Return Value on success
Upstream failed on Node	UpstreamFailed	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Net}</int></value></param> <param><value><int>{Node}</int></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param
Upstream failed cleared	UpstreamFailedCleared	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Net}</int></value></param> <param><value><int>{Node}</int></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>
Downstream failed on Node	DownstreamFailed	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Net}</int></value></param> <param><value><int>{Node}</int></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>
Downstream failed on Node	DownstreamFailedCleared	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Net}</int></value></param> <param><value><int>{Node}</int></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>
Node Controller failed on Node	NodeControllerFailed	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Net}</int></value></param> <param><value><int>{Node}</int></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>
Node Controller reboot on Node	NodeControllerReboot	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Net}</int></value></param> <param><value><int>{Node}</int></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>
Client failed on Node	ClientFailed	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Net}</int></value></param> <param><value><int>{Node}</int></value></param> <param><value><int>{Slot-of-client-card}</int></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>
Client failed on Node cleared	ClientFailedCleared	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Net}</int></value></param> <param><value><int>{Node}</int></value></param> <param><value><int>{Slot-of-client-card}</int></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>
Port went inactive	PortInactive	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Net}</int></value></param> <param><value><int>{Node}</int></value></param> <param><value><int>{Port}</int></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>
Port went active	PortActive	<param><value><string>{TransKey}</string></value></param> <param><value><int>{Net}</int></value></param> <param><value><int>{Node}</int></value></param> <param><value><int>{Port}</int></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>

Description	Remote Procedure	Parameters	Return Value on success
Connect to Artist net succesful after failure	ConnectArtistRestored	<param><value><string>{TransKey}</string></value></param> <param><value><string>{GatewayState}</string></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param
Connect to Artist net failure	ConnectArtistFailure	<param><value><string>{TransKey}</string></value></param> <param><value><string>{GatewayState}</string></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>
Gateway shutdown due to application closure or PC shutdown.	GatewayShutdown	<param><value><string>{TransKey}</string></value></param>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>
SIC failed on Artist-1024	SicFailed	<params> <param><value><string>{Transkey}</string></value></param> <param><value><struct> <member> <name>Bay</name> <value><i4>{Bay}</i4></value></member> <member> <name>Description</name> <value><string>{Description}</string></value></member> <member> <name>Net</name> <value><i4>{Net}</i4></value></member> <member> <name>Node</name> <value><i4>{Node}</i4></value></member> <member> <name>Path</name> <value><string>{Path}</string></value></member> <member> <name>Severity</name> <value><i4>{Severity}</i4></value></member> <member> <name>Status</name> <value><boolean>{Status}</boolean></value></member> <member> <name>Type</name> <value><string>SIC failed</string></value></member> </struct></value></param> </params>	<param><value><array><data> <value><string>{TransKey}</string></value> <value><int>{ErrorCode}</int></value> </data></array></value></param>

9.8 GetAlive

A KeepAlive mechanism can be enabled within the ‘Route Options’ in RRCS UI that triggers 'GetAlive' notifications on every notification channel being sent by RRCS every 3 seconds of message inactivity.

Please note that such 'GetAlive' messages must be answered by the Control System, otherwise RRCS disconnects the notification channel and removes the corresponding registration for events. This functionality can especially be used to overcome firewalls that disconnects connections after a certain amount of time of detected inactivity.

In case of Panel Spy, the GetAlive message frequency is 1 second.

Description	Remote Procedure	Parameters	Return Value
Ping	GetAlive	None	ignored by RRCS

9.9 Panel Spy

9.9.1 Overview

An external control system needs to perform the following two registration steps to be able to receive panel spy events from Artist-panels through RRCS:

The control system calls ‘RegisterForAllEvents’

The control system calls ‘ChangePanelSpyRegistry’ for at least one panel

Once these steps have been completed successfully, RRCS informs the external control system about the notification status for each panel (see ‘PanelSpyStateChange’) and calls the notification handlers if necessary (see ‘PanelSpyRotateEvent’, ‘PanelSpyKeyEvent’, ‘PanelSpyFuncKeyEvent’ and ‘PanelSpyNumKeyEvent’). RRCS watches the connection state to artist and the online state of all hardware components needed to be able to receive panel spy events and synchronizes the panel spy registration table with artist automatically. Therefore there is no need to reregister anything if for example a node-controller or a client-card reboots.

However, if RRCS cannot reach the external control system (if the external control system does not respond to the ‘GetAlive’ notification because, for instance, a network switch has failed) then no further notifications are sent. Therefore an external control system should periodically ask RRCS via ‘IsRegisteredForAllEvents’ whether the notification channel is still active or not and should restart the two registration steps on deactivation.

RRCS places no limit on the number of external control systems or on the number of panel spy receivers for each control system. If a control system registers itself multiple times for a single panel, it will receive multiple copies of the same panel notifications.

9.9.2 PanelSpyStateChange

Description	Remote Procedure	Parameters	Return Value

Notifies panel spy status changes for a list of panels	PanelSpyStateChange	`<param><value><string>{TransKey}</string></value></param>`
		`<param><value><array><data>`
		`{ panel spy state of 1st panel (see below)}`
		`{ panel spy state of 2nd panel (see below)}`
		`…`
		`</data></array></value></param>`
			ignored by RRCS

A single panel spy state is a xml-rpc-struct which contains the following members:

Name	Type	Parameter description
Node	Int	The panel’s node address
Port	int	The panel’s port address
Rotate	struct	Defines rotate event type status (see below)
Key	struct	Defines key event type status (see below)
FuncKey	struct	Defines function key event type status (see below)
NumKey	struct	Defines the numerical key event type status (see below)

The rotate/key/function key and numerical key status is defined as follows (xml-rpc-struct):

Name	Type	Parameter description
State	int	• 0=unregistered: Panel spy not registered • 1=busy: Panel spy registered: RRCS tries to activate the panel spy in artist. This is always the first notification-state after a successful ChangePanelSpyRegistry call • 2=active: The control system receives panel spy notifications • 3=error: The control system does nto receive panel spy notifications (however, RRCS still tries to activate the panel spy). else: undefined value
ErrorCode	int	Defines the error code. This struct member exists only if the state member indicates an error (=3)
ErrorDescription	string	A human readable error reason. This struct member exists only if the state member indicates an error (=3)

9.9.3 PanelSpyRotateEvent

Description	Remote Procedure	Parameters	Return Value
Rotary encoder change notification	PanelSpyRotateEvent	<param><value><string>{TransKey}</string></value></param> <param><value><struct>{RotateEventData}</struct</value></param>	ignored by RRCS

RotateEventData:

<table>
<tr><th>Name</th><th>Type</th><th>Parameter description</th></tr>
<tr><td>Node</td><td>int</td><td>The panel’s node address</td></tr>
<tr><td>Port</td><td>int</td><td>The panel’s port address</td></tr>
<tr><td>SubPanel</td><td>int</td><td>Specifies whether the event occured on the main panel (SubPanel = 0) or from an expansion panel.
If the event occured on an expansion panel a value of
- 1 means: 1st expansion panel
- 2 means: 2nd expansion panel
- ...
- 6 means: 6-th expansion panel
Other values are invalid</td></tr>
<tr><td>Key</td><td>int</td><td>Defines key number (< 0 means: primary rotary)</td></tr>
<tr><td>RotationTicks</td><td>int</td><td>The number of rotation ticks (positive values == rotate right)</td></tr>
<tr><td>FuncKeyStates</td><td>struct</td><td><table>
<tr><th>Name</th><th>Type</th><th>Parameter description</th></tr>
<tr><td>Shift</td><td>boolean</td><td>Shift page function key (key-no=0)</td></tr>
<tr><td>Hs</td><td>boolean</td><td>Headset function key (key-no=1)</td></tr>
<tr><td>Opt</td><td>boolean</td><td>Options function key (key-no=2)</td></tr>
<tr><td>Beep</td><td>boolean</td><td>Beep/F1 function key (key-no=3)</td></tr>
<tr><td>Norm</td><td>boolean</td><td>Norm/F2 function key (key-no=4)</td></tr>
</table></td></tr>
</table>

9.9.4 PanelSpyKeyEvent

Description	Remote Procedure	Parameters	Return Value
Key event notification	PanelSpyKeyEvent	<param><value><string>{TransKey}</string></value></param> <param><value><struct>{KeyEventData}</struct</value></param>	ignored by RRCS

KeyEventData:

Name	Type	Parameter description
KeyType	int	0 Standard key event, left position 1 Key event on standard key, right position. Only available in 1000 and 1100 series. 2 Rotary or primary rotary key event. 3 Standard key event, up position 4 Standard key event, down position
KeyAction	int	0 Key pressed (down) 1 Key released (up) 2 Double click event. (Currently not supported. Maybe supported in future releases) 3 Key long-press event. (Currently not supported. Maybe supported in future releases)
Node	int	The panel’s node address
Port	int	The panel’s port address
SubPanel	int	Specifies whether the event occured on the main panel (SubPanel = 0) or from an expansion panel. If the event occured on an expansion panel a value of - 1 means: 1st expansion panel - 2 means: 2nd expansion panel - ... - 6 means: 6-th expansion panel Other values are invalid
Key	int	Defines key number (< 0 means: primary rotary)
IsKeyLatched	boolean	Tells whether the key is in latching state or not

<table>
<tr><th>Name</th><th>Type</th><th>Parameter description</th></tr>
<tr><td>FuncKeyStates</td><td>struct</td><td>
<table>
<tr><th>Name</th><th>Type</th><th>Parameter description</th></tr>
<tr><td>Shift</td><td>boolean</td><td>Shift page function key (key-no=0)</td></tr>
<tr><td>Hs</td><td>boolean</td><td>Headset function key (key-no=1)</td></tr>
<tr><td>Opt</td><td>boolean</td><td>Options function key (key-no=2)</td></tr>
<tr><td>Beep</td><td>boolean</td><td>Beep/F1 function key (key-no=3)</td></tr>
<tr><td>Norm</td><td>boolean</td><td>Norm/F2 function key (key-no=4)</td></tr>
</table>
</td></tr>
</table>

9.9.5 PanelSpyFuncKeyEvent

Description	Remote Procedure	Parameters	Return Value
Function key event notification	PanelSpyFuncKeyEvent	`<param><value><string>{TransKey}</string></value></param>` `<param><value><struct>{FuncKeyEventData}</struct</value></param>`	ignored by RRCS

FuncKeyEventData:

Name	Type	Parameter description
FuncKeyAction	Int	0 function key pressed (down) 1 function key released released (up) 2 Auto-off: The function key has been turned off automatically (e.g. via a timer)
Node	int	The panel’s node address
Port	int	The panel’s port address
FuncKeyNo	int	0 Shift 1 Hs 2 Opt 3 Beep 4 Norm
FuncKeyStates	struct	**Name** – **Type** – **Parameter description** Shift – boolean – Shift page function key (key-no=0) Hs – boolean – Headset function key (key-no=1) Opt – boolean – Options function key (key-no=2) Beep – boolean – Beep/F1 function key (key-no=3) Norm – boolean – Norm/F2 function key (key-no=4)

9.9.6 PanelSpyNumKeyEvent

Description	Remote Procedure	Parameters	Return Value
Numerical key event notification	PanelSpyNumKeyEvent	`<param><value><string>{TransKey}</string></value></param>` `<param><value><struct>{NumKeyEventData}</struct</value></param>`	ignored by RRCS

NumKeyEventData:

Name	Type	Parameter description
NumKeyAction	Int	0 Numerical key pressed (down) 1 Numerical key released released (up)
Node	int	The panel's node address
Port	int	The panel's port address
SubPanel	int	Specifies whether the event occured on the main panel (SubPanel = 0) or from an expansion panel. If the event occured on an expansion panel a value of - 1 means: 1st expansion panel - 2 means: 2nd expansion panel - ... - 6 means: 6-th expansion panel Other values are invalid

Name	Type	Parameter description
NumKeyNo	int	1 M1 2 M2 3 M3 4 M4 5 M5 6 M6 7 M7 8 M8 35 '#' 42 '*' 48 '0' 49 '1' 50 '2' 51 '3' 52 '4' 53 '5' 54 '6' 55 '7' 56 '8' 57 '9' 67 clear = 'C' character 80 'P' 82 Redial = 'R' character 83 Speaker = 'S' character

<table>
<tr><th>Name</th><th>Type</th><th>Parameter description</th></tr>
<tr><td>FuncKeyStates</td><td>struct</td><td><table>
<tr><th>Name</th><th>Type</th><th>Parameter description</th></tr>
<tr><td>Shift</td><td>boolean</td><td>Shift page function key (key-no=0)</td></tr>
<tr><td>Hs</td><td>boolean</td><td>Headset function key (key-no=1)</td></tr>
<tr><td>Opt</td><td>boolean</td><td>Options function key (key-no=2)</td></tr>
<tr><td>Beep</td><td>boolean</td><td>Beep/F1 function key (key-no=3)</td></tr>
<tr><td>Norm</td><td>boolean</td><td>Norm/F2 function key (key-no=4)</td></tr>
</table></td></tr>
</table>

10 Redundancy

RRCS is designed to run in a redundant environment. The Gateway software has two states “Working” and “Standby”. The state is checked and controlled by an external Control System. To achive that, RRCS provides a set of commands.. These “Redundency” commands are supported on both (Working and Standby) gateways at anytime.

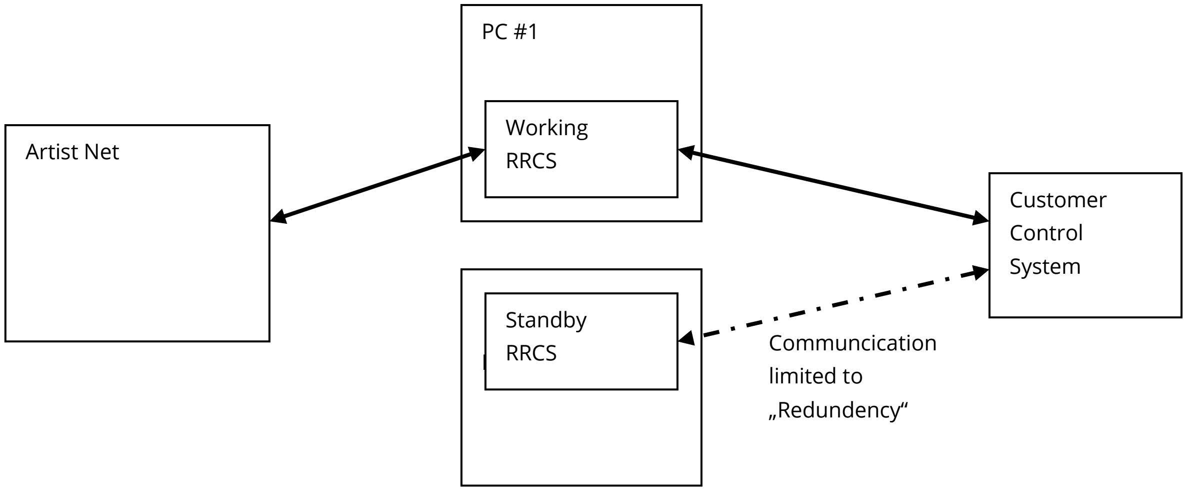
“GetState” to query the Gateway state (Working/Standby).

“SetStateWorking” and “SetStateStandby” to switchover the Gateway.

“GetAlive” to ckeck that the gateway is still up and running.

“GetArtistConnectionState” to query the network connection state between the gateway and the Artist net.

“ConnectArtistFailure” and “ConnectArtistFailureRestored” to report the Artist connection state.



Possible scenarios that require a switchover:

The Working Gateway PC has a power failure and does not to answer to “GetAlive” requests.

The network between the Customer Control System and the Working network PC fails and the gateway does not to answer to “GetAlive” requests.

The Working Gateway cannot connect to Artist net and reports “ConnectArtistFailure”.

A switchover to the the Standby Gateway only makes sense, if it’s in a good condition. That means the Standby Gateway answers to “GetAlive” requests and the network connection to Artist is ok. If the Control System initiates a switchover the Working gateway should be set to Standby (if possible), before the Standby gateway is set to Working. The case that there are two Working gateways at the same time must be avoided, because this could cause undefined results in the Artist network.

11 Communicaction Examples

11.1 Example for an incoming request to RRCS

This is an example for setting a Crosspoint request between source port 35 on node 2 and destination port 69 on node 5. Only the XML data is shown.

```
<?xml version="1.0"?>
<methodCall>
   <methodName>SetXp</methodName>
   <params>
      <param><value><string>C0002817191</string></value></param>
      <param><value><int>1</int></value></param>
      <param><value><int>2</int></value></param>
      <param><value><int>19</int></value></param>
      <param><value><int>1</int></value></param>
      <param><value><int>5</int></value></param>
      <param><value><int>3</int></value></param>
   </params>
</methodCall>

Response:
<?xml version="1.0"?>
<methodResponse>
   <params>
      <param>
         <value><array><data>
            <value><string>C0002817191</string></value>
            <value><int>0</int></value>
         </data></array></value>
      </param>
   </params>
</methodResponse>
```

11.2 Example for an outgoing request from RRCS

This is an example for an “Upstream Failed” event.

```
<?xml version="1.0"?>
<methodCall>
   <methodName>UpstreamFailed</methodName>
   <params>
      <param>
         <value><string>R1947584733</string></value>
      </param>
      <param>
         <value><int>2</int></value></param>
      </params>
</methodCall>

Response:
<?xml version="1.0"?>
<methodResponse>
   <params>
      <param>
         <value><array><data>
            <value><string>R1947584733</string></value>
            <value><int>0</int></value>
         </data></array></value>
      </param>
   </params>
</methodResponse>
```

12 Performance

The time to perform an individual remote procedure call is related to several variables. The remote procedure calls can be split into four categories.

12.1 Querying information

The performance of RPCs querying system information is related to the network and PC processing performance. If the PC and network are idle, these RPCs are typically answered within 10ms.

These RPCs are querying information:

```
GetXpStatus
GetAllActiveXps
GetAllActiveXpsCount
GetActiveXpsRange
GetXpVolume
GetPortAlias
GetInputGain
GetOutputGain
GetGpInputState
GetGpOutputState
GetAllCaps
GetAllLogicSources
GetObjectList
```

12.2 Pure RRCS communication

Some RPCs are purely related to RRCS internal states and are related to the network and PC processing performance. If the PC and network are idle, these RPCs are typically answered within 1ms.

```
GetState
GetVersion
SetStateWorking
SetStateStandby
RegisterForEvents
UnregisterForEvents
RegisterForGpInputChange
UnregisterForGpInputChange
RegisterForGpOutputChange
UnregisterForGpOutputChange
RegisterForEventsEx
UnregisterForEventsEx
```

12.3 Changing system states

The performance of RPCs querying system information is related to the network performance, PC processing performance and usage of the Artist system. If the PC, network and Artist are idle, these RPCs are typically answered within 50ms.

These RPCs are changing the system state:

```
SetXp
SetXpPrio
SetXpDestructive
KillXp
SetXpVolume
SetInputGain
SetOutputGain
SetGpOutput
SetLogicSourceState
```

12.4 Changing system configuration

The performance of RPCs querying system information is related to the network performance, PC processing performance, the system configuration size and the usage of the Artist system. If the PC, network and Artist are idle, these RPCs are typically answered within several seconds. The response time will be much longer, if other PCs change the configuration at the same time.

These RPCs are changing the system configuration:

```
SetPortAlias
ConfigurationChange
```

12.5 Unexpected Delay on consecutive ConfigurationChange Requests on hybrid Artist Rings

In rare situations unexpected delays can occur if 2 consecutive requested configuration changes are made in hybrid rings (mixed setups with Artist-32/64/128 and Artist-1024) via RRCS.

It is strongly recommended in general, but especially if you observe such situations frequently, to implement one or both of the following two options, depending on which solution fits best to your use cases:

The ConfigurationChange API is a list of ConfigurationChange operations. Each operation can be of different type and aim different configuration objects.
It is recommended to use this API whenever several context-dependent operations are to be carried out, such as re-configuring a key, a port or a bigger part of the system trigger by one user action.

The BufferConfigurationChange API in conjunction with ApplyConfigurationChange API works like the ConfigurationChange API, but you can use RRCS as a ConfigurationChange operation cache/queue and control the time when all enqueued operations get applied within the RRCS-driver implementation, probably by a timer.
It is recommended to use this API when option one does not fit to your use cases or you already have implemented option one and still observe the above-mentioned situation, but in general whenever you expect high frequently requested context-independent configuration changes.
In between of multiple calls of the BufferConfigurationChange API and ApplyConfigurationChange API, all other operational and configurational APIs can still be called, for instance to ensure time-critical operations are performed with higher priority.
Please Note: When requesting configurational operations in between, the timer for applying buffered configuration changes should be adapted to not run into the same issue again.